Department of
Mechanical Engineering

Laboratory Report

Name of Course: Computational Fluid Dynamics & Heat Transfer

Course Code: 20ME406P

Semester: VII

Branch: Mechanical Engineering

Batch (Division): Division 1

Name of Student: Ronit Chakraborty

Roll number: 23BME049

Academic Year: 2026-27

20ME406P Computational Fluid Dynamics and Heat Transfer Lab

Certificate

This is to certify that Mr./Miss. Ronit Chakraborty (Roll number: 23BME049) of Semester VII (AY: 2026-27) - B.Tech. Mechanical Engineering has satisfactorily completed his/her laboratory work in **Computational Fluid Dynamics and Heat Transfer Laboratory** for the term ending in December.

Date:

Signature of Teacher

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Sr. No.	Title of experiment	Date of experiment	Date of evaluation	Page no.	Marks/ Grade	Teacher's signature	CO
1	External flow analysis over NACA 0012 airfoil for different angle of attacks.						CO1, CO4, CO5, CO6
2	Fluid flow and heat transfer analysis in a double pipe heat exchanger.						CO2, CO4, CO5, CO6
3	Internal fluid flow and heat transfer study in a centrifugal pump.						CO1, CO2, CO4, CO5
4	Analysis of Karman-Vortex over a cylinder (Vortex shedding).						CO1, CO4, CO5, CO6
5	Couette flow analysis for both explicit and implicit formulation (Parabolic equation).						CO4, CO5, CO6
6	Heat conduction in 2D flat plate with explicit and implicit formulation (Elliptic equation).						CO2, CO4, CO5, CO6
7	1D wave propagation in a still lake (Hyperbolic equation).						CO4, CO5, CO6
8	2D simulation of Rayleigh–Taylor instability (VOF method).						CO1, CO4, CO5, CO6
9	Lid driven cavity flow analysis utilizing vorticity-streamfunction framework.						CO1, CO4, CO5, CO6
10	Study of different turbulent models to analyse the flow in a pipe for various Reynolds number.						CO1, CO4, CO5, CO6
11	Simulation of a generic convection-diffusion transport equation with forced/natural convention over flat plat/in pipe.						CO2, CO4, CO5, CO6

Course Information

Subject Code	Subject	Department	Program
20ME406P	Computational Fluid Dynamics and Heat Transfer Lab	Mechanical Engineering	B.Tech.

Teaching and Examination Scheme

L	T	P	C	Hrs/Week	LW	LE/Viva	Total Marks	
0	0	2	1	2	25	25	50	

Course Objectives

1. To utilize computational tools to understand fluid-flow problems.
2. To employ computational tools to comprehend heat-transfer problems.
3. To apply numerical schemes to solve governing equations of physical systems.
4. To simulate forced and natural convection with internal and external flows.
5. To validate numerical results using available analytical, experimental, or benchmark data.

Course Outcomes

On completion of the course, students will be able to:

CO1	Understand computational software tools used to analyse fluid-flow problems.
CO2	Utilize computational tools to comprehend heat-transfer problems.
CO4	Classify and evaluate the physics of problems and apply appropriate discretization schemes.
CO5	Analyse and understand the results through post-processing for a given problem.
CO6	Compare simulation results with available experimental, analytical, or benchmark results.

General Instructions

1. Students shall compile the supplied C++17 source files using a standard GNU C++ compiler.
2. Every experiment shall include source-code execution, ParaView visualization, residual/history interpretation, and a short written result discussion.
3. The exact syllabus experiment titles are retained. The implementation section inside each experiment defines the numerical model used for the laboratory exercise.
4. No result shall be reported without grid size, Reynolds number or governing non-dimensional parameter, boundary conditions, and convergence or time-step information.
5. VTK files are used for visualization only. CSV files are the primary numerical output for plots and comparisons.
6. Each student shall complete the compulsory assignment. Individual assignments may be separately assigned by faculty.

Compilation Standard

```
g++ -O3 -std=c++17 source_file.cpp -o executable_name
```

Recommended References

1. S. V. Patankar, *Numerical Heat Transfer and Fluid Flow*, Taylor and Francis.

2. H. K. Versteeg and W. Malalasekera, *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*, Pearson.
3. J. H. Ferziger, M. Peric, and R. L. Street, *Computational Methods for Fluid Dynamics*, Springer.
4. U. Ghia, K. N. Ghia, and C. T. Shin, High-Re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method, *Journal of Computational Physics*, 1982.
5. C. Hirsch, *Numerical Computation of Internal and External Flows*, Butterworth-Heinemann.

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Common Build and Smoke-Test Procedure

The full package contains `build_all.sh` and `smoke_test_all.sh`. Use them before distributing the code to students.

```
chmod +x build_all.sh smoke_test_all.sh
./build_all.sh
./smoke_test_all.sh
```

Successful smoke tests demonstrate compilation and file generation. They do not replace grid-convergence studies or benchmark comparison.

1 Experiment 1: External flow analysis over NACA 0012 airfoil for different angle of attacks.

Aim

To analyse external flow over a NACA 0012 airfoil at different angles of attack using an incompressible finite-volume pressure-correction solver with an immersed airfoil mask and fluid-only VTK output.

Governing Equations and Physical Model

The governing equations are the steady incompressible Navier-Stokes equations in conservative finite-volume form:

$$\nabla \cdot \mathbf{u} = 0, \qquad \rho \nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{S}_{IB}.$$

The immersed-boundary penalty source $\mathbf{S}_{IB}$ enforces nearly zero velocity inside the NACA 0012 solid region. The airfoil is represented by the standard thickness distribution

$$y_t = 5tc \left(0.2969\sqrt{x/c} - 0.1260x/c - 0.3516(x/c)^2 + 0.2843(x/c)^3 - 0.1015(x/c)^4 \right).$$

Boundary Conditions

Uniform velocity is prescribed at the inlet. Zero-gradient flow is used at the outlet. Slip/far-field treatment is used at top and bottom boundaries. No-penetration/no-slip behavior is imposed through the immersed solid mask.

Numerical Method

Cell-centred finite-volume advection-diffusion with pressure correction. The output grid removes airfoil-intersecting cells so streamlines in ParaView do not pass through the solid body.

Stability, Boundedness, and Convergence Requirements

The iteration is pseudo-steady. Under-relaxation of velocity and pressure is required. Coarse grids cannot produce accurate lift or drag; qualitative pressure and wake interpretation is the intended laboratory outcome.

Compilation and Execution

```
g++ -O3 -std=c++17 code/exp01_airfoil_fvm_simple.cpp -o exp01
./exp01_airfoil_fvm_simple 120 60 100 5 300 80
```

Expected Output Files

```
airfoil_final.vtk
airfoil_wall.vtk
history.csv
```

Benchmark or Verification Requirement

Check qualitative stagnation region, acceleration over the airfoil surface, wake deflection with angle of attack, and sensitivity to grid refinement. Do not claim validated aerodynamic coefficients unless compared against a body-fitted benchmark.

Compulsory Assignment

Run the case for angle of attack 0, 5, and 10 degrees at Re = 100. Plot the pressure field and streamline pattern. Report how stagnation, wake deflection, and upper/lower surface acceleration change with angle of attack.

Record in Lab Manual

Students must submit: governing equation, computational domain, boundary conditions, mesh/grid size, input parameters, residual or history plot, ParaView contour/streamline figure, numerical table, and a short conclusion.

Parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Grid X	120
Grid Y	60
Reynolds Number	100
Angle of attack	-5,0,5

Governing Equation

$$\nabla \cdot \mathbf{u} = 0,$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u}.$$

Boundary Condition: Uniform flow, Uniform far-field velocity on the upper and lower bounaries, no-slip condition.

1) For -5 degree angle of attack

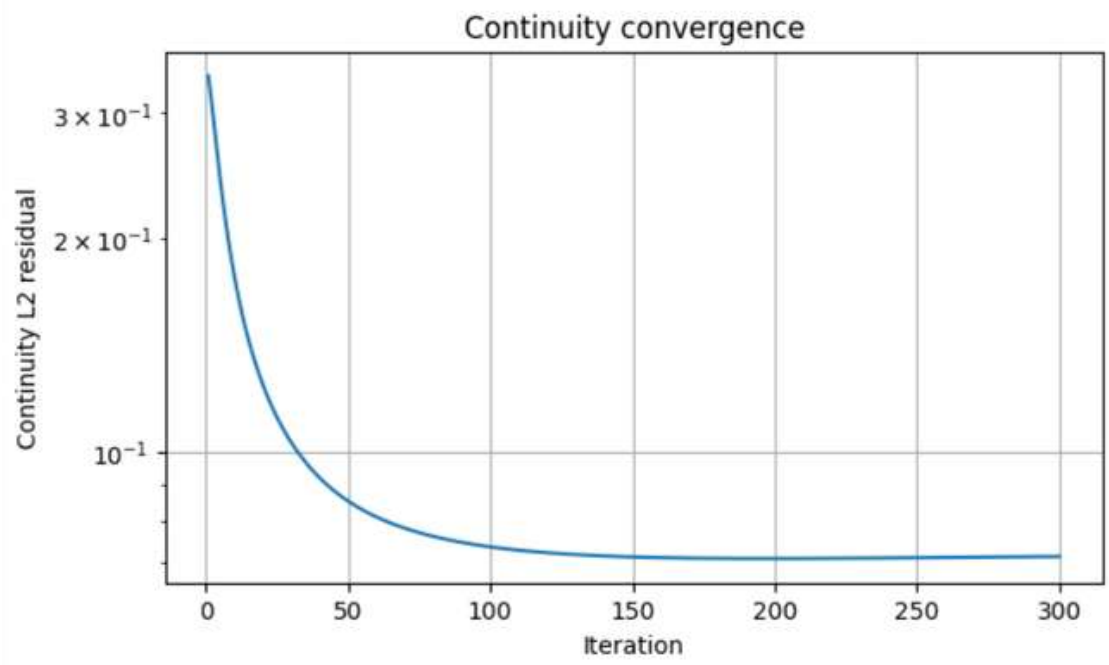


Fig1: Convergence with iterations

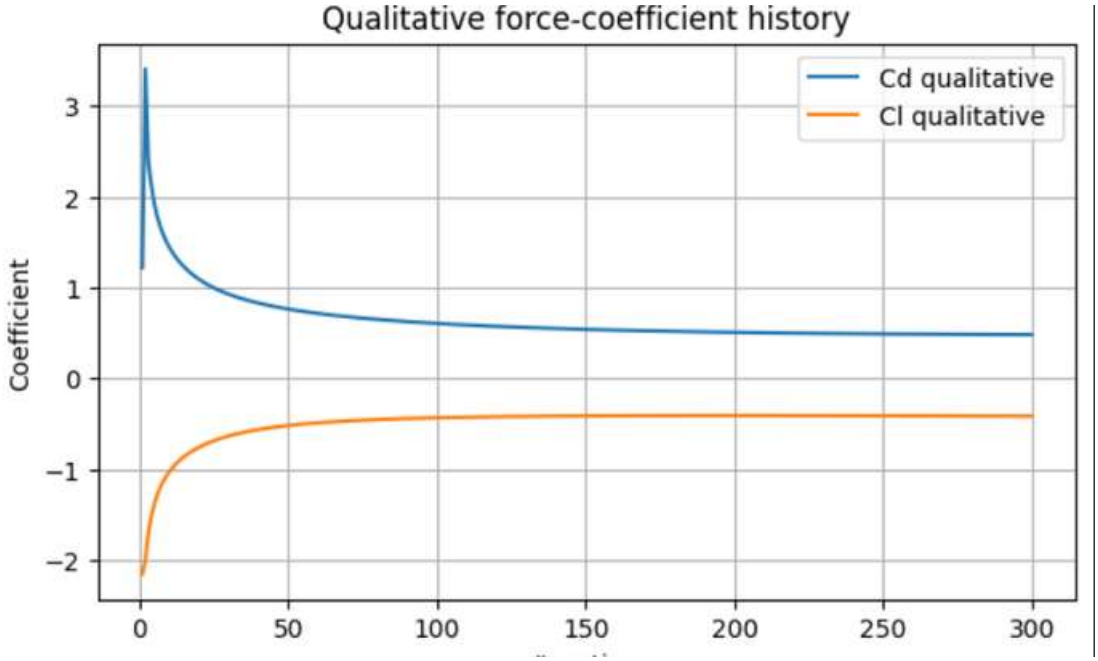


Fig2: Coefficient of drag and lift for -5 degree of attack

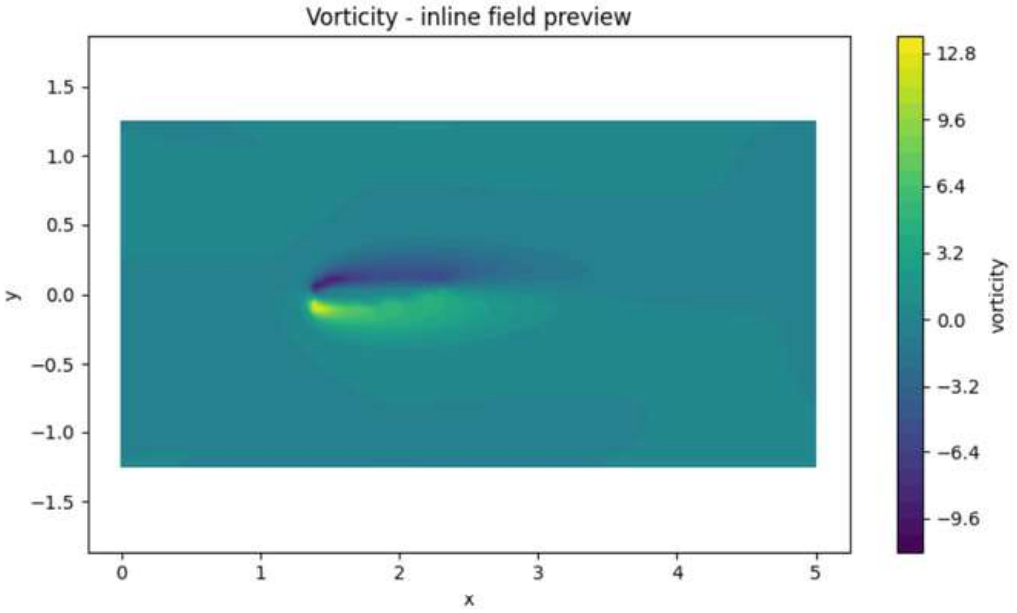


Fig3: Voritcity field wake visualization

2) For 0 degree angle of attack

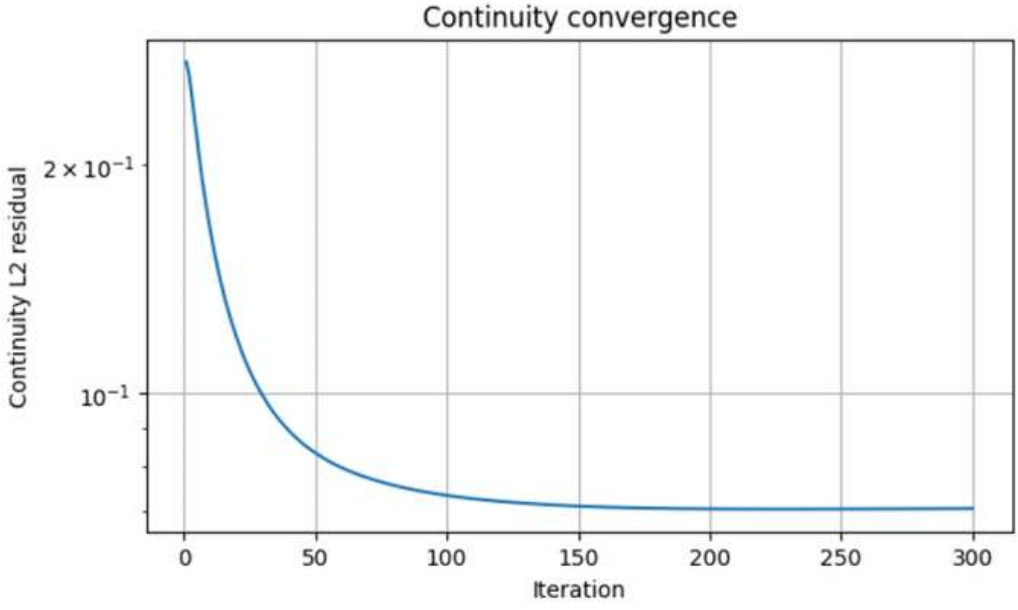


Fig4: Convergence with iterations

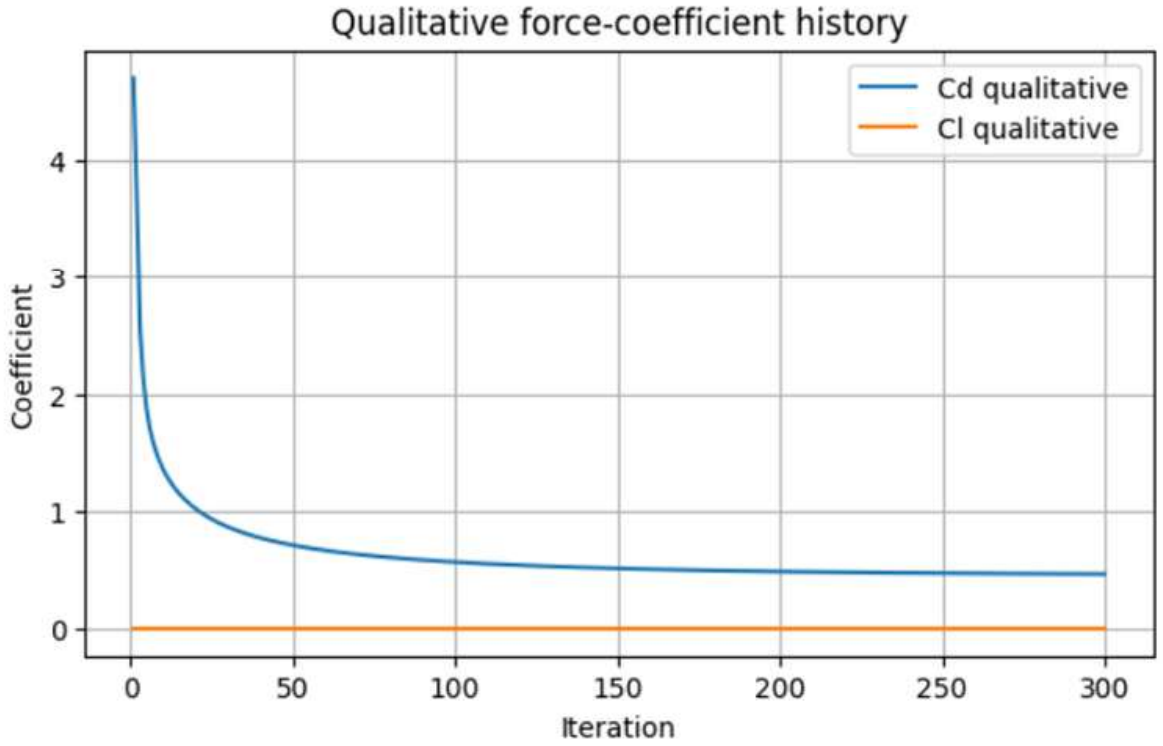


Fig5: Coefficient of drag and lift for 0 degree of attack

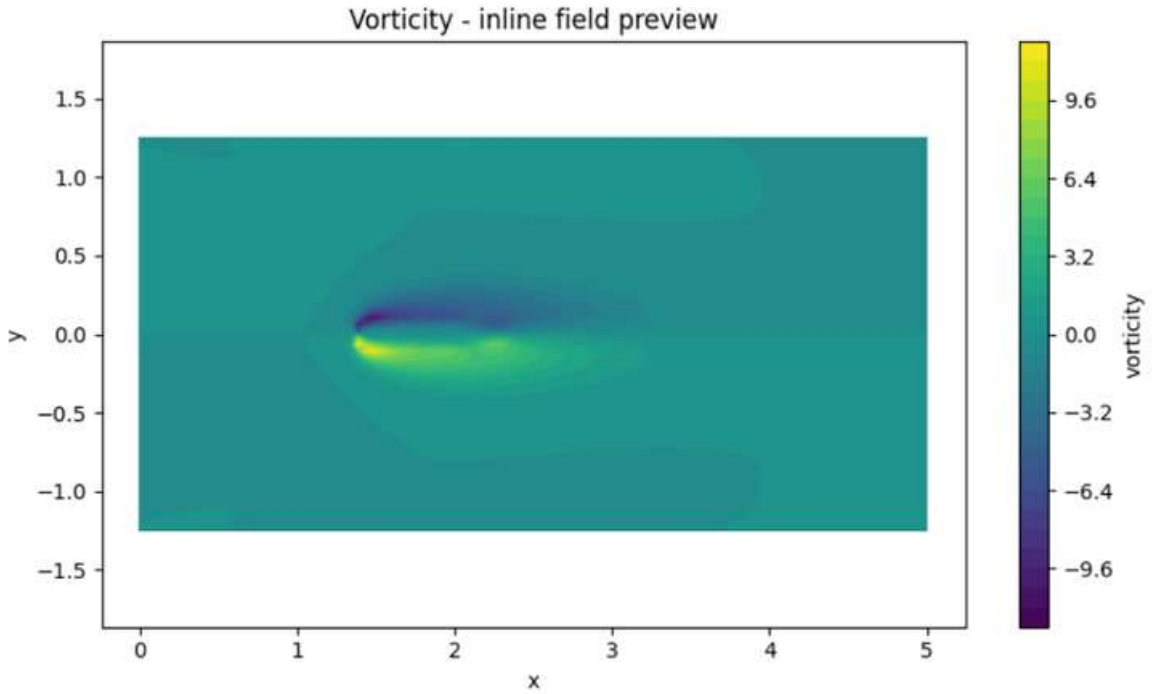


Fig6: Voritcity field wake visualization

3) For 5 degree angle of attack

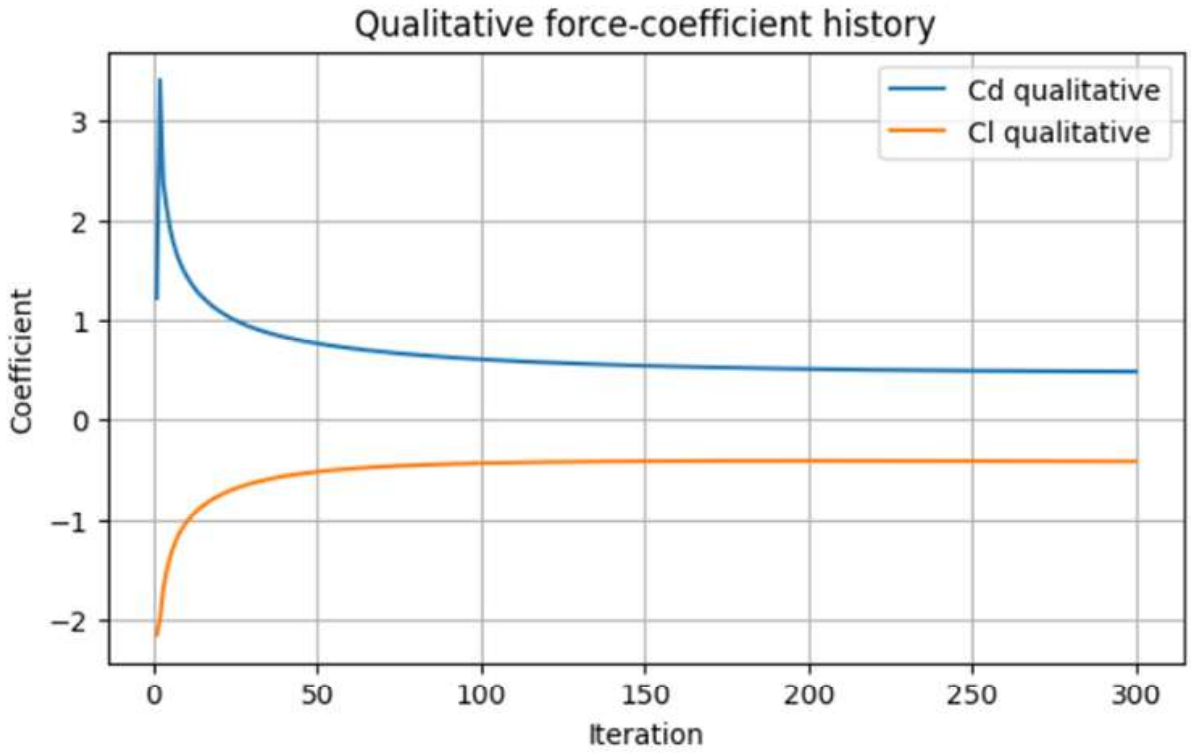


Fig7: Coefficient of drag and lift for 5 degree of attack

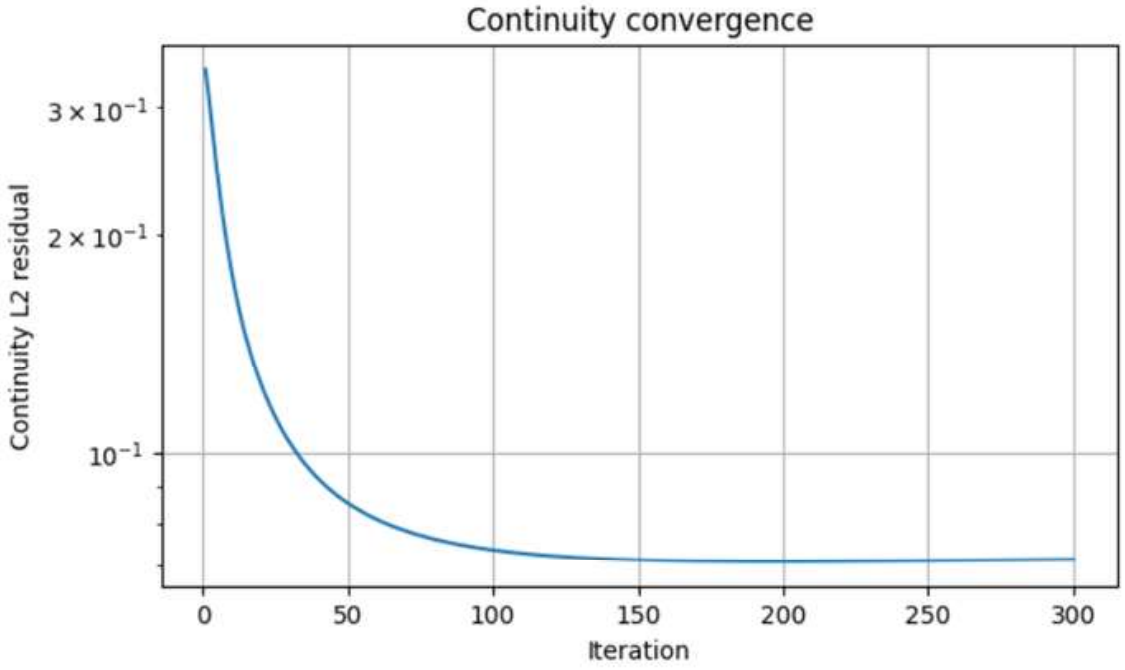


Fig8: Convergence with iterations

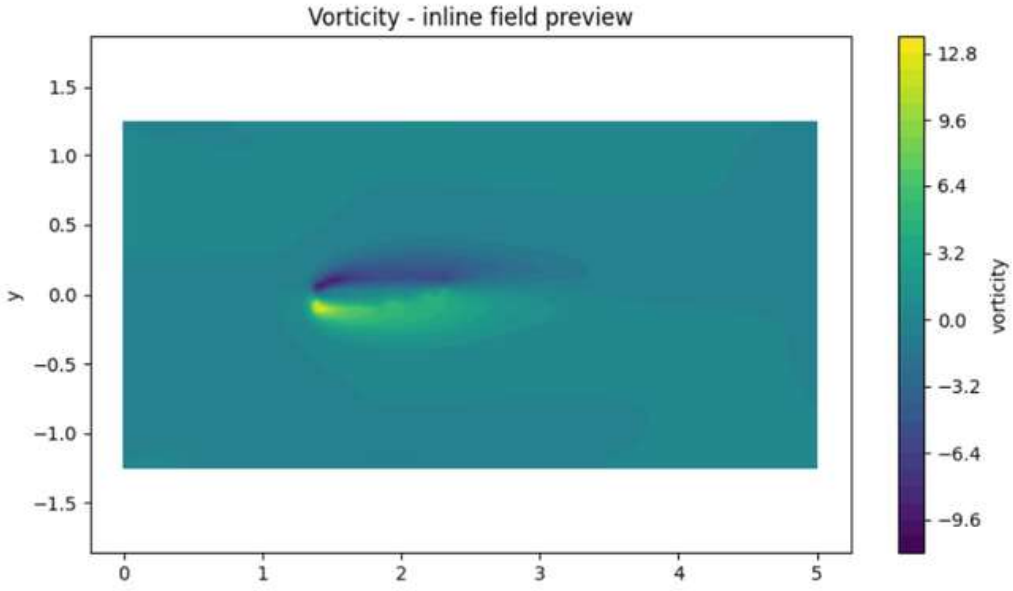


Fig9: Voritcity field wake visualization

Result:

S. No.	Angle of Attack	Drag Coefficient at Convergence	Lift Coefficient at Convergence
1	-5	0.45	0.4
2	0	0.5	0
3	5	0.45	0.39

Conclusion

The external flow analysis over a NACA 0012 airfoil was analysed for angles of attack of -5,0,5, at a Reynolds number of 100. With increasing iterations, it approached a stable value which indicates convergence of the numerical solution. The drag coefficient also stabilized as the simulation progressed, while the lift coefficient varied with the angle of attack. The vorticity contours illustrated the wake formation behind the airfoil. Overall, the experiment demonstrated the application of computational fluid dynamics in analyzing flow behavior and aerodynamic force coefficients around an airfoil.

2 Experiment 2: Fluid flow and heat transfer analysis in a double pipe heat exchanger.

Aim

To analyse conjugate heat transfer between two fluid streams separated by a conducting wall in a rectangular double-pipe heat-exchanger analogue.

Governing Equations and Physical Model

The fluid energy equation is

$$\rho c_p \mathbf{u} \cdot \nabla T = k_f \nabla^2 T.$$

The wall conduction equation is

$$\nabla \cdot (k_s \nabla T) = 0.$$

At both fluid-solid interfaces,

$$T_f = T_s, \qquad k_f \frac{\partial T_f}{\partial n} = k_s \frac{\partial T_s}{\partial n}.$$

Boundary Conditions

Hot and cold inlet temperatures are specified. Outlets use zero-gradient temperature. Fluid-solid interfaces enforce no-slip velocity for interpretation and conjugate heat-transfer continuity for temperature. External top and bottom are adiabatic.

Numerical Method

Finite-volume convection-diffusion is solved over a three-zone grid. Fluid velocities are prescribed Poiseuille profiles. The thermal problem is iterated until temperature residual reduction.

Stability, Boundedness, and Convergence Requirements

Cell Peclet number governs numerical diffusion. If $|Pe_c| > 2$, central differencing becomes oscillatory; bounded upwind treatment is safer for undergraduate execution.

Compilation and Execution

```
g++ -O3 -std=c++17 code/exp02_double_pipe_cht_fvm.cpp -o exp02
./exp02_double_pipe_cht_fvm 160 20 6 20 100 100 20 1 3000 7 1 0 1 1
```

Expected Output Files

```
double_pipe_cht.vtk
history.csv
summary.txt
```

Benchmark or Verification Requirement

Compare energy balance between hot-fluid heat loss, cold-fluid heat gain, and wall-conducted heat. Mesh refinement should reduce imbalance.

Compulsory Assignment

Compare co-current and counter-current configurations for identical inlet temperatures and flow rates. Report hot outlet temperature, cold outlet temperature, heat-transfer rate, and effectiveness.

Record in Lab Manual

Students must submit: governing equation, computational domain, boundary conditions, mesh/grid size, input parameters, residual or history plot, ParaView contour/streamline figure, numerical table, and a short conclusion.

Parameters

Parameters	Value
Re for Hot fluid	300,50
Re for Cold Fluid	300,50
Flow type	Parallel Flow
Wall to fluid ratio	10,50,100

Boundary Conditions: The heat exchanger is divided into 3 different zones namely Hot Zone, Cold Zone and Wall zone. External boundaries are adiabatic, and continuity of temperature and heat flux at fluid-solid interfaces.

1) Parallel Flow (Wall to fluid ratio = 10, Re_hot=Re_cold=300)

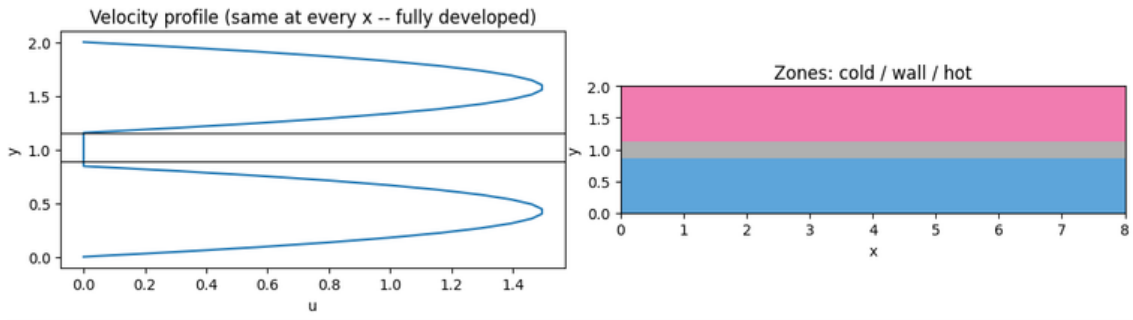


Fig1: Velocity Profile

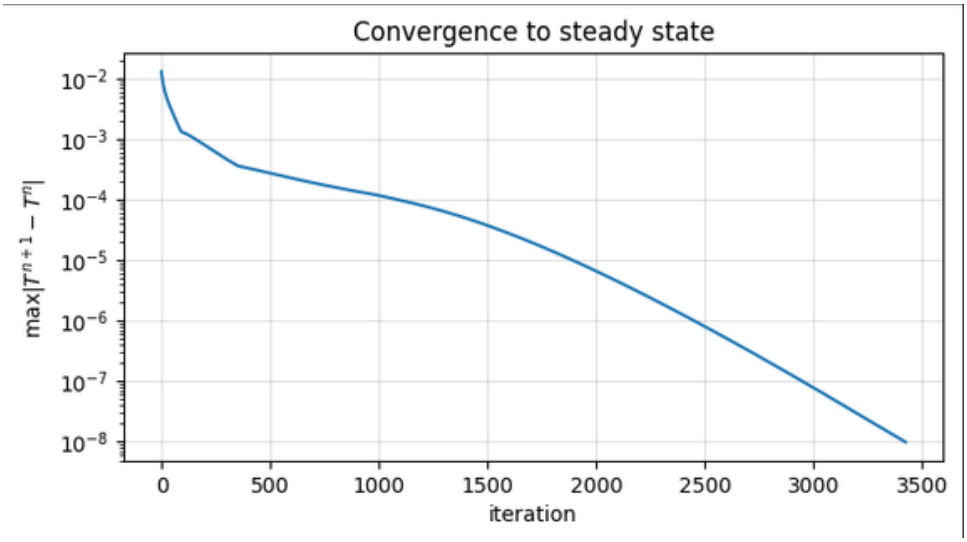


Fig2: Convergence to steady state

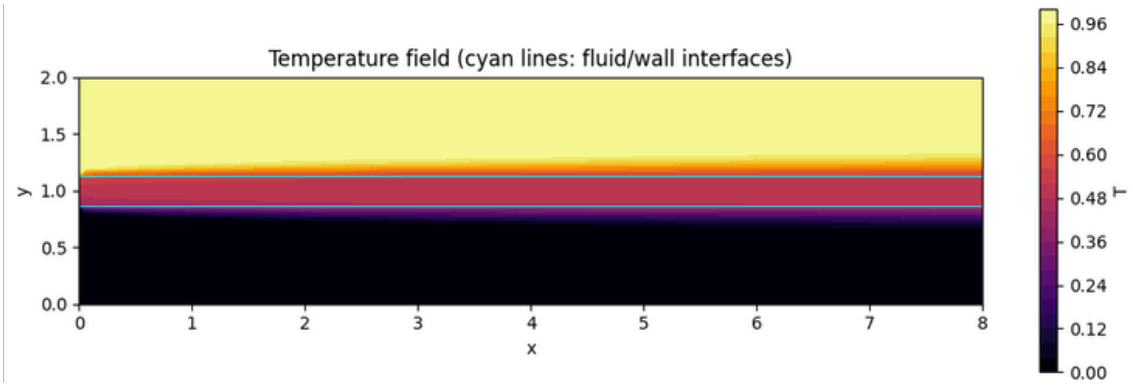


Fig3: Temperature Field in all zones with thermal boundary layer

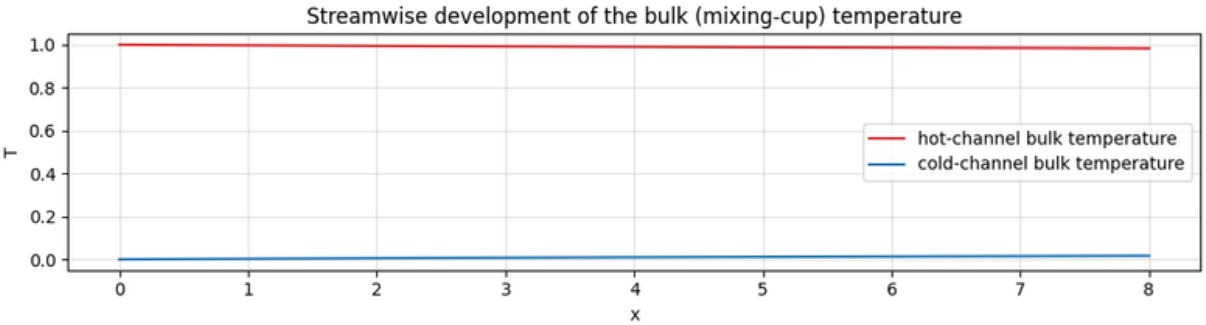


Fig4: Temperature vs Length profile for hot and cold fluid

2) Parallel Flow (Wall to fluid ratio = 50, Re_hot=Re_cold=50)

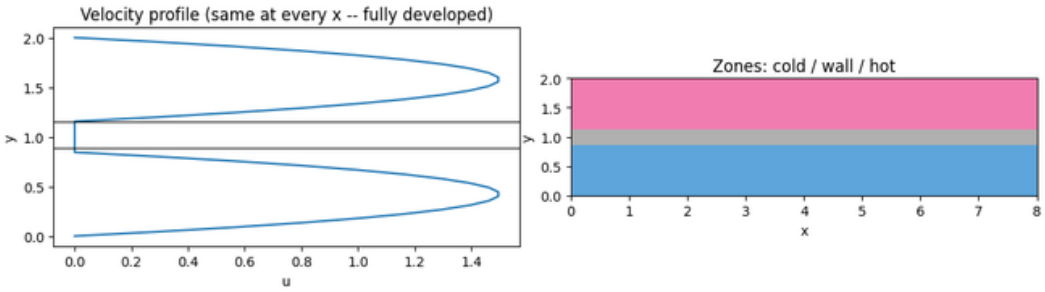


Fig5: Velocity Profile

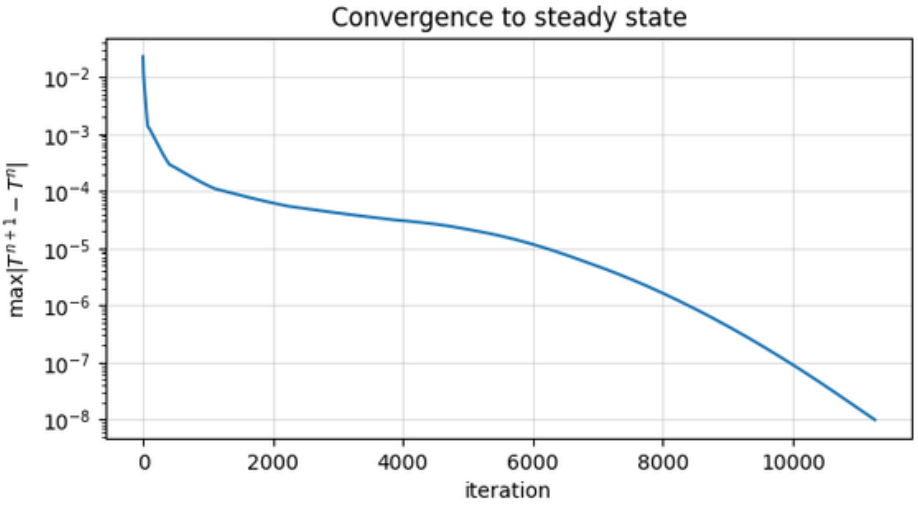


Fig6: Convergence to steady state

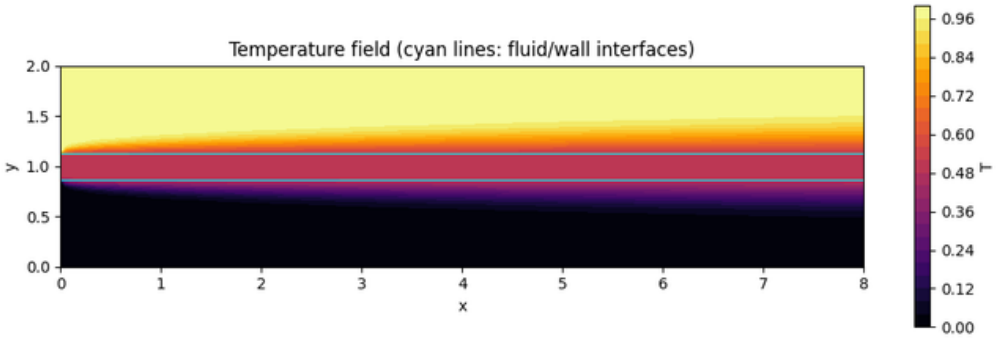


Fig7: Temperature Field in all zones with thermal boundary layer

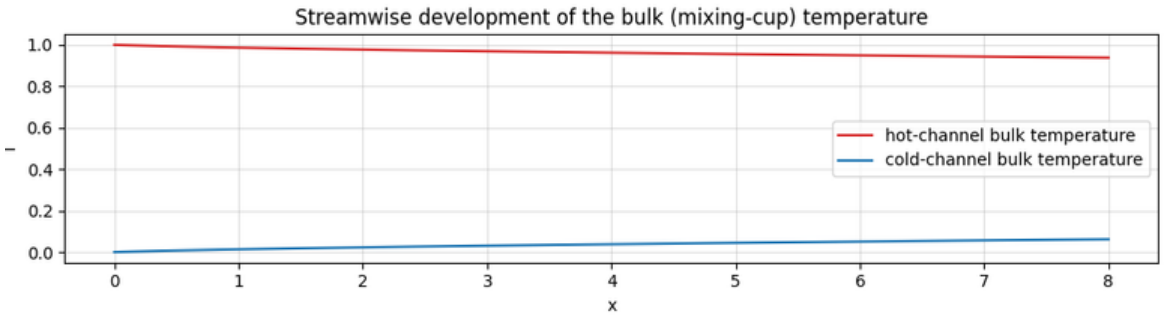


Fig8: Temperature vs Length profile for hot and cold fluid

Result:

Reynolds Number	Wall to Fluid Ratio	Hot Temp. Inlet	Hot Temp. outlet	Cold Temp. Outlet
300	10	1	0.97	0.02
50	50	1	0.94	0.06

Conclusion: The hot-stream inlet temperature was maintained at Th=1.0, while the cold-stream inlet temperature was maintained at Tc=0. For parallel flow with a wall-to-fluid ratio of 10 and Reynolds numbers of 300 for both hot and cold streams, the hot outlet temperature remained approximately 0.97, while the cold outlet temperature increased slightly from its inlet value. When the wall-to-fluid ratio was increased to 50 and the Reynolds numbers of both streams were reduced to 50, the hot outlet temperature decreased to approximately 0.94, and the cold outlet temperature increased to approximately 0.06. The temperature contours showed heat transfer from the hot stream through the conducting wall to the cold stream. The temperature residual decreased with increasing iterations, reaching approximately 10^-8, indicating convergence toward a steady-state solution. Overall, the results demonstrate the influence of flow conditions and wall thermal properties on the conjugate heat transfer performance of a double-pipe heat exchanger.

3 Experiment 3: Internal fluid flow and heat transfer study in a centrifugal pump.

Aim

To study internal flow and heat transfer in a centrifugal-pump passage using a 2D annular single-rotating-frame representation.

Governing Equations and Physical Model

In a rotating frame, the incompressible momentum equation is

$$\rho(\mathbf{u}_r \cdot \nabla)\mathbf{u}_r = -\nabla p + \mu\nabla^2\mathbf{u}_r - 2\rho\boldsymbol{\Omega} \times \mathbf{u}_r - ho\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}).$$

The centrifugal contribution gives the radial pressure-rise scale

$$\Delta p \approx \frac{1}{2}\rho\Omega^2(r_2^2 - r_1^2).$$

The temperature equation is

$$\rho c_p \mathbf{u} \cdot \nabla T = k\nabla^2 T + \dot{q}_{loss}.$$

Boundary Conditions

The inner circular boundary represents pump-eye inlet. The outer circular boundary represents impeller outlet. The annular wall temperature is prescribed through a heat-transfer coefficient.

Numerical Method

Polar-grid finite-volume scalar transport with analytically consistent radial through-flow, swirl, centrifugal pressure-rise field, and wall/loss heating. VTK guide files explicitly show inlet, outlet, blade guides, and flow-path guides.

Stability, Boundedness, and Convergence Requirements

The interpretation of velocity magnitude must separate radial velocity, tangential velocity, absolute velocity, and relative velocity. Tangential velocity naturally increases with radius.

Compilation and Execution

```
g++ -O3 -std=c++17 code/exp03_centrifugal_pump_srf.cpp -o exp03
./exp03_centrifugal_pump_srf 80 160 2000 500 300 330 1200 0.85 120 0.04
```

Expected Output Files

```
pump_annulus_srf.vtk
pump_boundaries.vtk
pump_flow_path_guides.vtk
pump_summary.txt
```

Benchmark or Verification Requirement

Pressure rise should scale approximately with Ω^2. Pump head should scale as $\Delta p/(hog)$.

Compulsory Assignment

Run 1000, 2000, and 3000 rpm cases. Plot pressure rise and pump head versus rpm. Verify whether the numerical trend follows an approximately quadratic relationship.

Record in Lab Manual

Students must submit: governing equation, computational domain, boundary conditions, mesh/grid size, input parameters, residual or history plot, ParaView contour/streamline figure, numerical table, and a short conclusion.

Parameters:

Parameter	Value
RPM	1000,2000
Re	700,1000
Temp. Inet	300
Temp. Wall	330
Iterations	2000

Boundary Conditions: Specified Inlet Temperature, radial inflow at the eye, outward radial transport, wall heat exchange represented by distributed source.

1) For pump at Re=700, RPM=1000

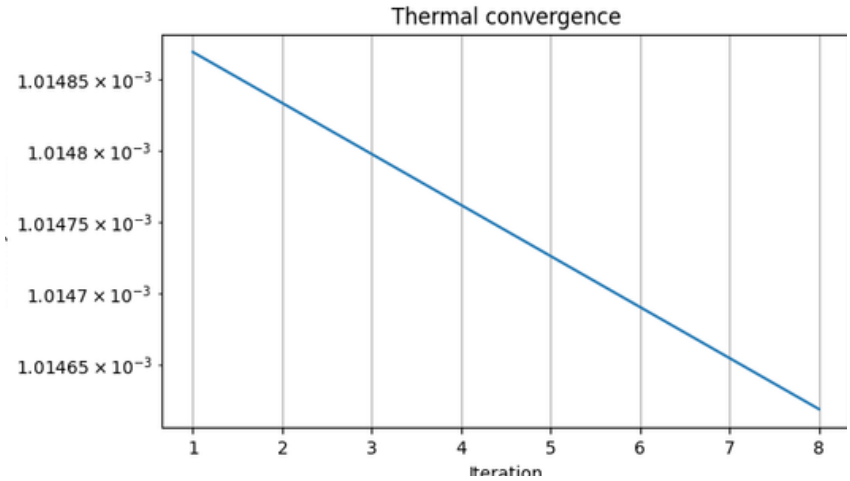


Fig1: Thermal convergence

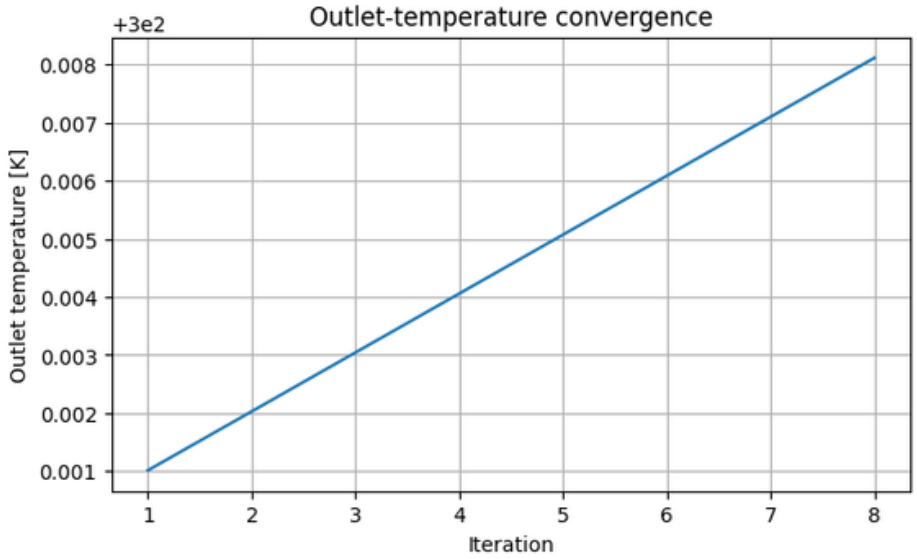


Fig2: Outlet temperature convergence

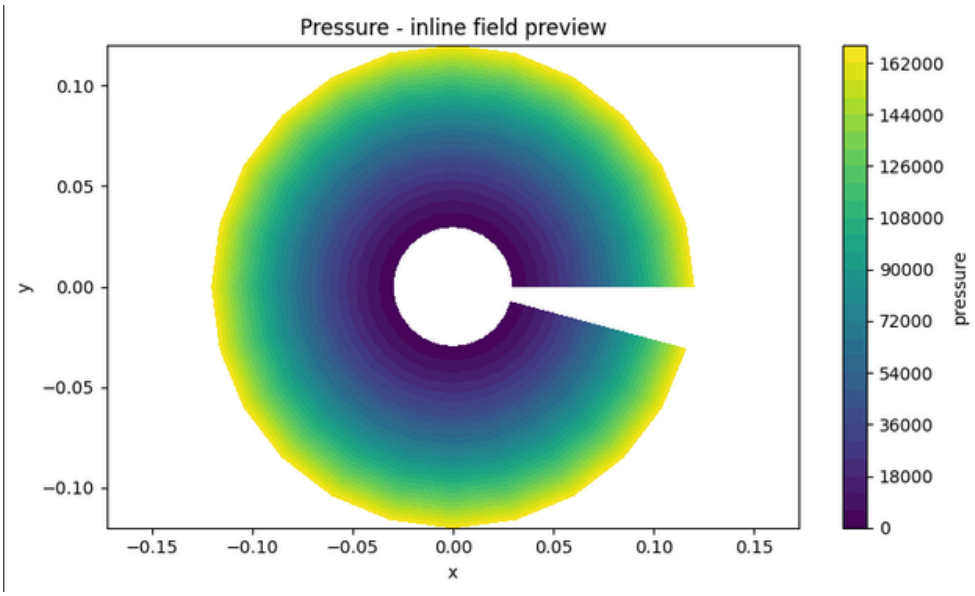


Fig3: Pressure field

2) For pump at Re=1000, RPM=2000

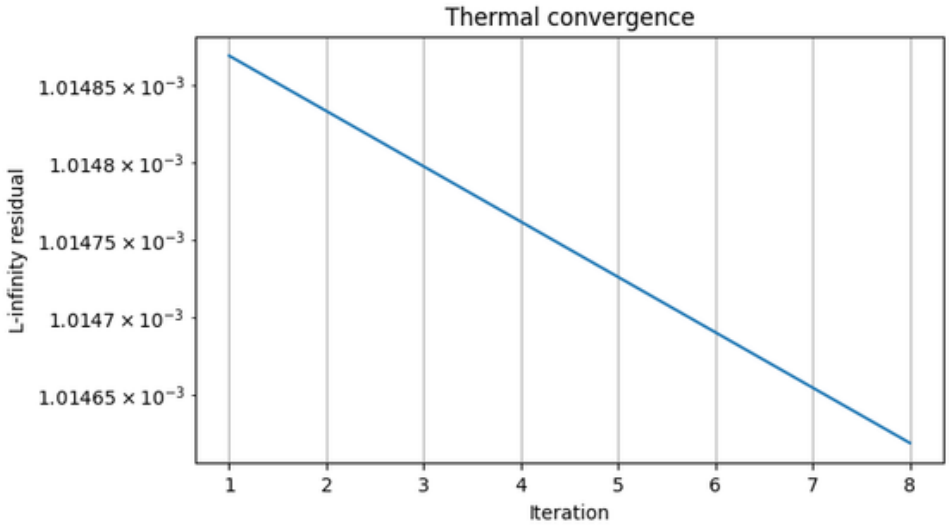


Fig4: Thermal convergence

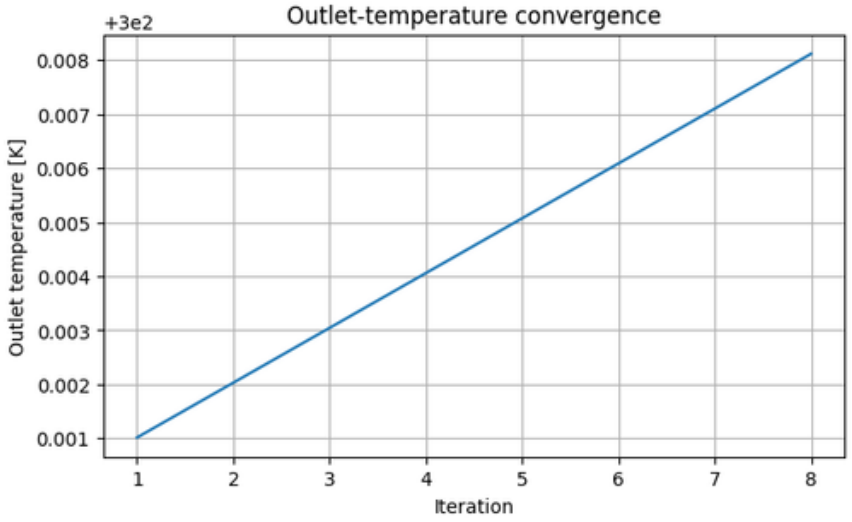


Fig5: Outler temperature convergence

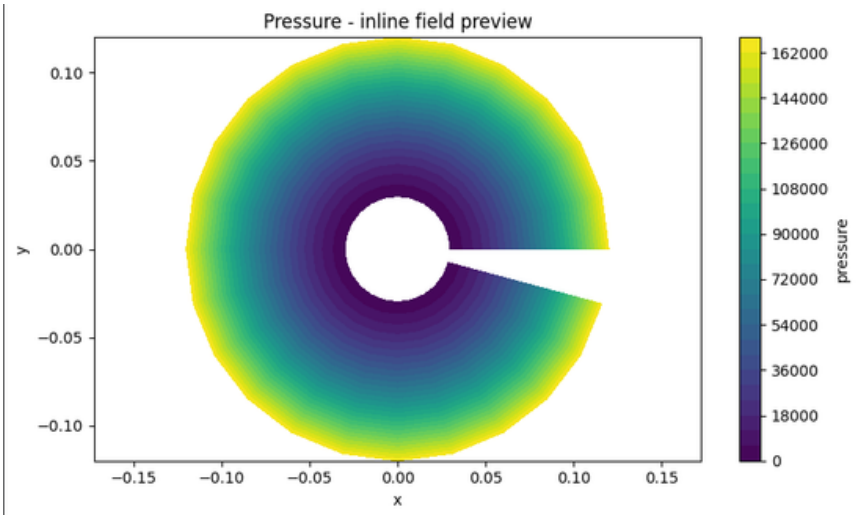


Fig6: Pressure field

Result

Pump Speed	Re	T_in	T_out	Thermal Residual	Max Pressure (Pa)
1000	700	300	300.1	1.01x10^(-3)	162000
2000	1000	300	300.12	1.01x10^(-3)	162000

Conclusion: The internal fluid flow and heat transfer in a centrifugal pump were studied using a two-dimensional annular rotating-frame model for different pump speeds and Reynolds numbers. The inlet temperature was maintained at 300 K, while the wall temperature was set to 330 K. The thermal convergence plots showed a gradual decrease in the thermal residual, indicating a reduction in the numerical error during the iterative solution. The outlet temperature increased slightly from the inlet temperature, demonstrating heat transfer from the heated wall to the fluid. The pressure contours illustrated the radial pressure distribution within the pump passage, with higher pressure observed toward the outer radius. Increasing the pump speed is expected to increase the centrifugal pressure rise, approximately following the square of rotational speed under comparable conditions.

4 Experiment 4: Analysis of Karman-Vortex over a cylinder (Vortex shedding).

Aim

To simulate unsteady vortex shedding over a circular cylinder and estimate the shedding frequency using a transient projection finite-volume solver.

Governing Equations and Physical Model

The incompressible equations are

$$\nabla \cdot \mathbf{u} = 0, \quad \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u}.$$

Vorticity is computed from

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}.$$

Boundary Conditions

Uniform inflow is imposed at the inlet. Convective/zero-gradient outlet is used downstream. Top and bottom boundaries are far-field slip boundaries. Cylinder cells enforce no-slip through a solid mask.

Numerical Method

Explicit time advancement with pressure projection. VTK snapshots are written at specified intervals. Lift/drag histories provide a proxy for vortex-shedding oscillations.

Stability, Boundedness, and Convergence Requirements

The time step must satisfy both convective CFL and viscous diffusion limits: $\Delta t \lesssim C\Delta x/U$ and $\Delta t \lesssim C_d \Delta x^2/u$.

Compilation and Execution

```
g++ -O3 -std=c++17 code/exp04_cylinder_karman_vortex.cpp -o exp04
./exp04_cylinder_karman_vortex 181 81 100 1200 45 300
```

Expected Output Files

```
cylinder_vortex_final.vtk
cylinder_wall.vtk
cylinder_history.csv
```

Benchmark or Verification Requirement

For Re = 100, expected Strouhal number is approximately 0.16 to 0.17 after transient decay on adequately resolved grids.

Compulsory Assignment

Run Re = 80, 100, and 150. Extract lift coefficient history after the initial transient and estimate Strouhal number from peak spacing.

Record in Lab Manual

Students must submit: governing equation, computational domain, boundary conditions, mesh/grid size, input parameters, residual or history plot, ParaView contour/streamline figure, numerical table, and a short conclusion.

Parameters:

Parameter	Value
nx	350
ny	350
Re	5000
outer steps	2000
iterations	200

Boundary Conditions: Prescribed low-mach inlet velocity, open outlet treatment, wall teatment on uper /lower boundary, no-slip bounce back at cylinder.

1) Output for Re=5000

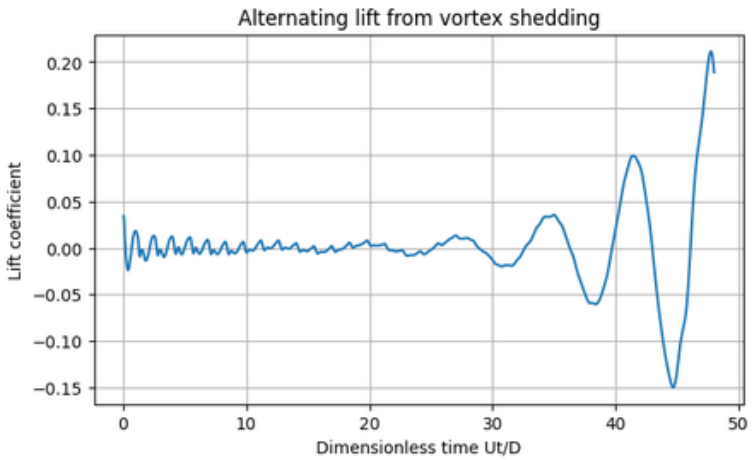


Fig1: Alternating lift

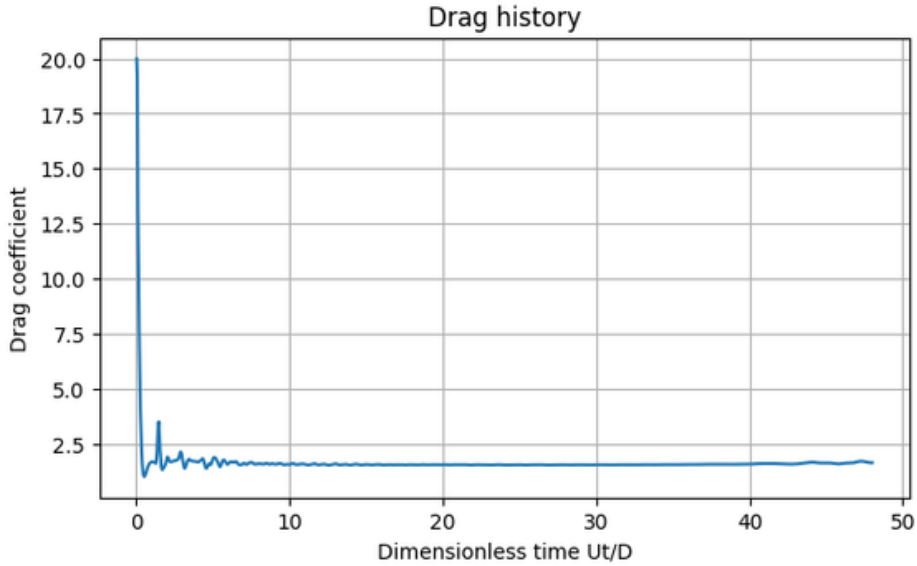


Fig2: Drag History

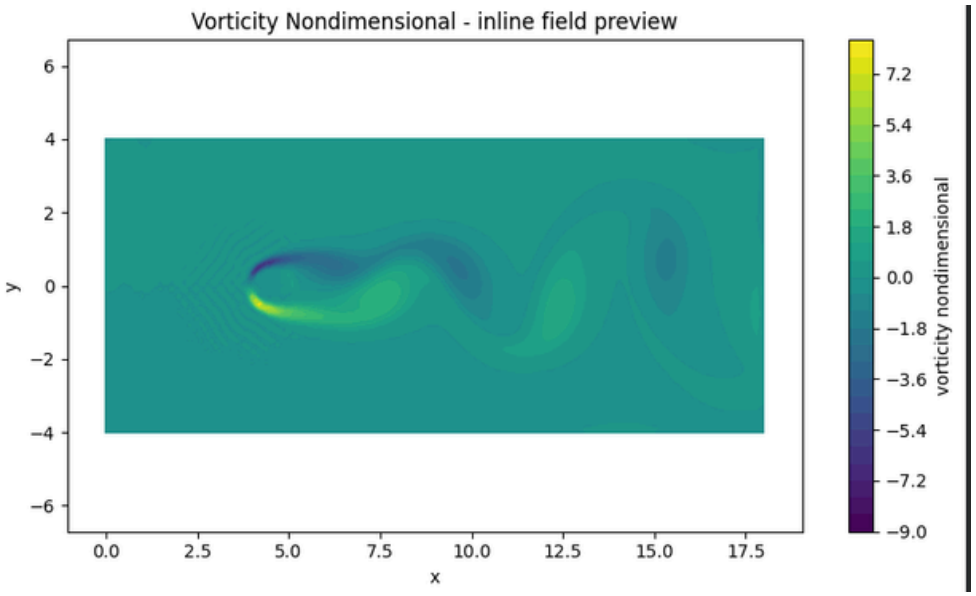


Fig3: Vorticity inline Field view

Conclusion: The flow around a circular cylinder was numerically investigated using computational fluid dynamics at a Reynolds number of 5000, with a computational grid of 350 × 350 and 2000 outer time steps. The lift coefficient initially exhibited small oscillations, followed by a significant increase in oscillation amplitude toward the end of the simulation, indicating the development of unsteady flow and vortex shedding. The drag coefficient showed a sharp initial transient before settling at ~1.3, with minor fluctuations at later. The vorticity contour reveals that alternating regions of positive and negative vorticity downstream of the cylinder, forming a characteristic vortex street. These observations demonstrate the influence of inertial effects on wake formation and aerodynamic forces at a higher Reynolds number. However, the increasing amplitude of the lift fluctuations suggests that a stable periodic state may not have been fully established within the simulated duration. Further simulation with sufficient time steps and verification of numerical stability would help obtain more reliable results.

5 Experiment 5: Couette flow analysis for both explicit and implicit formulation (Parabolic equation).

Aim

To solve transient Couette flow using explicit FTCS and implicit backward-Euler formulations for the parabolic diffusion equation.

Governing Equations and Physical Model

For start-up Couette flow,

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2}, \quad u(0,t) = 0, \quad u(H,t) = U.$$

The steady solution is $u/U = y/H$. The transient analytical series is used for error computation.

Boundary Conditions

Bottom plate is stationary. Top plate moves with constant speed U. Initial velocity is zero inside the fluid.

Numerical Method

Explicit FTCS and implicit backward-Euler TDMA are solved on the same grid and compared against the analytical series.

Stability, Boundedness, and Convergence Requirements

FTCS requires $Fo = u\Delta t/\Delta y^2 \leq 0.5$. Backward Euler is unconditionally stable for diffusion but becomes first-order dissipative in time for large Fo.

Compilation and Execution

```
g++ -O3 -std=c++17 code/exp05_couette_explicit_implicit.cpp -o exp05
./exp05_couette_explicit_implicit 81 1.0 1.0 0.01 2.0 0.45 5.0 50
```

Expected Output Files

```
couette_field.vtk
couette_profiles.csv
couette_summary.txt
```

Benchmark or Verification Requirement

Use the analytical transient Couette solution for relative L2 error.

Compulsory Assignment

Run $Fo = 0.25$, 0.45, and 0.75 for explicit FTCS. Identify the stable and unstable cases and compare against the implicit solution.

Record in Lab Manual

Students must submit: governing equation, computational domain, boundary conditions, mesh/grid size, input parameters, residual or history plot, ParaView contour/streamline figure, numerical table, and a short conclusion.

Parameter:

Parameter	Value
Explicit Value (Fo)	0.55,0.45,0.25
Implicit Value (Fo)	0.5,0.5,0.5

Boundary Conditions: Initially stationary fluid, lower plate fixed, upper plate impulsively set to speed U,Constant Pressure

1) Explicit (Fo- 0.55)

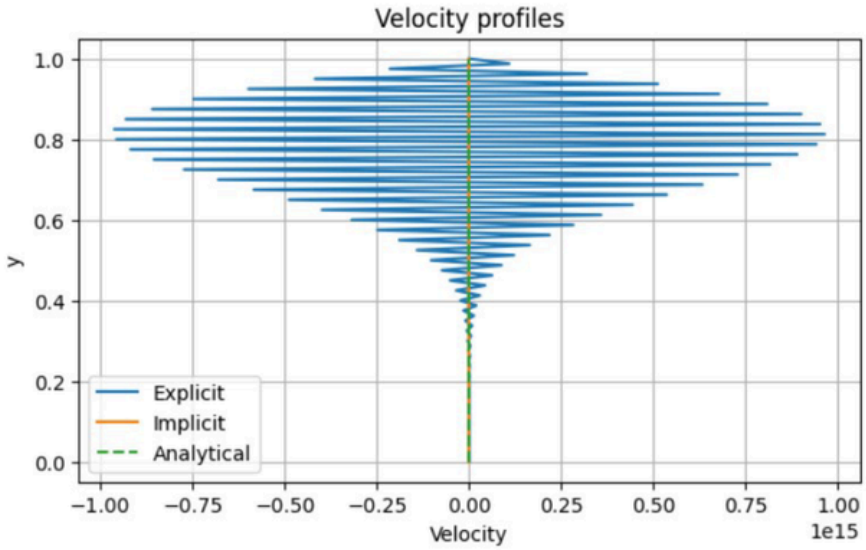


Fig1: Velocity Profile

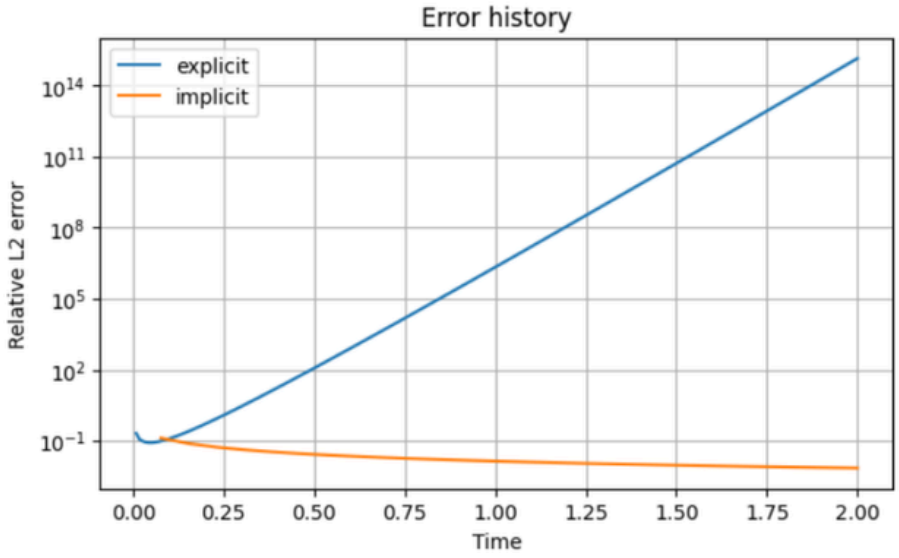


Fig2: Error Plot

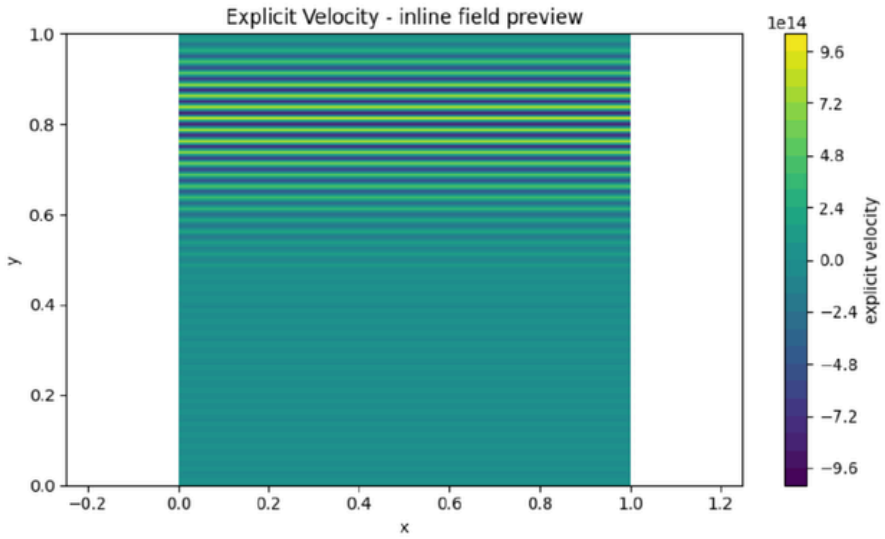


Fig3: Velocity Field

2) Explicit (Fo- 0.45)

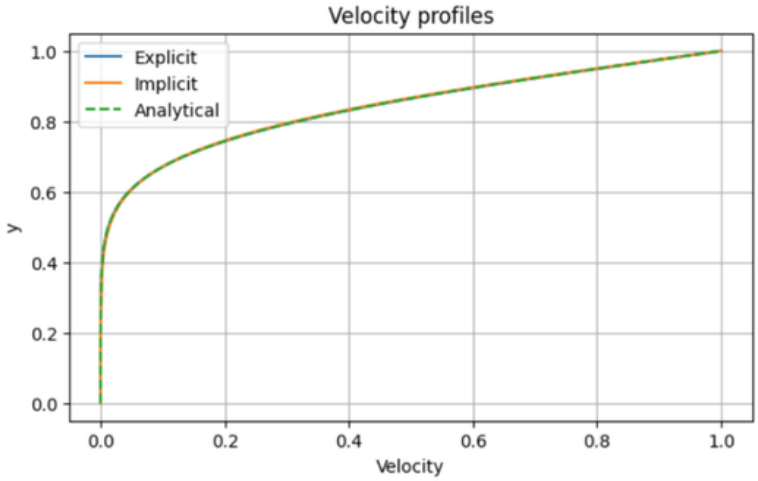


Fig4: Velocity Profile

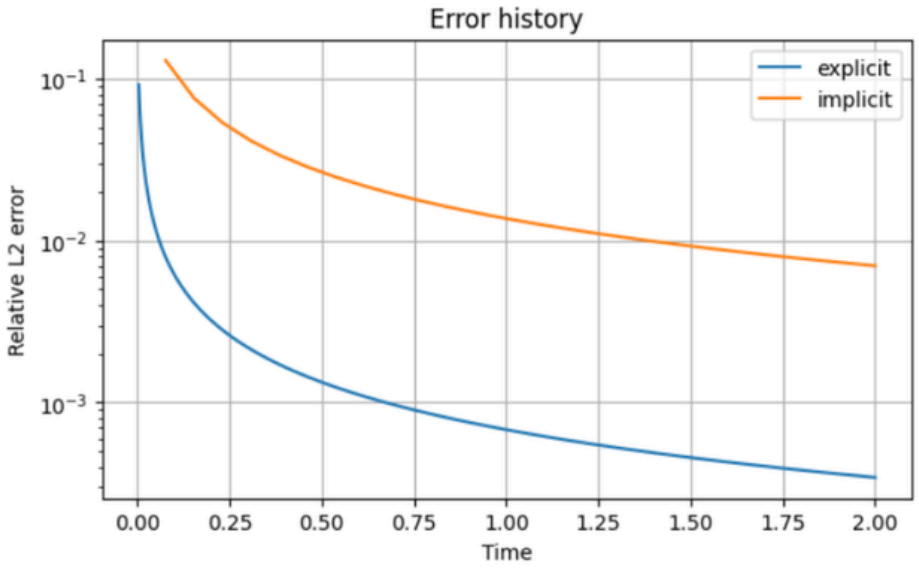


Fig5: Error Plot

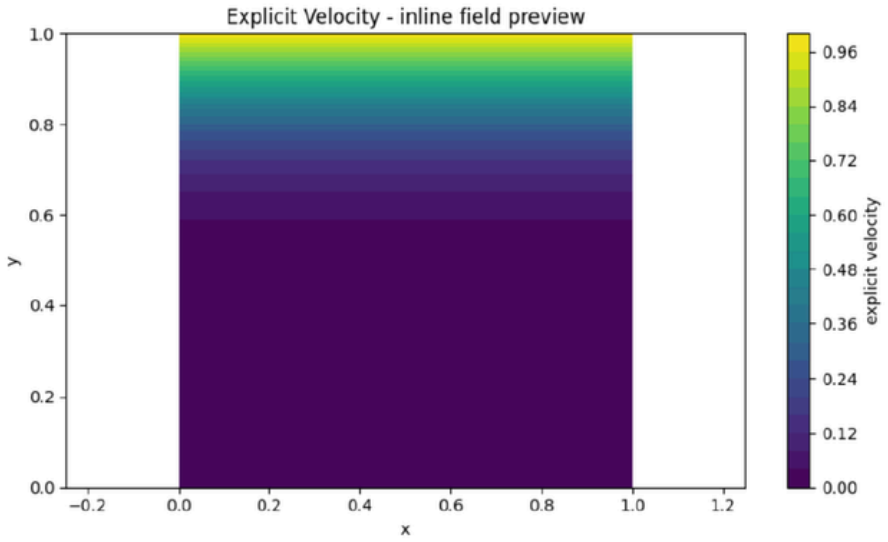


Fig6: Velocity Field

3) Explicit (Fo- 0.25)

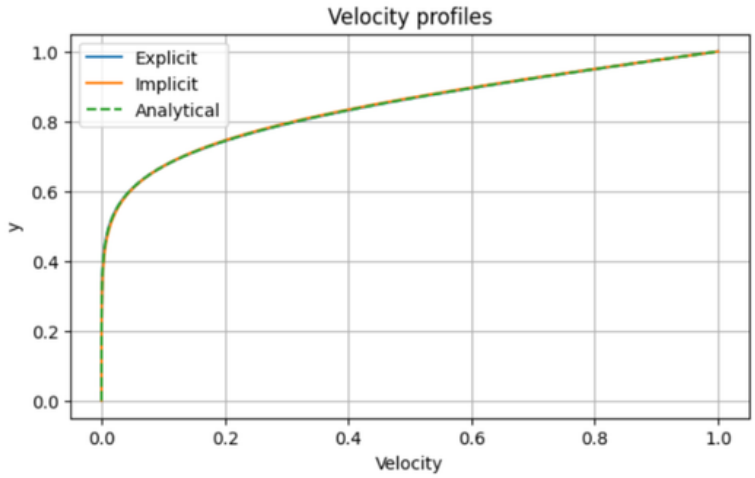


Fig7:Velocity Profile

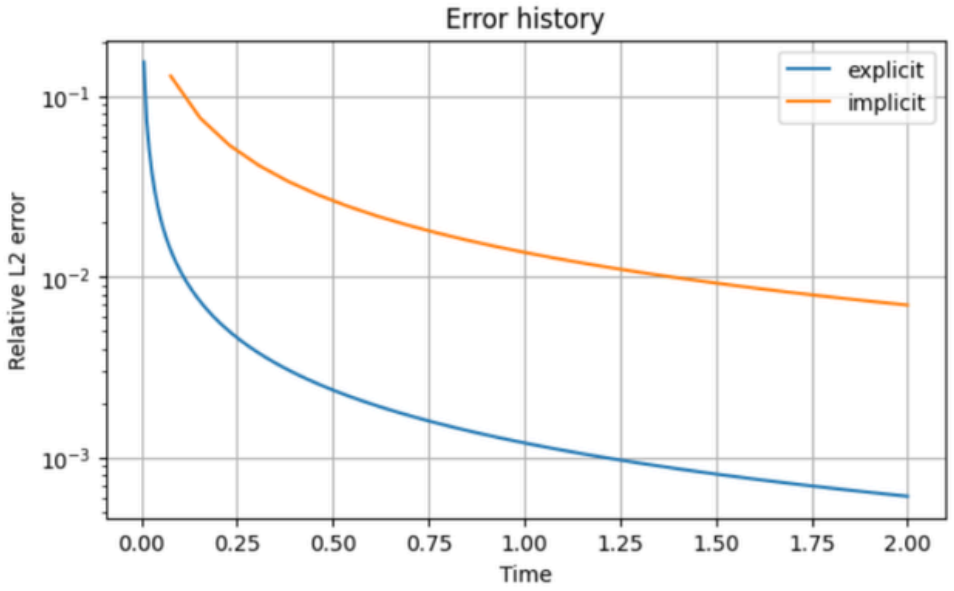


Fig8: Error Plot

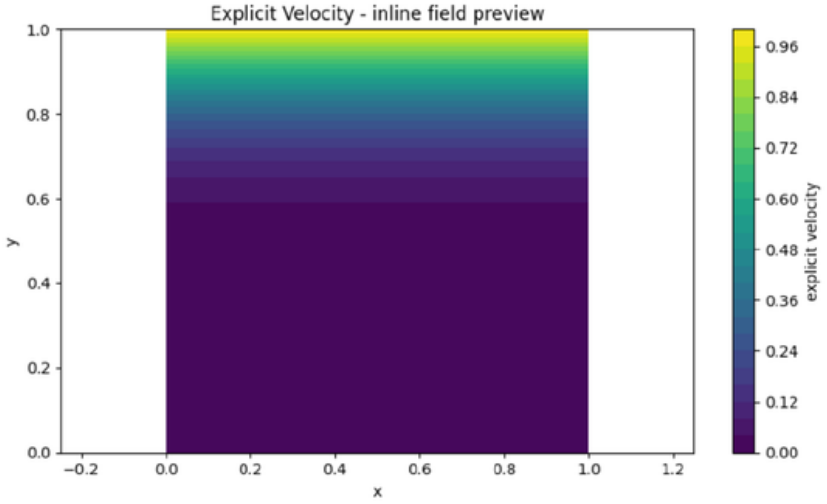


Fig9: Velocity Field

Conclusion: The transient Couette flow was analyzed using explicit FTCS and implicit numerical schemes and compared with the analytical solution. The velocity profiles show the development of fluid motion between a stationary lower plate and an upper plate moving at a constant velocity. For the explicit scheme with Fo=0.55, the velocity profile exhibits severe oscillations and the relative error increases significantly, indicating numerical instability. In contrast, for Fo=0.45 and Fo=0.25, the explicit solutions remain stable and closely follow the analytical profile. The implicit scheme remains stable for the tested cases, although its error decreases more gradually than that of the stable explicit cases.

6 Experiment 6: Heat conduction in 2D flat plate with explicit and implicit formulation (Elliptic equation).

Aim

To solve the 2D steady heat-conduction equation in a flat plate using point-wise and line-wise iterative formulations for the elliptic equation.

Governing Equations and Physical Model

The governing equation is Laplace's equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0.$$

For the boundary condition $T(0, y) = T_0 \sin(\pi y/H)$, $T(L, y) = 0$, $T(x, 0) = 0$, $T(x, H) = 0$, the analytical solution is

$$T(x, y) = T_0 \frac{\sinh(\pi(L - x)/H)}{\sinh(\pi L/H)} \sin(\pi y/H).$$

Boundary Conditions

Left boundary follows a sinusoidal temperature profile. Right, bottom, and top boundaries are held at zero temperature.

Numerical Method

Jacobi iteration represents point-wise explicit algebraic relaxation. Line-by-line TDMA represents an implicit algebraic line solve.

Stability, Boundedness, and Convergence Requirements

Elliptic steady problems do not have physical explicit/implicit time stability. Convergence rate depends on spectral radius and relaxation factor.

Compilation and Execution

```
g++ -O3 -std=c++17 code/exp06_plate_conduction_explicit_implicit.cpp -o exp06
./exp06_plate_conduction_explicit_implicit 81 61 1.0 1.0 100 10000 1e-8 1.55 1000
```

Expected Output Files

```
plate_conduction_2d.vtk
plate_conduction_profiles.csv
plate_conduction_summary.txt
```

Benchmark or Verification Requirement

Compare numerical temperature field against the analytical Laplace solution.

Compulsory Assignment

Compare Jacobi and line-TDMA convergence for 41x41, 81x61, and 121x81 grids. Report iterations, residual, and relative L2 error.

Record in Lab Manual

Students must submit: governing equation, computational domain, boundary conditions, mesh/grid size, input parameters, residual or history plot, ParaView contour/streamline figure, numerical table, and a short conclusion.

Parameters

Parameter	Value
Nodes in X	200, 50,750
Nodes in Y	200, 20,800
Boundary temperature	100

Boundary Conditions: Right end and Top end is at 0 deg C

1) Nx=Ny=200

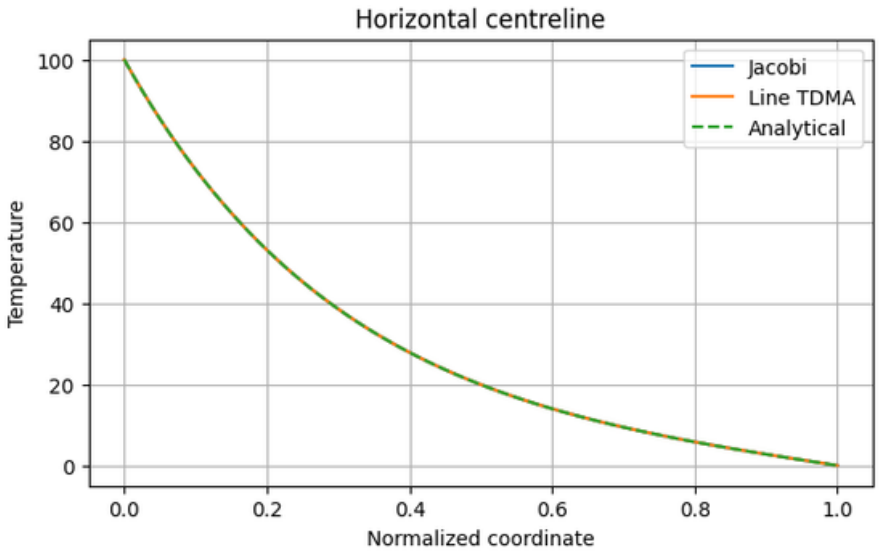


Fig1: Temperature Gradient

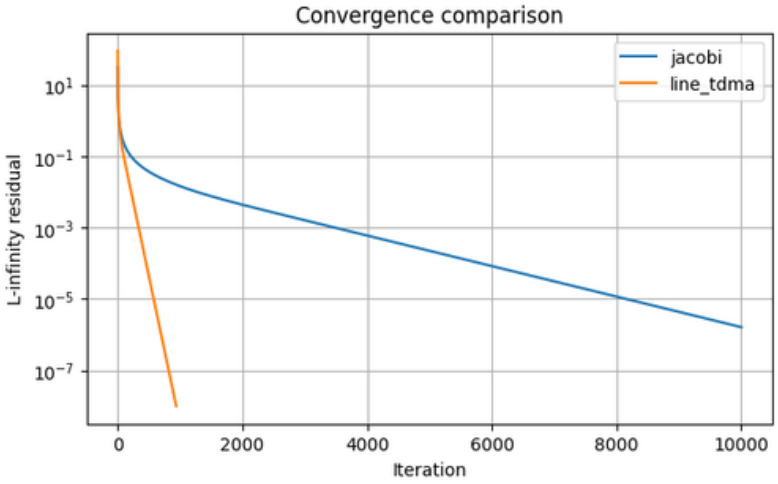


Fig2: Convergence using different methods

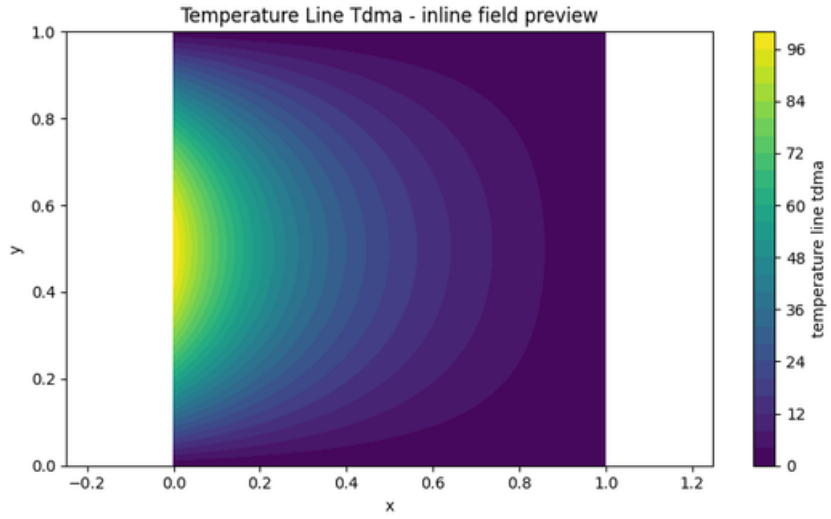


Fig3: Temperature Field

2) Nx=50, Ny=20

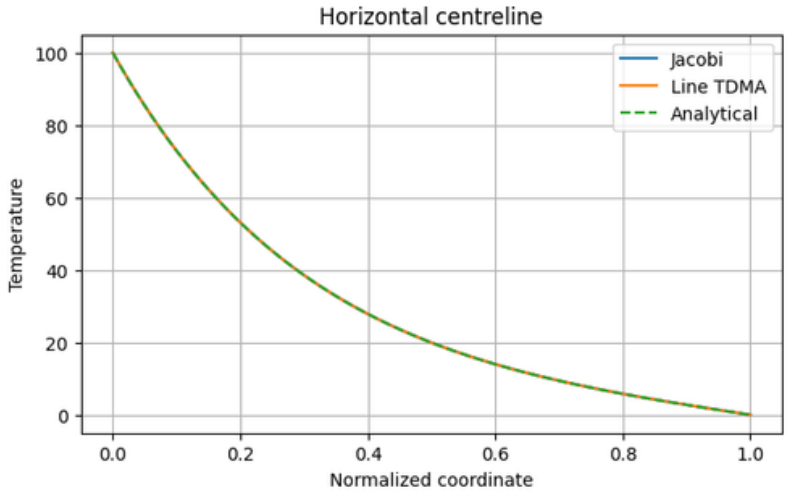


Fig4: Temperature Gradient

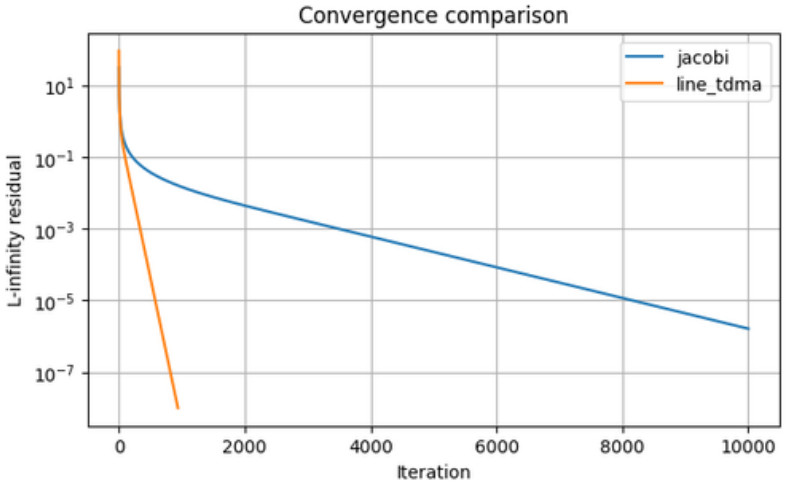


Fig5: Convergence using different methods

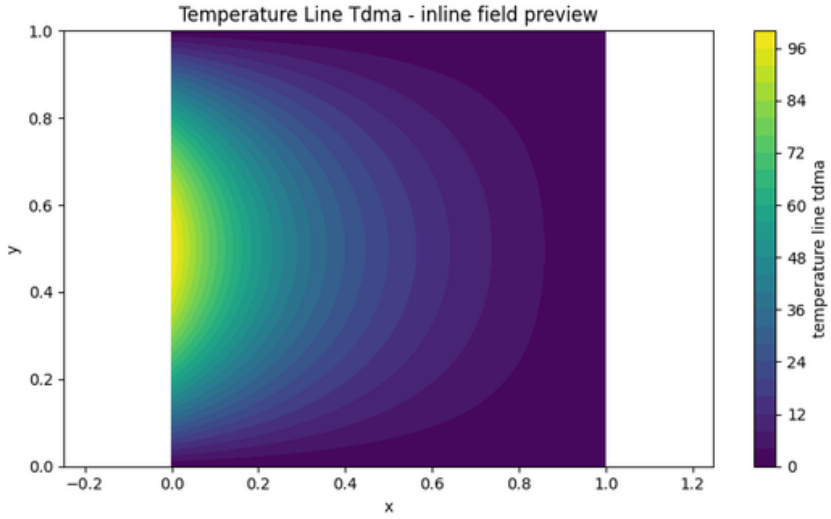


Fig6: Temperature Field

3) Nx=750, Ny=800

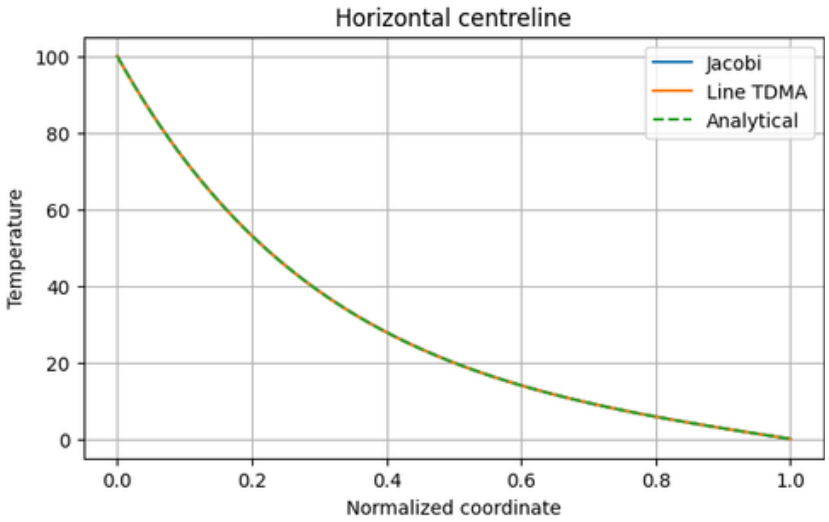


Fig7: Temperature Gradient

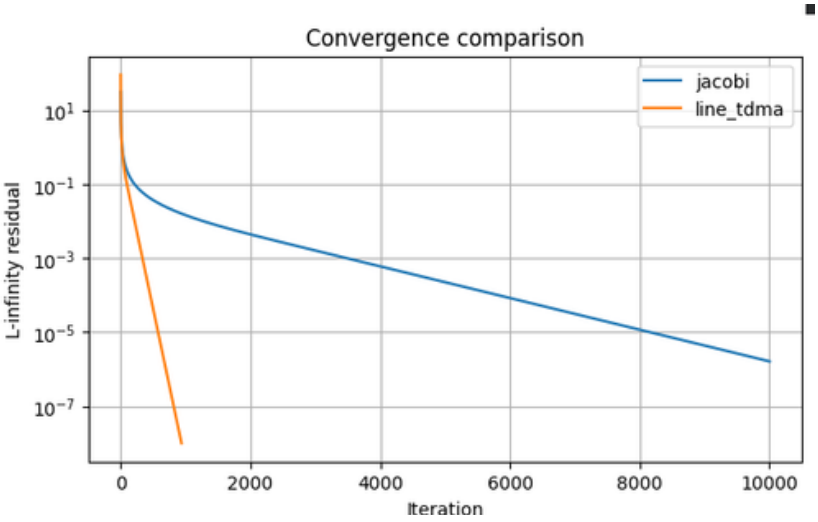


Fig8: Convergence using different methods

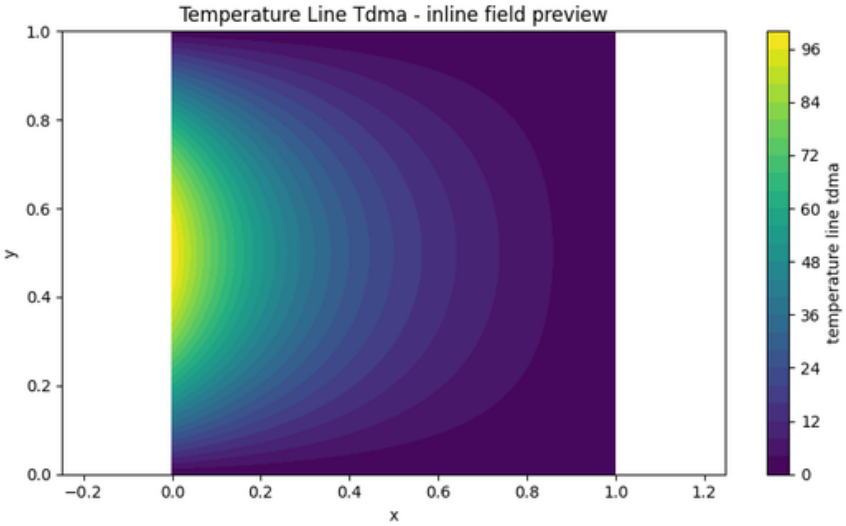


Fig9: Temperature Field

Result Table:

Grid	Temperature Distribution
50x20	Temperature decreases from the heated left boundary toward the zero-temperature right boundary
200x200	Smooth temperature gradient across the plate
750x800	Smoothly varying temperature field with better spatial resolution

Conclusion: The two-dimensional steady-state heat conduction in a flat plate was analyzed by solving Laplace's equation using the Jacobi iterative method and the line-wise TDMA method. The temperature profiles obtained using both numerical methods closely matched the analytical solution, demonstrating their ability to predict the temperature distribution in the plate. The temperature contours showed that the temperature was highest near the heated left boundary and gradually decreased toward the right boundary, where the temperature was maintained at zero. The convergence plots indicated that Line TDMA reduced the residual much faster than the Jacobi method, requiring fewer iterations to achieve convergence. This difference arises because Jacobi updates the temperature point by point, whereas Line TDMA solves along grid lines.

One-Dimensional Wave Propagation in a Still Lake

Aim: To compare numerical diffusion, dispersion, stability, and conservation for first-order upwind and Lax Wendroff schemes.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	401
Grid points, ny	20
Amplitude	0.02
T_end	2
CFL	0.4, 0.8, 1.1
Gravity	9.81
N_snapshots	180

Governing Equation:

For small perturbations of depth H,

$$\frac{\partial \eta}{\partial t} + H\frac{\partial u}{\partial x} = 0,$$
$$\frac{\partial u}{\partial t} + g\frac{\partial \eta}{\partial x} = 0,$$

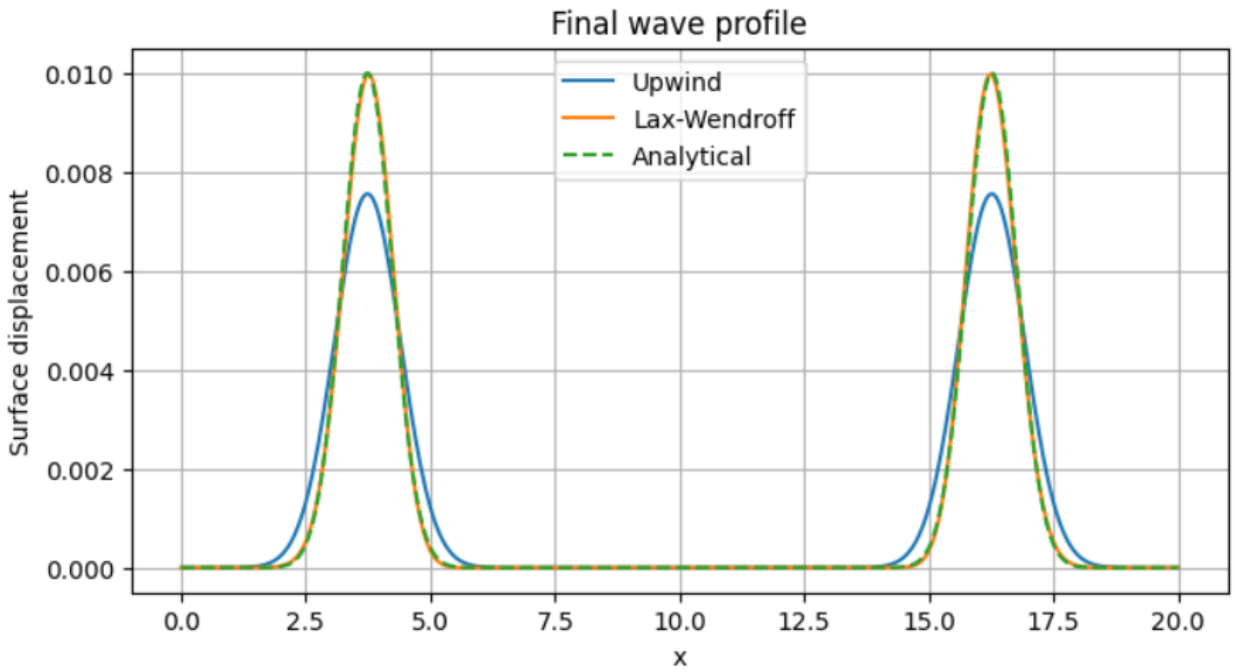
with wave speed $c = \sqrt{gH}$ and $CFL = c\Delta t/\Delta x$.

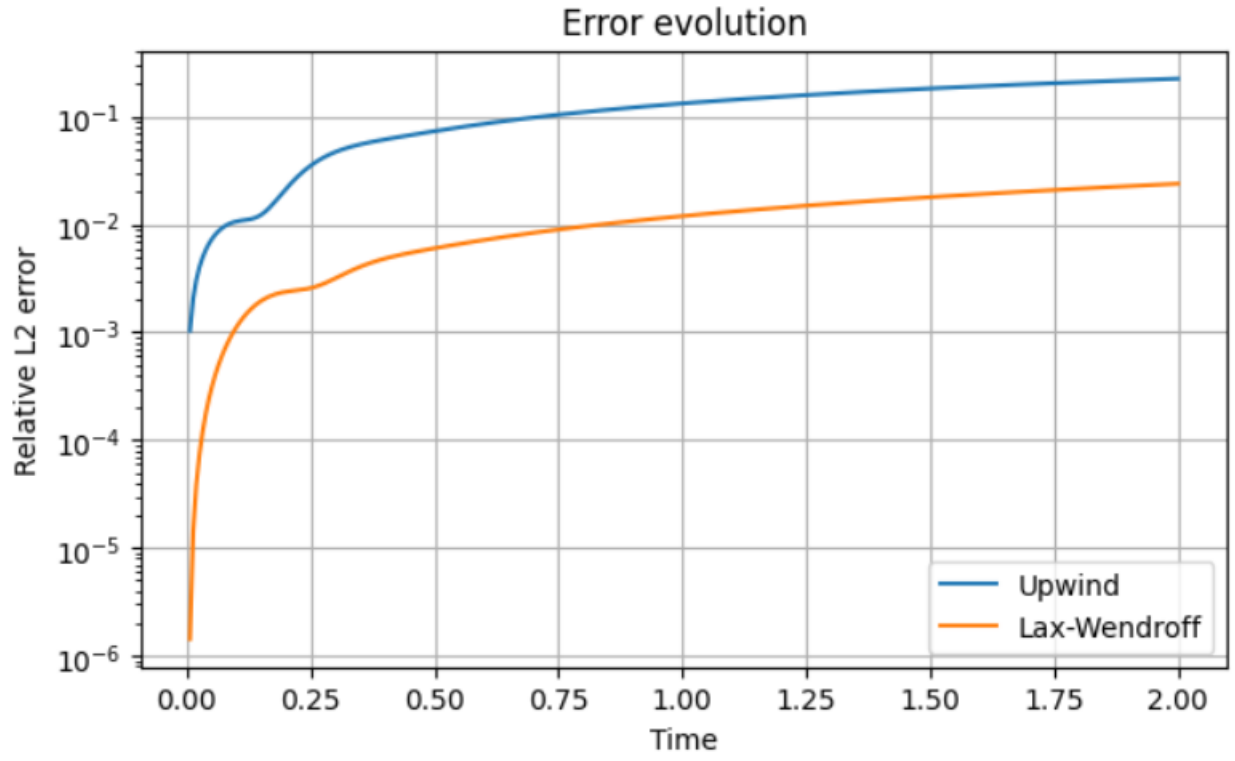
Boundary Conditions: Periodic one-dimensional domain with an initially localized Gaussian surface perturbation

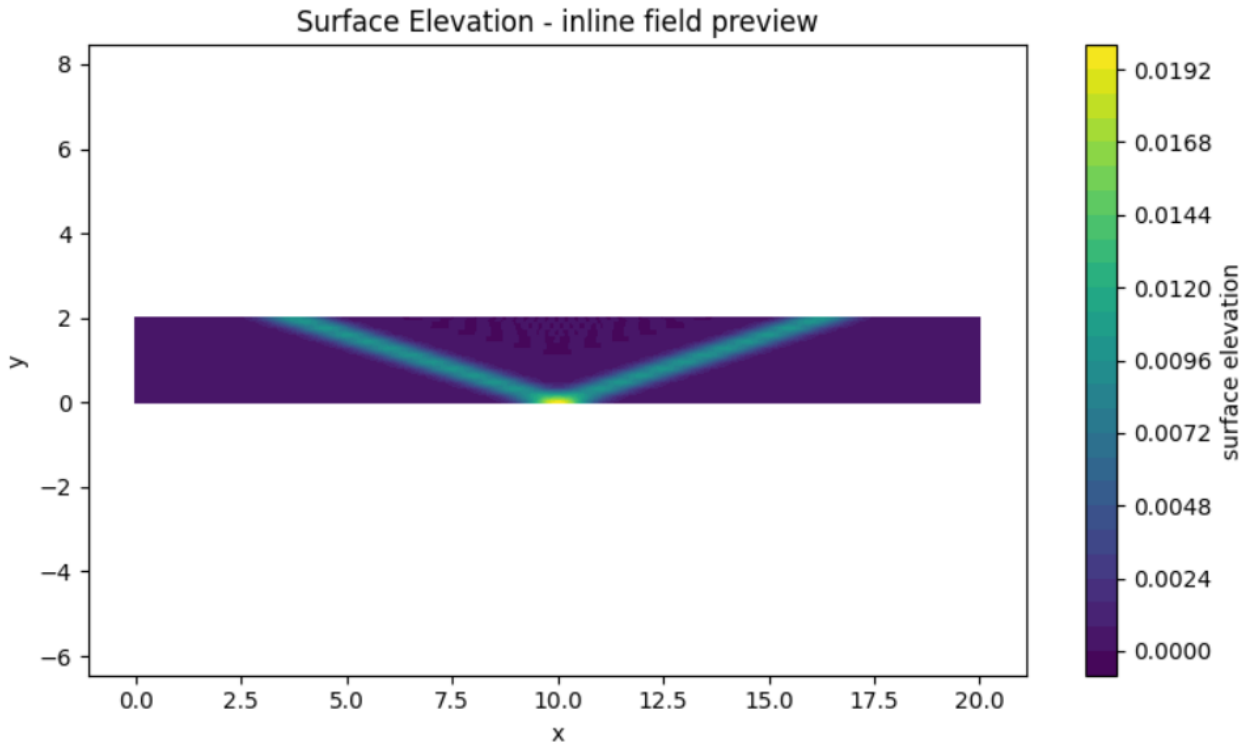
Numerical Method: Conservative first-order upwind and second-order Lax-Wendroff time marching on the linearized shallow water system.

Observations: Wave amplitude, phase error, relative L2 error, mass conservation, numerical diffusion, oscillations, and CFL dependence at CFL values 0.4, 0.8, and 1.1. Comparison stability, amplitude loss, phase error, and conservation for both schemes.

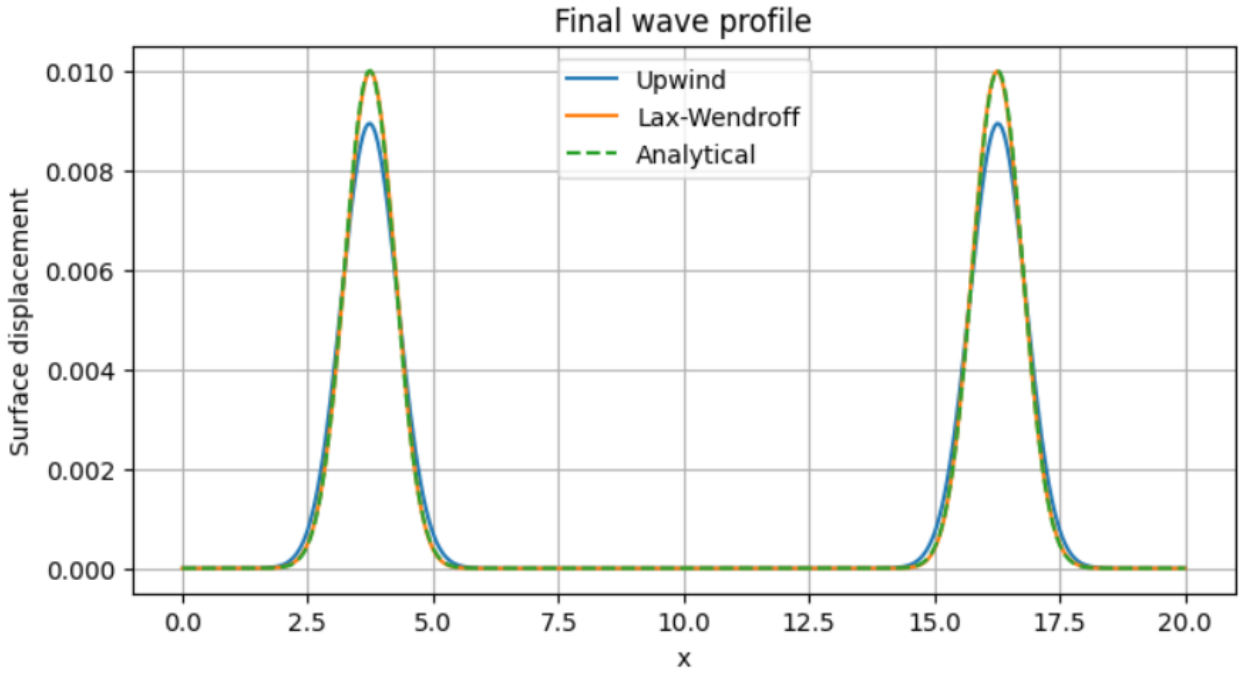
For CFL number 0.4

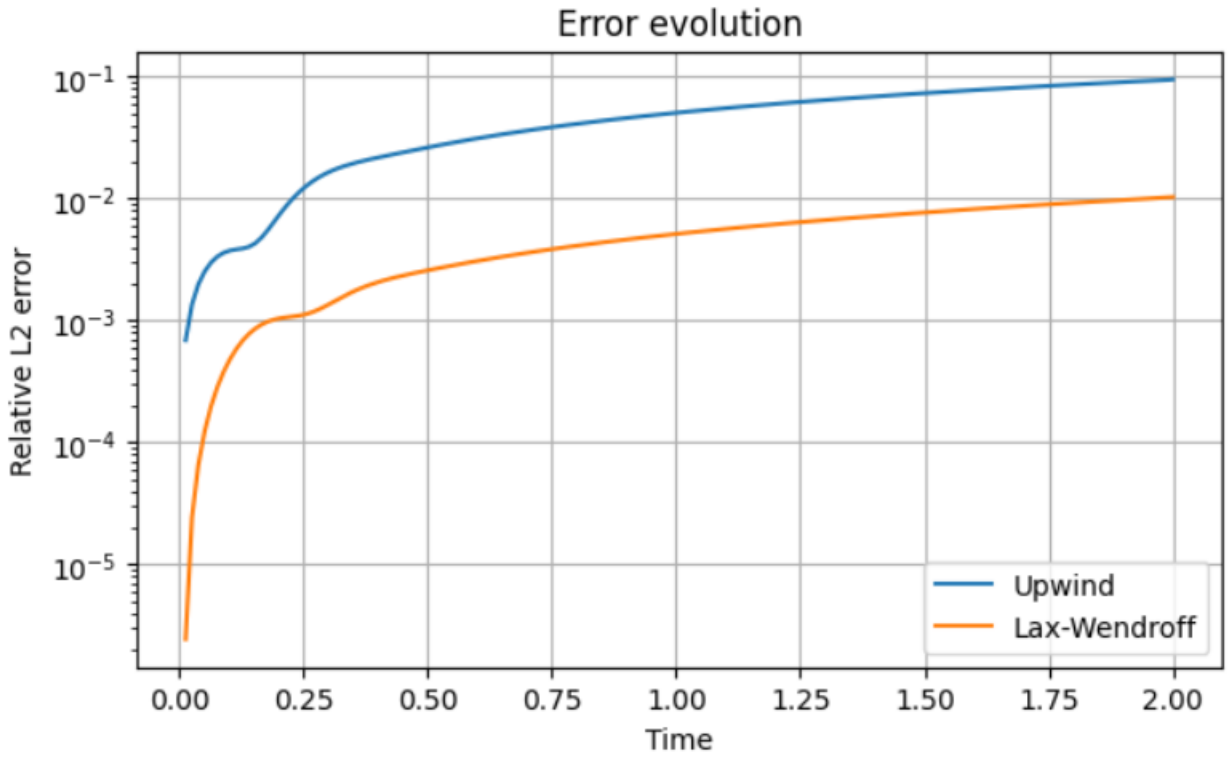


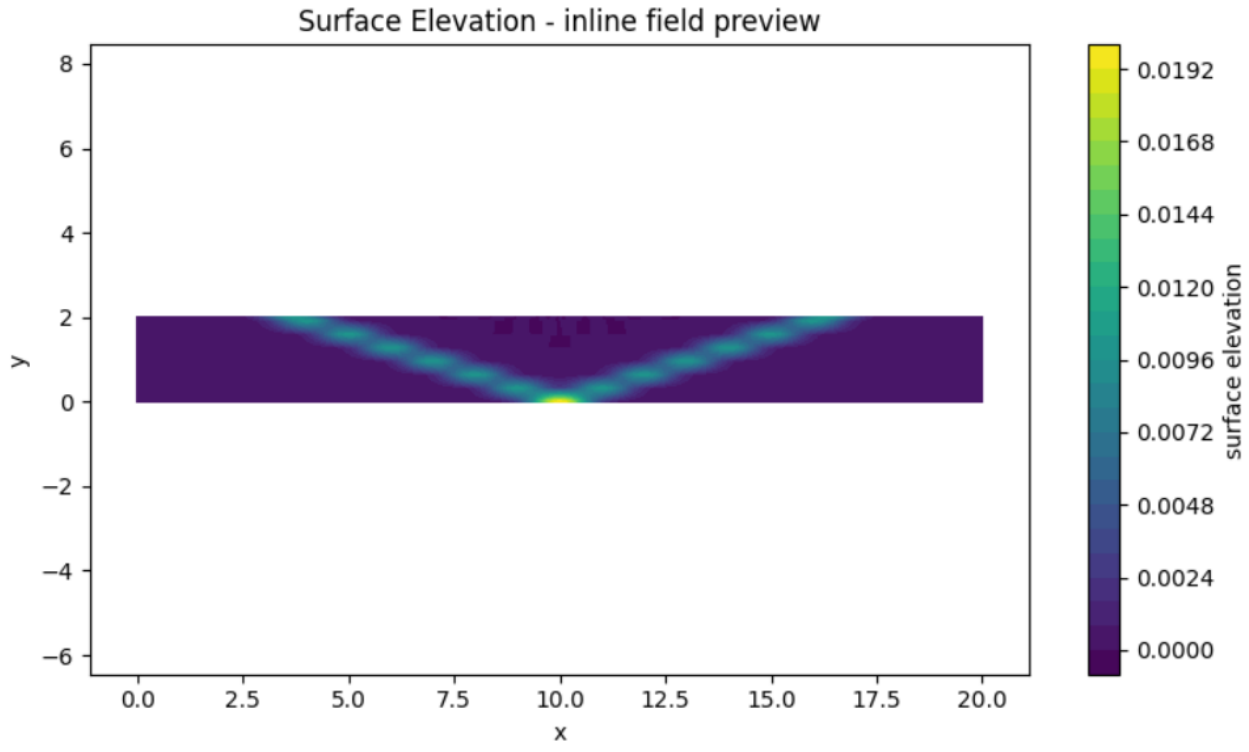




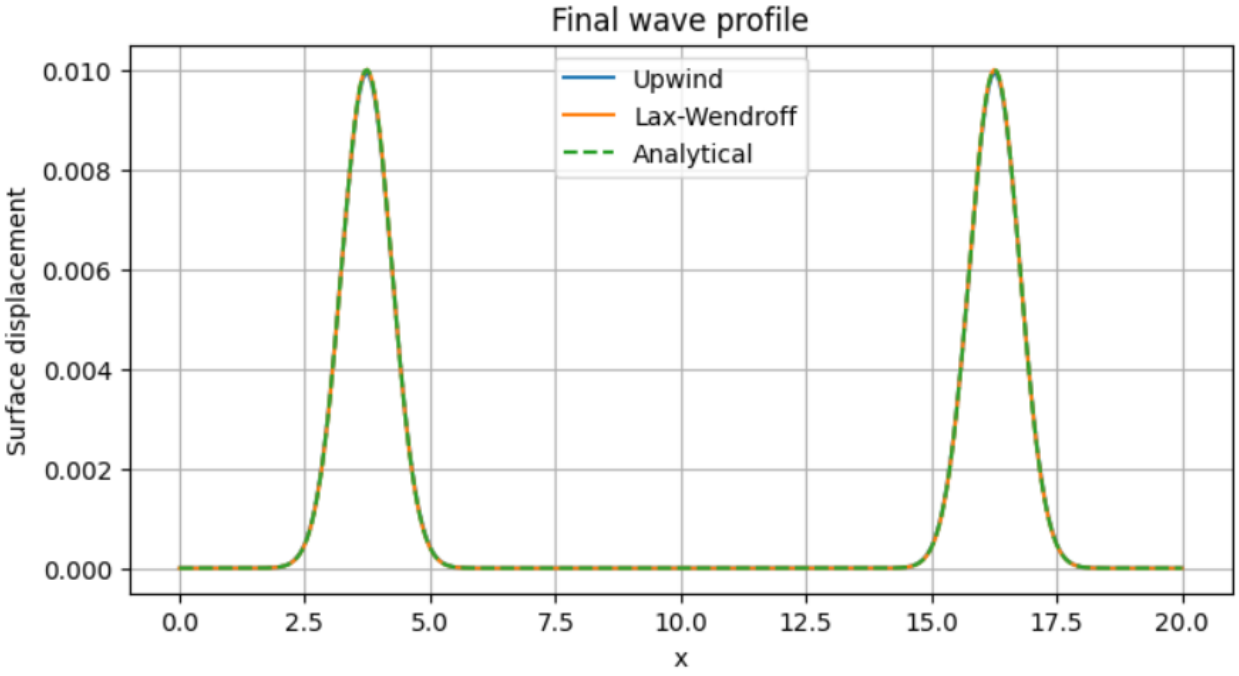
For CFL number 0.8

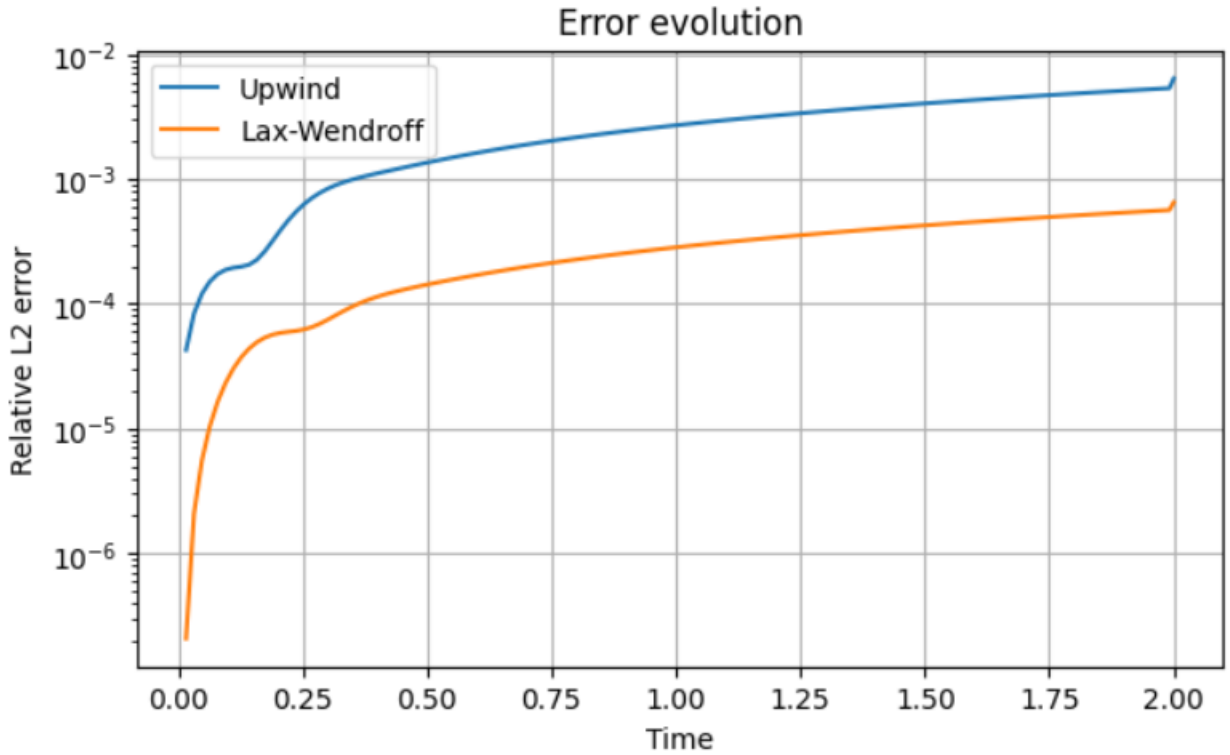


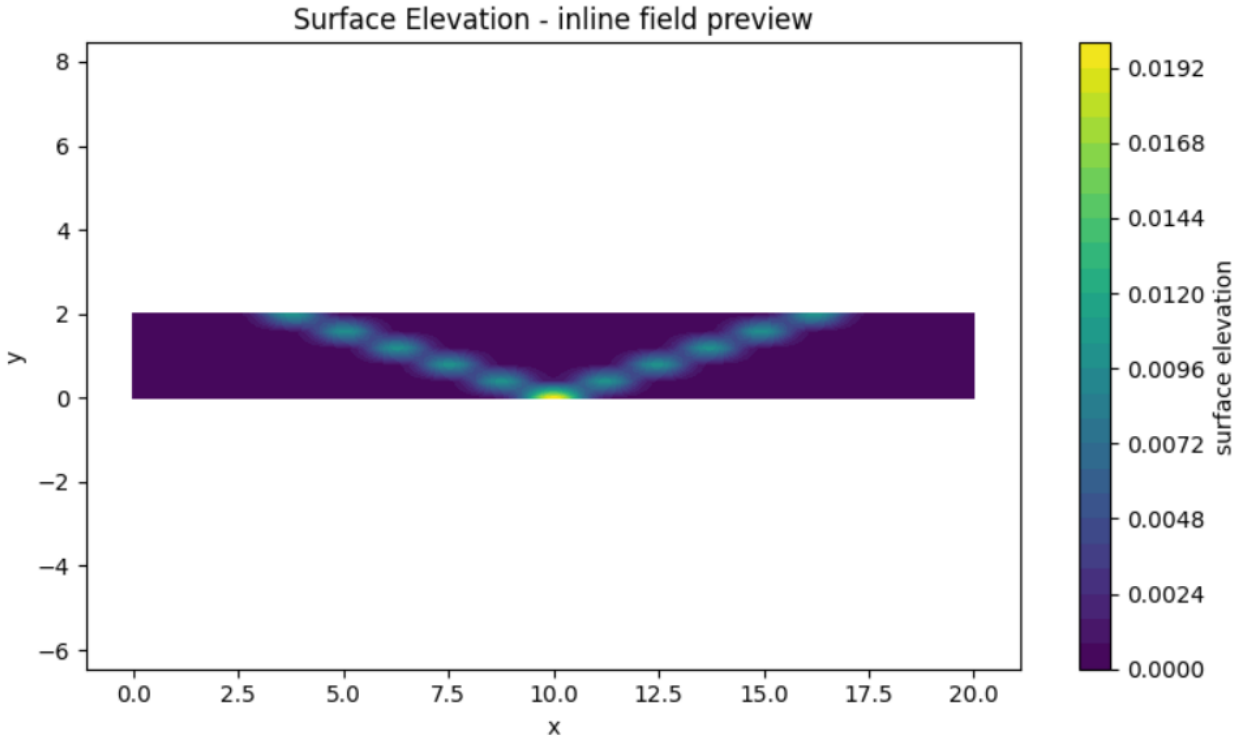

Error evolution
Relative L2 error
Time
Upwind
Lax-Wendroff


Surface Elevation - inline field preview
y
x
surface elevation

For CFL number 1.1







Results:

Parameter	**CFL = 0.4**	**CFL = 0.8**	**CFL = 1.1 (nominal)**
Pulse centres at t = 2 (x)	≈ 3.75 and 16.25	≈ 3.75 and 16.25	≈ 3.75 and 16.25
Analytical peak amplitude (each pulse)	≈ 0.0100	≈ 0.0100	≈ 0.0100
Upwind peak amplitude at t = 2	≈ 0.0076	≈ 0.0090	≈ 0.0100
Upwind amplitude loss	≈ 24%	≈ 10%	negligible
Lax–Wendroff peak amplitude at t = 2	≈ 0.0100	≈ 0.0100	≈ 0.0100
Upwind relative L^2 error at t = 2	$\approx 2.3 \times 10^{-1}$	$\approx 9.5 \times 10^{-2}$	$\approx 6 \times 10^{-3}$
Lax–Wendroff relative L^2 error at t = 2	$\approx 2.4 \times 10^{-2}$	$\approx 1.0 \times 10^{-2}$	$\approx 6 \times 10^{-4}$

Parameter	**CFL = 0.4**	**CFL = 0.8**	**CFL = 1.1 (nominal)**
Error ratio (upwind / Lax–Wendroff)	≈ 10	≈ 10	≈ 10
Phase error (peak position)	none visible	none visible	none visible
Spurious oscillations	none visible	none visible	none visible
Growth of error in time	smooth, monotonic	smooth, monotonic	smooth, monotonic
Mass conservation (conservative form, periodic domain)	conserved (by construction)	conserved (by construction)	conserved (by construction)

Conclusion:

The one-dimensional linearised shallow-water wave problem was solved on a periodic domain using the conservative first-order upwind scheme and the second-order Lax–Wendroff scheme, at CFL numbers of 0.4, 0.8 and 1.1, and the results were compared with the analytical solution. The initial Gaussian surface perturbation of amplitude 0.02 splits into two equal pulses of amplitude ≈ 0.010 that travel in opposite directions. At $t = 2$ the pulse centres are at $x \approx 3.75$ and $x \approx 16.25$, i.e. about 6.25 from the origin at $x = 10$, which corresponds to a wave speed of $c \approx 3.1$. This is consistent with $c = \sqrt{(gH)}$ for $g = 9.81$ and a depth of about 1. The surface-elevation field plots show the two symmetric wavefronts leaving the source in all three cases.

Numerical diffusion of the upwind scheme: The first-order upwind solution is strongly diffusive at low CFL. At CFL = 0.4 the pulse peak falls from 0.0100 to approximately 0.0076, an amplitude loss of about 24%, and the base of the pulse is visibly broadened. At CFL = 0.8 the peak is approximately 0.0090 (about 10% loss), and at the nominal CFL = 1.1 the upwind profile is indistinguishable from the analytical one. The corresponding relative L^2 errors at $t = 2$ are approximately 2.3×10^{-1}, 9.5×10^{-2} and 6×10^{-3}. The error grows steadily with time because the smearing accumulates as the pulses travel. This behaviour follows from the leading truncation error of upwind being a diffusive term, with an effective numerical viscosity of about $(c\Delta x/2)(1 - \mathrm{CFL})$, which becomes smaller as CFL approaches 1.

Accuracy of the Lax–Wendroff scheme: The second-order Lax–Wendroff solution retains a peak amplitude of approximately 0.0100 and overlaps the analytical profile in all three cases. Its relative L^2 errors at $t = 2$ are approximately 2.4×10^{-2}, 1.0×10^{-2} and 6×10^{-4}, about one order of magnitude smaller than those of upwind at every CFL value. The leading error of Lax–Wendroff

is dispersive and scales with $(1 - \mathrm{CFL}^2)$; the reduction of the error by a factor of about 2.4 between CFL = 0.4 and 0.8 agrees with the ratio $(1 - 0.4^2) / (1 - 0.8^2) \approx 2.3$. No phase error or trailing oscillations were visible in the plots. This is expected for a smooth Gaussian pulse resolved by roughly 25 grid points across its width, where dispersive errors remain small. Lax–Wendroff can produce oscillations near sharp gradients, but the present smooth case does not test this.

Effect of CFL number and stability: For both schemes the error decreases as the Courant number approaches unity, because at CFL = 1 the discrete update reproduces the exact characteristic shift. The explicit stability limit for both schemes is $\mathrm{CFL} \leq 1$. At the nominal CFL = 1.1, however, the plots show no growth of oscillations or error: both error histories continue the downward trend seen from CFL = 0.4 to 0.8. This indicates that the solver kept the effective Courant number at or just below unity, for example by limiting the time step, rather than running at 1.1. The nominal CFL = 1.1 case therefore does not demonstrate the instability that theory predicts for $\mathrm{CFL} > 1$, which would appear as exponentially growing oscillations. Demonstrating the stability limit would require a run without a time-step limiter.

Conservation and overall comparison: Both schemes are written in conservative form on a periodic domain, so the total surface elevation (mass) is conserved by construction. The amplitude loss of the upwind solution is therefore a redistribution of the same volume into lower, wider pulses, not a loss of mass. Overall, Lax–Wendroff is about ten times more accurate than upwind at the same CFL, with no visible dispersion at this resolution, while upwind is robust but strongly diffusive unless CFL is close to 1. The accuracy of both schemes improves as CFL approaches 1, and the stability bound $\mathrm{CFL} \leq 1$ governs the choice of time step for these explicit methods.

Rayleigh-Taylor Instability using a VOF-Style Method

Aim: To study the growth of a perturbed heavy-over-light interface while monitoring phase-fraction boundedness and conservation.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	96
Grid points, ny	192
nu	0.0008
T_end	2
Atwood number	0.2, 0.5, 0.8
Gravity	1
Poisson_iteration	60

Governing Equation:

The transported phase indicator α satisfies

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) = 0,$$

while the Boussinesq momentum equation contains a buoyancy term proportional to the Atwood number

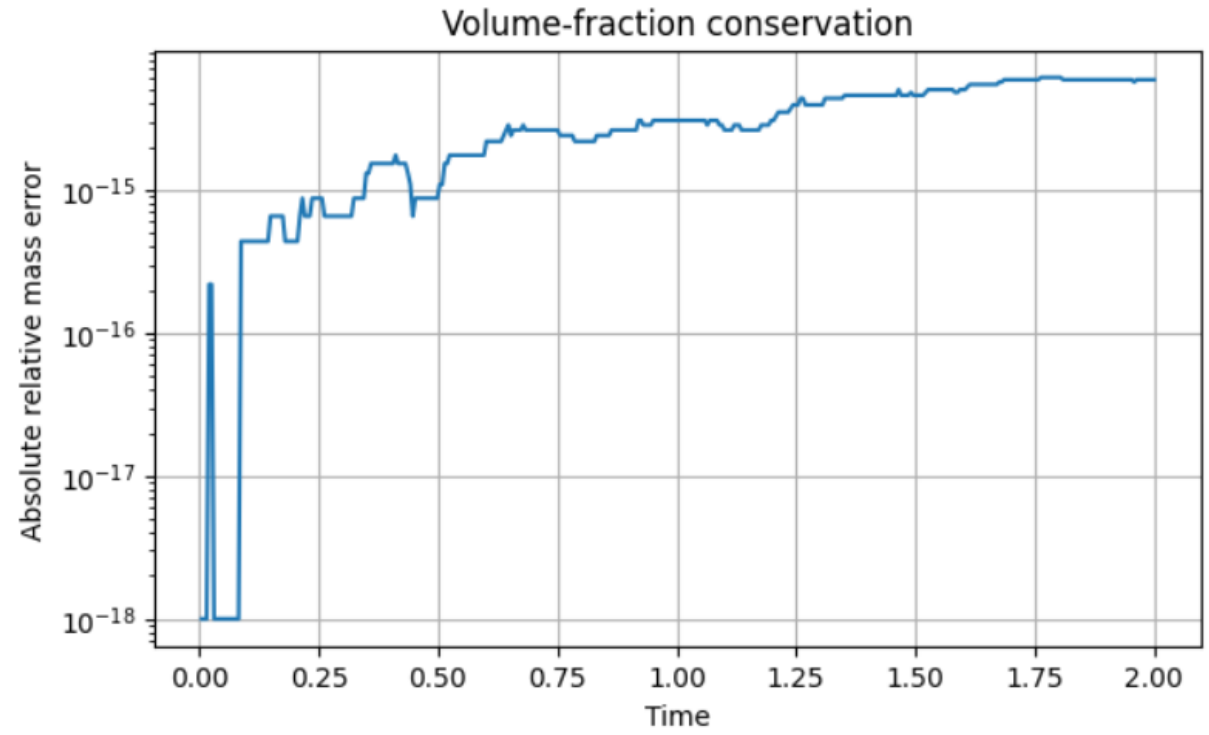
$$A = \frac{\rho_h - \rho_l}{\rho_h + \rho_l}.$$

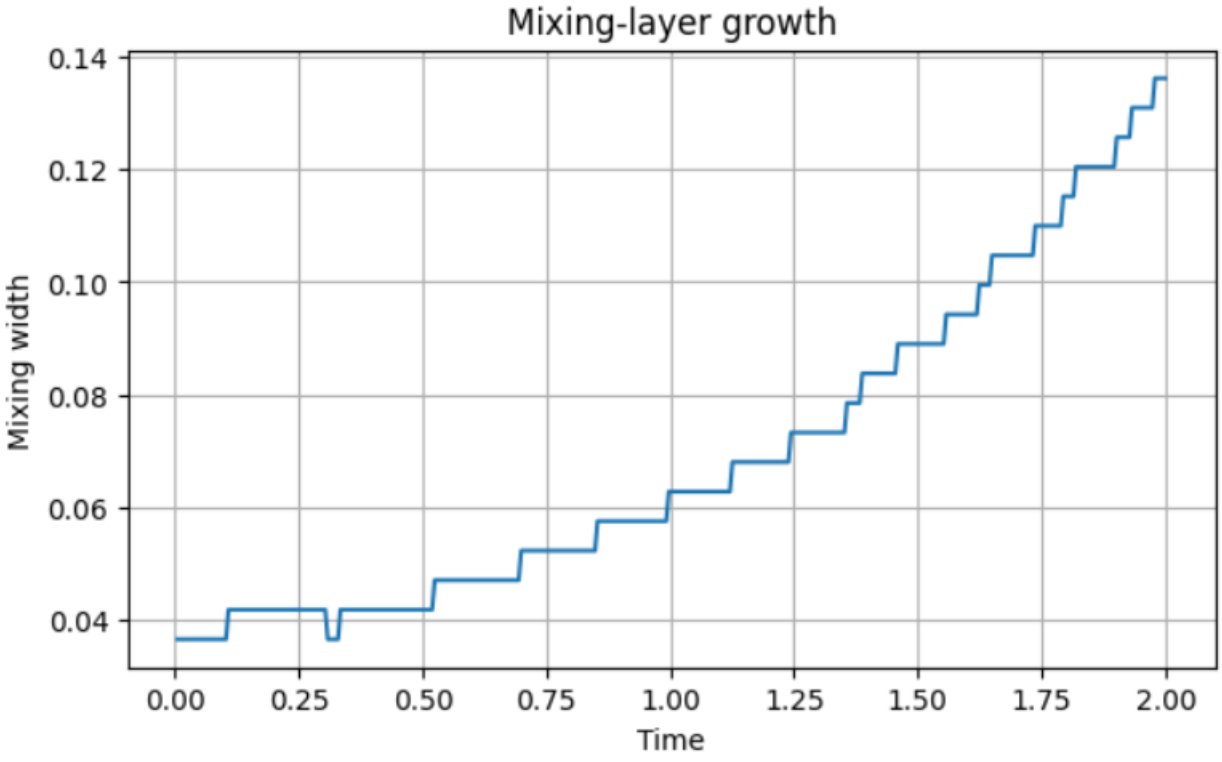
Boundary Conditions: No-slip impermeable box boundaries; initially perturbed interface with heavier fluid above lighter fluid.

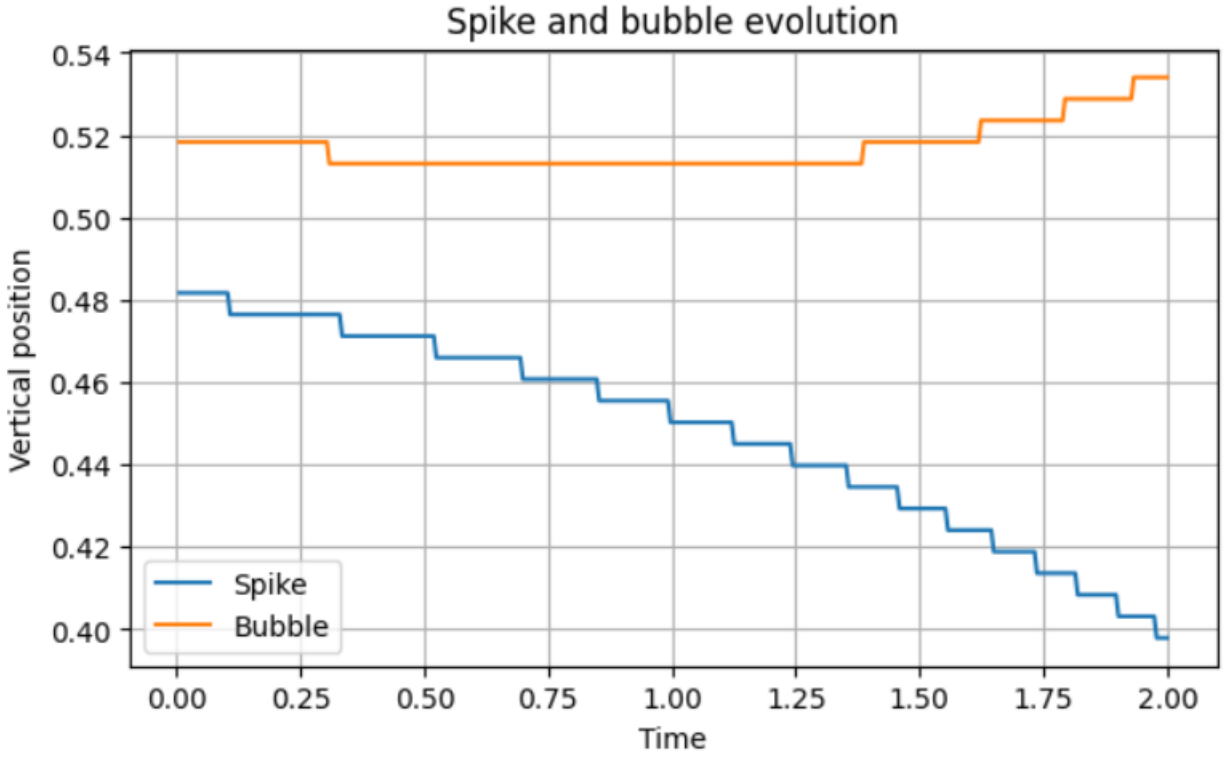
Numerical Method: Projection flow solver coupled to conservative upwind phase-fraction transport, clipping, and conservative redistribution to maintain $0 \leq \alpha \leq 1$.

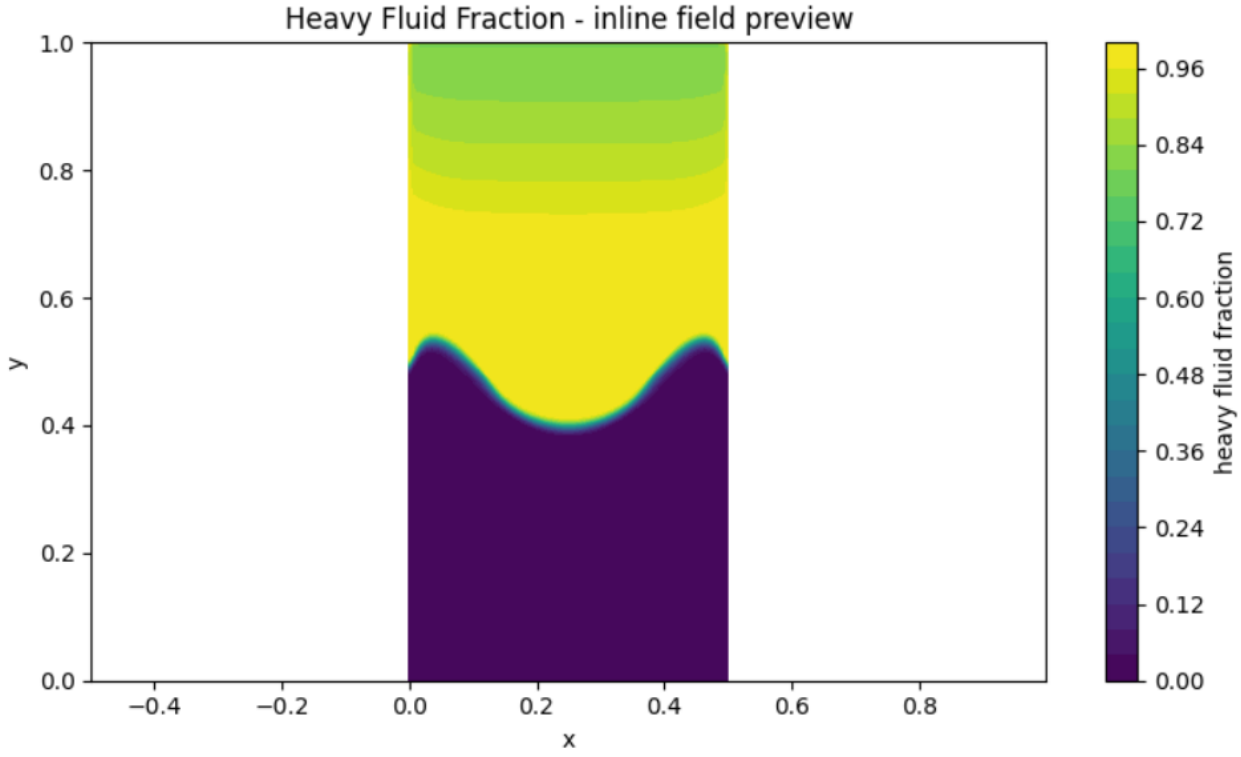
Observations: Spike and bubble positions, mixing width, divergence, mass error, minimum and maximum phase fraction, and interface shape at Atwood numbers 0.2, 0.5, and 0.8. Compare spike, bubble, mixing width, boundedness, and conservation.

For Atwood number 0.2:

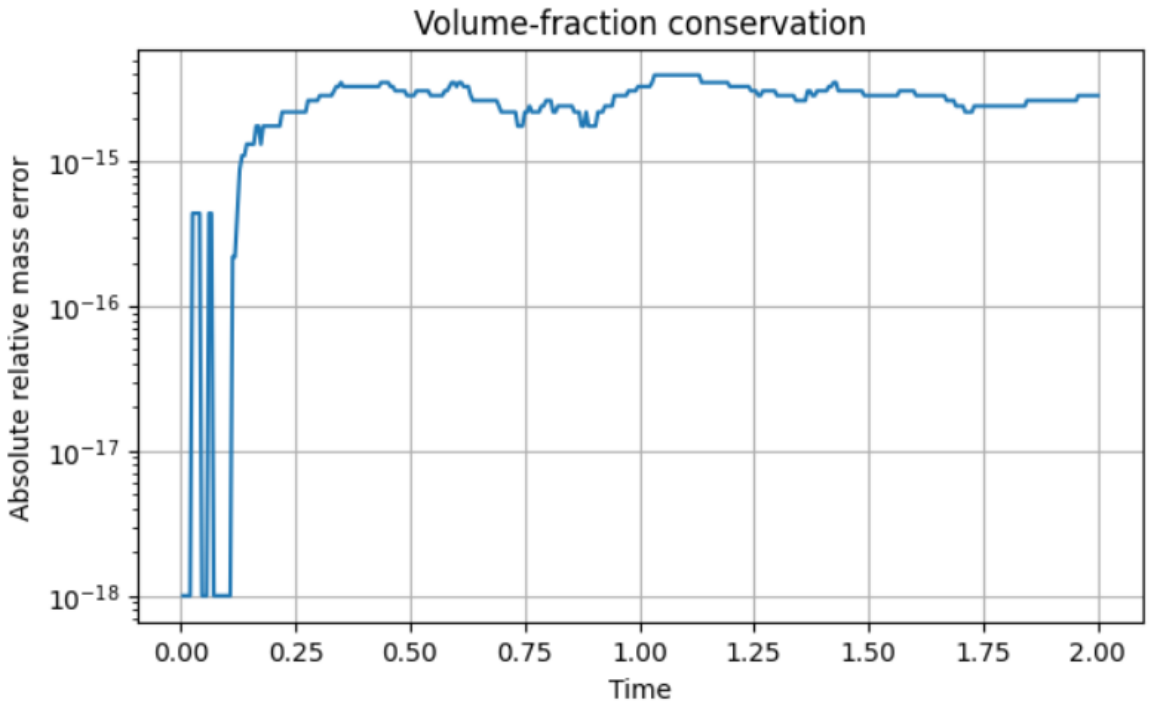


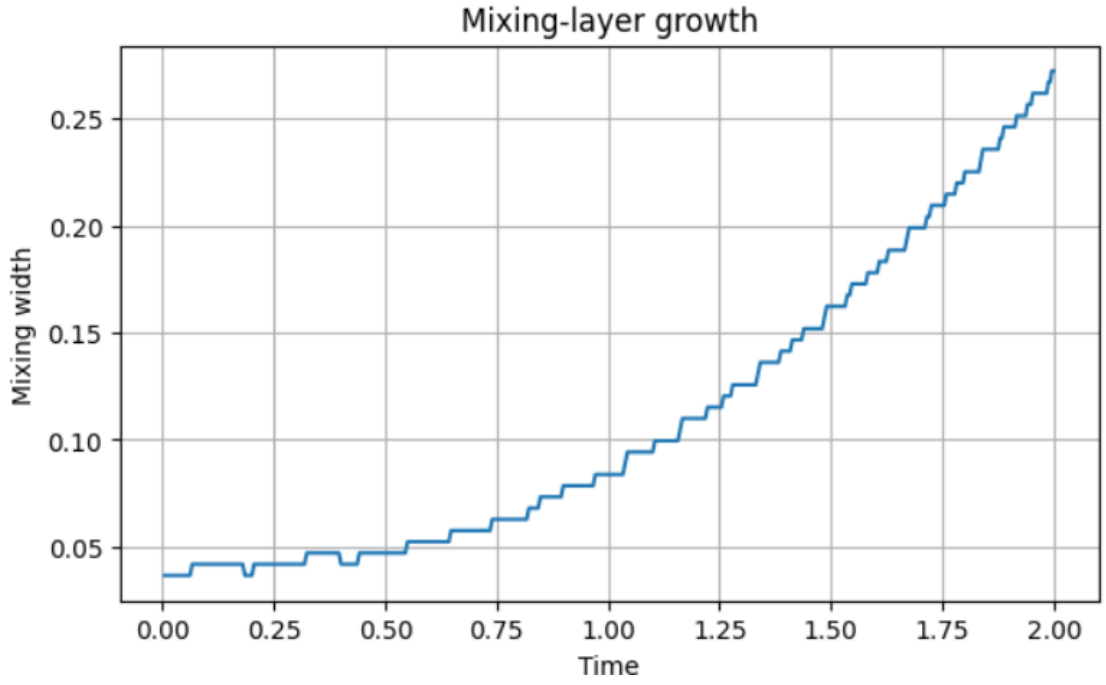


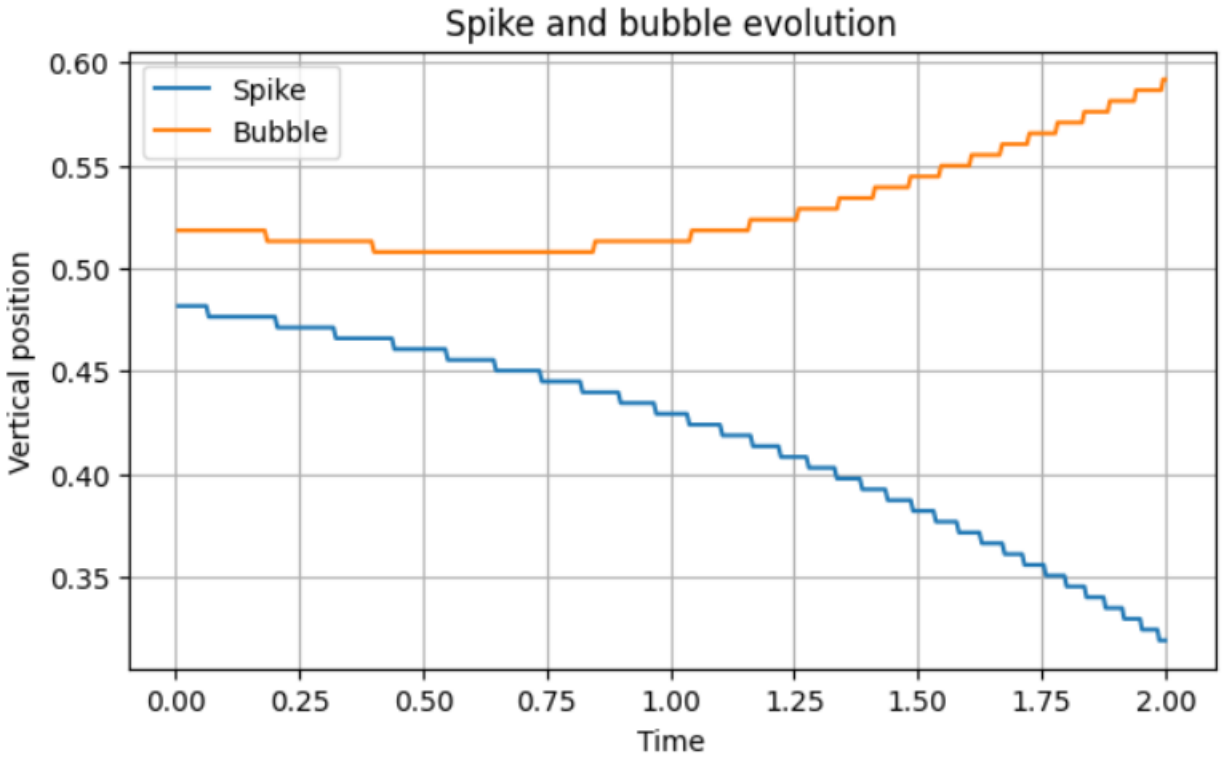

Spike and bubble evolution
Vertical position
Time
Spike
Bubble
0.54
0.52
0.50
0.48
0.46
0.44
0.42
0.40
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

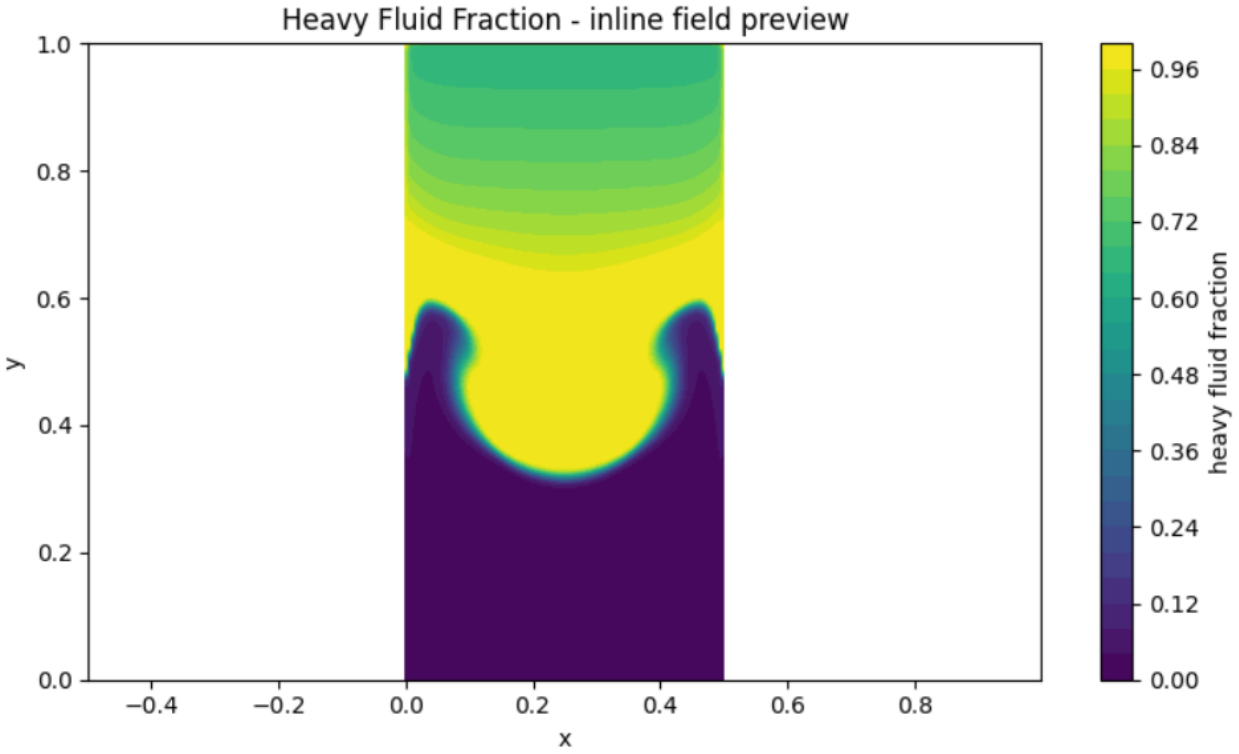

Heavy Fluid Fraction - inline field preview
x
y
heavy fluid fraction
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

For Atwood number 0.5:

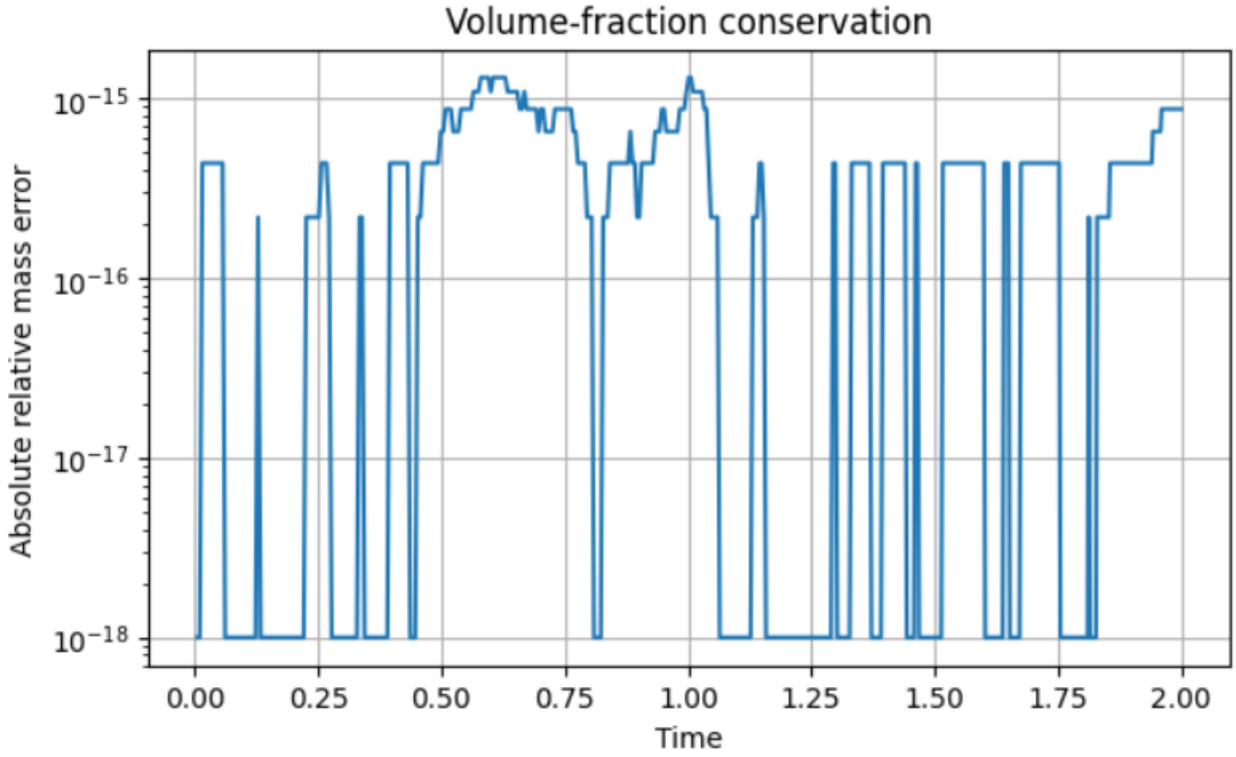


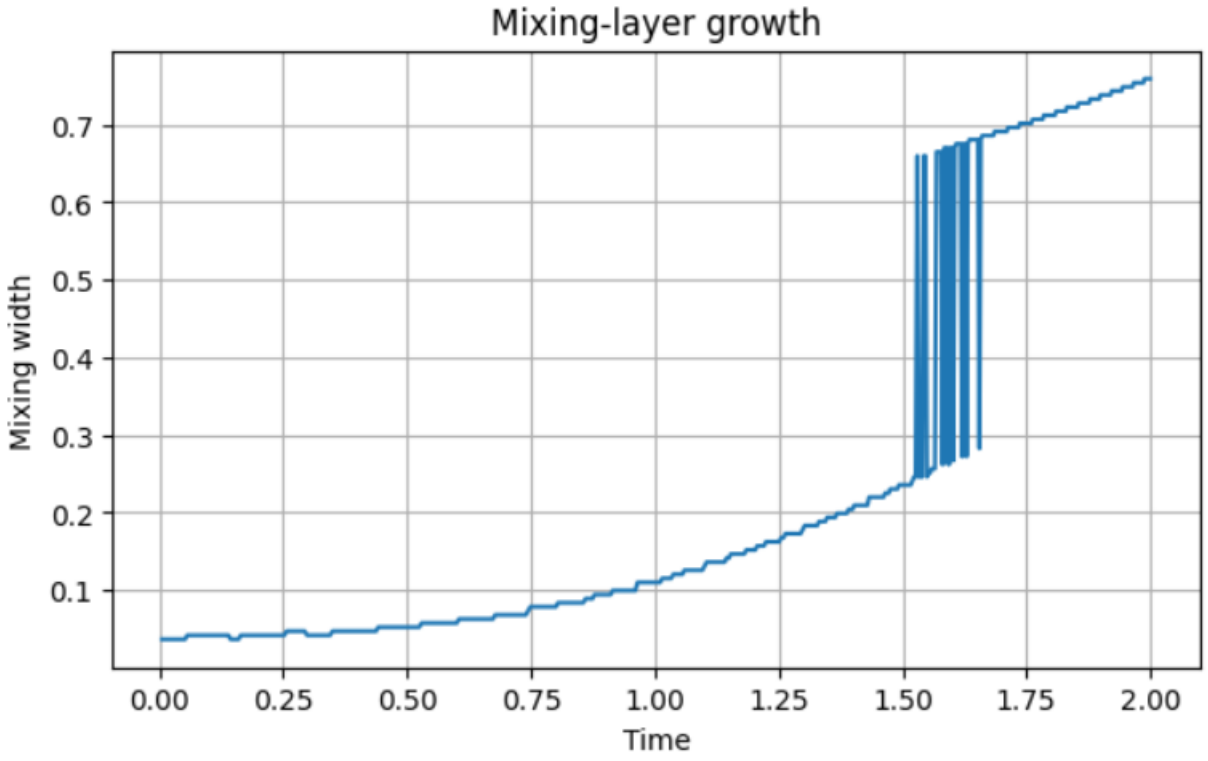


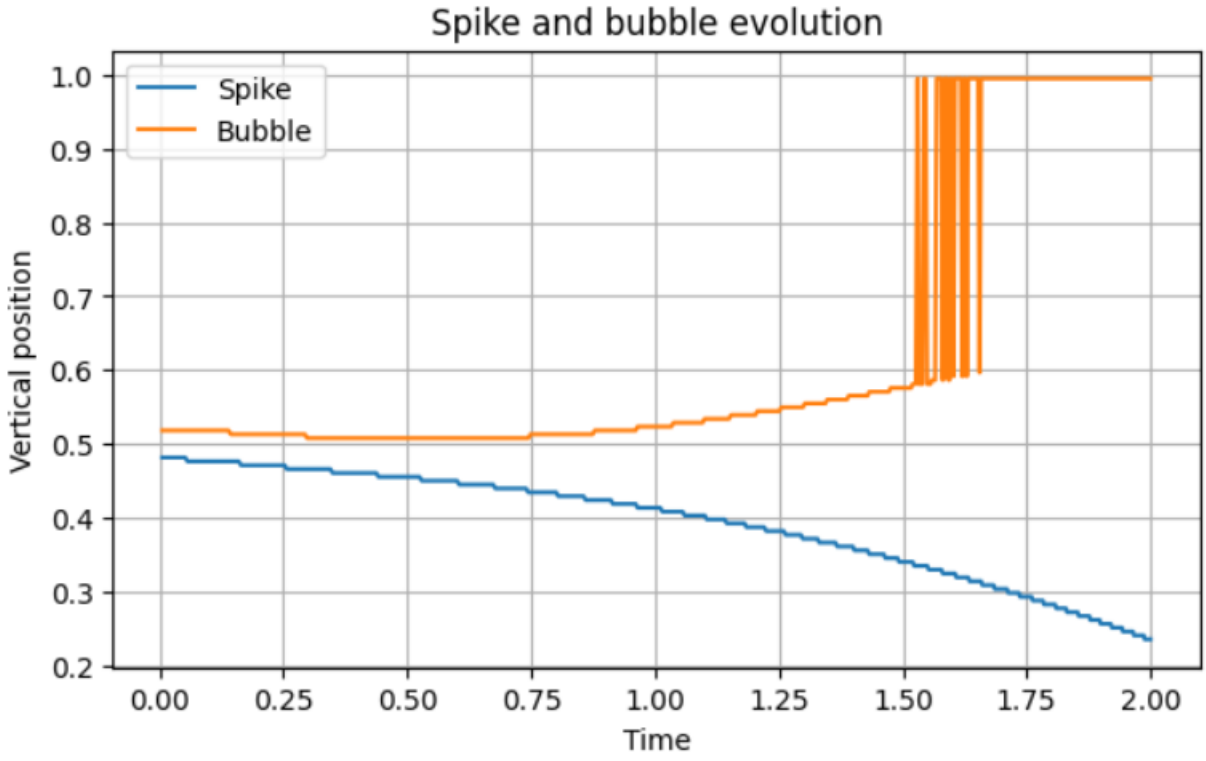

Spike and bubble evolution
Spike
Bubble
Vertical position
Time
0.60
0.55
0.50
0.45
0.40
0.35
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

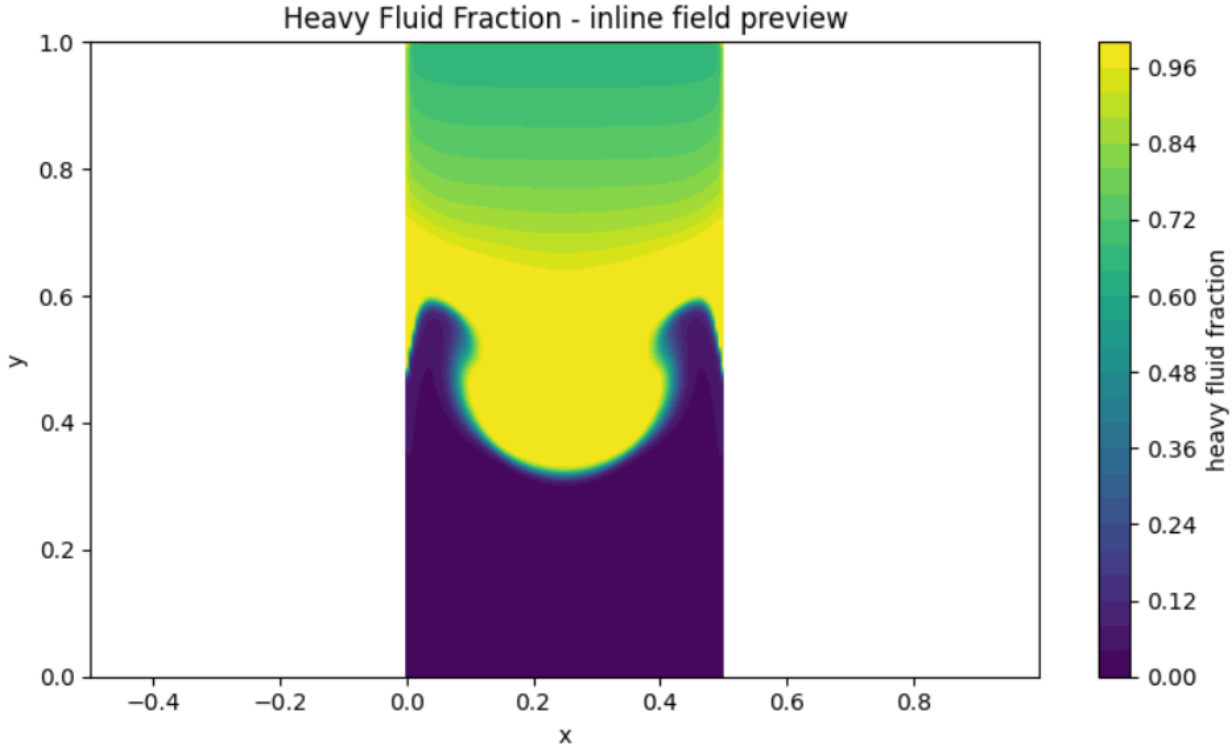

Heavy Fluid Fraction - inline field preview
y
x
heavy fluid fraction
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

For Atwood number 0.8:






Spike and bubble evolution
Spike
Bubble
Vertical position
Time
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00


Heavy Fluid Fraction - inline field preview
y
x
heavy fluid fraction
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

Results:

Parameter	**A = 0.2**	**A = 0.5**	**A = 0.8**
Interface at t = 0 (spike / bubble position, y)	≈ 0.48 / 0.52	≈ 0.48 / 0.52	≈ 0.48 / 0.52
Spike position at t = 2 (y)	≈ 0.398	≈ 0.319	≈ 0.235
Spike descent from t = 0 to 2	≈ 0.084	≈ 0.163	≈ 0.247
Bubble position at t = 2 (y)	≈ 0.534	≈ 0.592	≈ 0.58 at t ≈ 1.5, then reads 1.0 (artefact)
Mixing width at t = 0	≈ 0.037	≈ 0.037	≈ 0.037
Mixing width at t = 2	≈ 0.136	≈ 0.272	≈ 0.76 (affected by artefact)
Net mixing-width growth	≈ 0.099 (at t = 2)	≈ 0.235 (at t = 2)	≈ 0.20 (at t = 1.5)
Maximum absolute relative mass error	$\approx 6 \times 10^{-15}$	$\approx 4 \times 10^{-15}$	$\approx 1.3 \times 10^{-15}$
Jumps in bubble position / mixing width	none	none	t ≈ 1.5–1.65

Conclusion:

The Rayleigh–Taylor instability of a single-mode perturbed heavy-over-light interface in a no-slip box was simulated using a projection flow solver with conservative upwind phase-fraction transport, at Atwood numbers of 0.2, 0.5 and 0.8. In all cases the heavy fluid above penetrates downward as a spike at mid-width, while the lighter fluid rises along the side walls as bubbles. The interface starts at y ≈ 0.5 with the spike at y ≈ 0.48 and the bubble at y ≈ 0.52. By t = 2 the spike has descended to y ≈ 0.398, 0.319 and 0.235 for A = 0.2, 0.5 and 0.8 respectively, i.e. by about 0.084, 0.163 and 0.247. The bubble has risen to y ≈ 0.534 for A = 0.2 and y ≈ 0.592 for A = 0.5.

Effect of Atwood number on growth: Because the buoyancy term scales with the Atwood number, spike descent, bubble rise and mixing width all increase with A. The mixing width, which equals the bubble position minus the spike position in the plotted curves, grows from ≈ 0.037 at t = 0 to

≈ 0.136 (A = 0.2) and ≈ 0.272 (A = 0.5) at t = 2. The net growth of ≈ 0.099 and ≈ 0.235 is in the ratio of about 2.4, close to the Atwood-number ratio of 2.5. Growth is slow at first and accelerates for A = 0.2, the net growth at t = 2 is about 3.8 times that at t = 1, close to quadratic in time. While for A = 0.5 the ratio is about 5. In every case the spike moves faster than the bubble, and the bubble lags most at A = 0.2.

Conservation and boundedness: The absolute relative mass error of the volume fraction stays at round-off level, below approximately 6×10^{-15}, 4×10^{-15} and 1.3×10^{-15} for A = 0.2, 0.5 and 0.8, because both the transport and the redistribution are conservative. The phase fraction stays within $0 \leq \alpha \leq 1$ in the field plots. However, conservation of the total phase fraction does not guarantee a physical field. In the field plots the heavy-fluid fraction in the upper layer decreases progressively towards the top wall, to about 0.8 for A = 0.2 and about 0.6 for A = 0.5 at t = 2, although no interface is present there. For an incompressible flow with an exactly divergence-free velocity this region should remain at $\alpha = 1$. This points to a residual velocity divergence acting through the conservative transport term, and/or to the clipping and redistribution step spreading phase fraction, so the boundedness is partly enforced by redistribution and not obtained from accurate transport alone.

Lid-Driven Cavity using Vorticity-Stream Function

Aim: To solve the incompressible cavity problem and analyze the primary recirculation using vorticity and stream function.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	65
Grid points, ny	65
Raynold's number	100,400
U_lid	2
CFL	0.8
Tolerance	1e-0.8
Max_steps	2000
SOR	1.7

Governing Equation:

$$\nabla^2 \psi = -\omega,$$

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x},$$

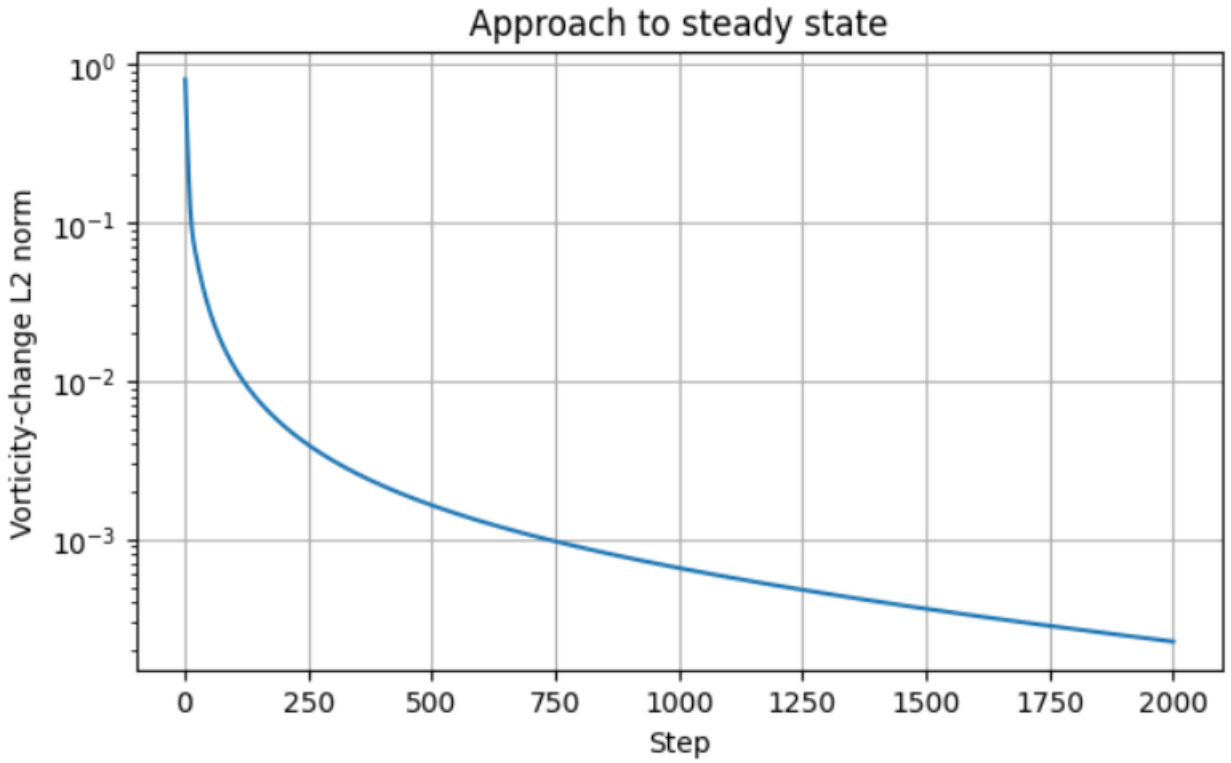
$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + v\frac{\partial \omega}{\partial y} = \frac{1}{Re}\nabla^2 \omega.$$

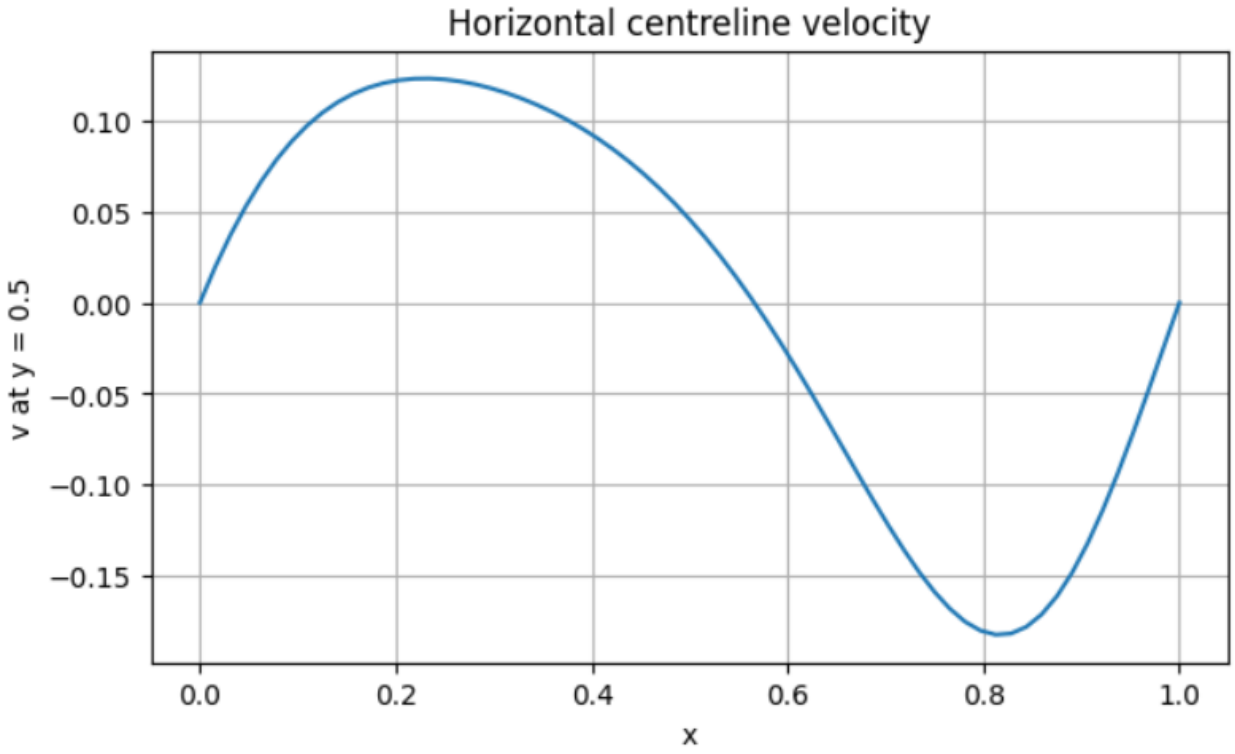
Boundary Conditions: Unit square; stationary side and bottom walls; top wall moving at prescribed lid speed; zero Stream Function on all walls.

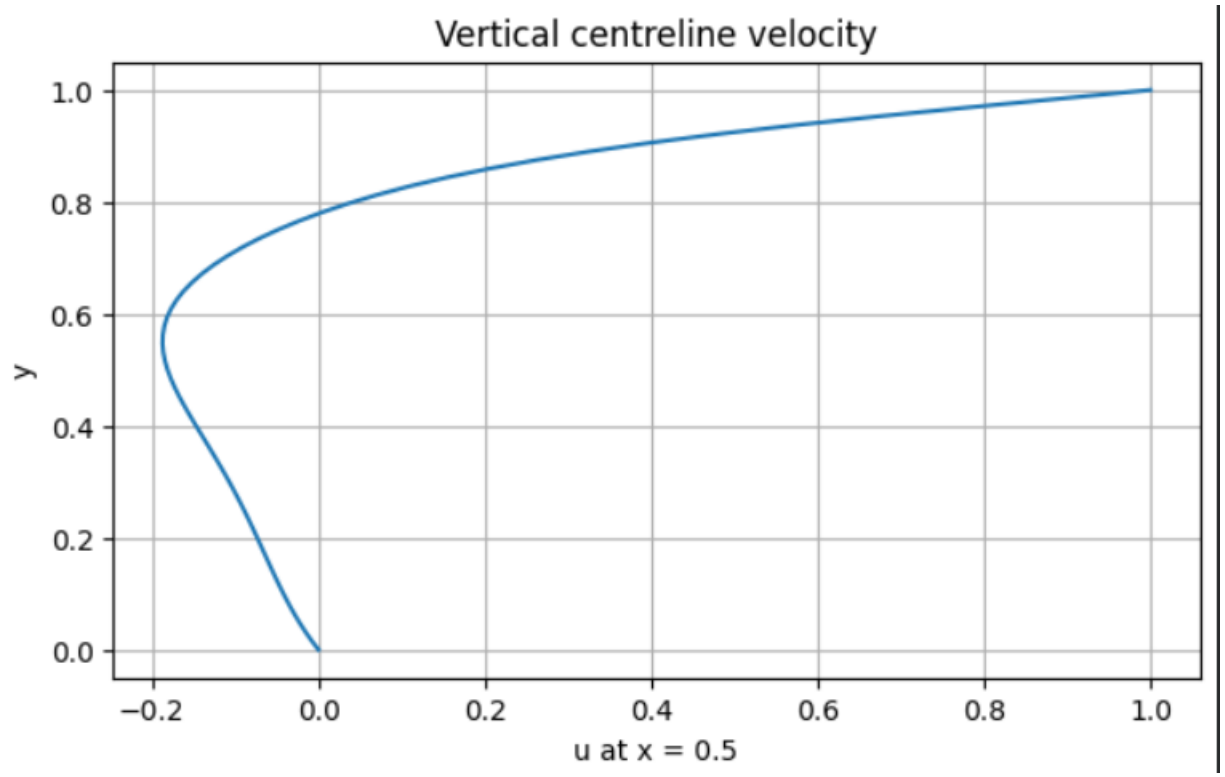
Numerical Method: Explicit upwind vorticity transport, red-black SOR solution of the Stream Function Poisson equation, and Thom-type wall-vorticity conditions.

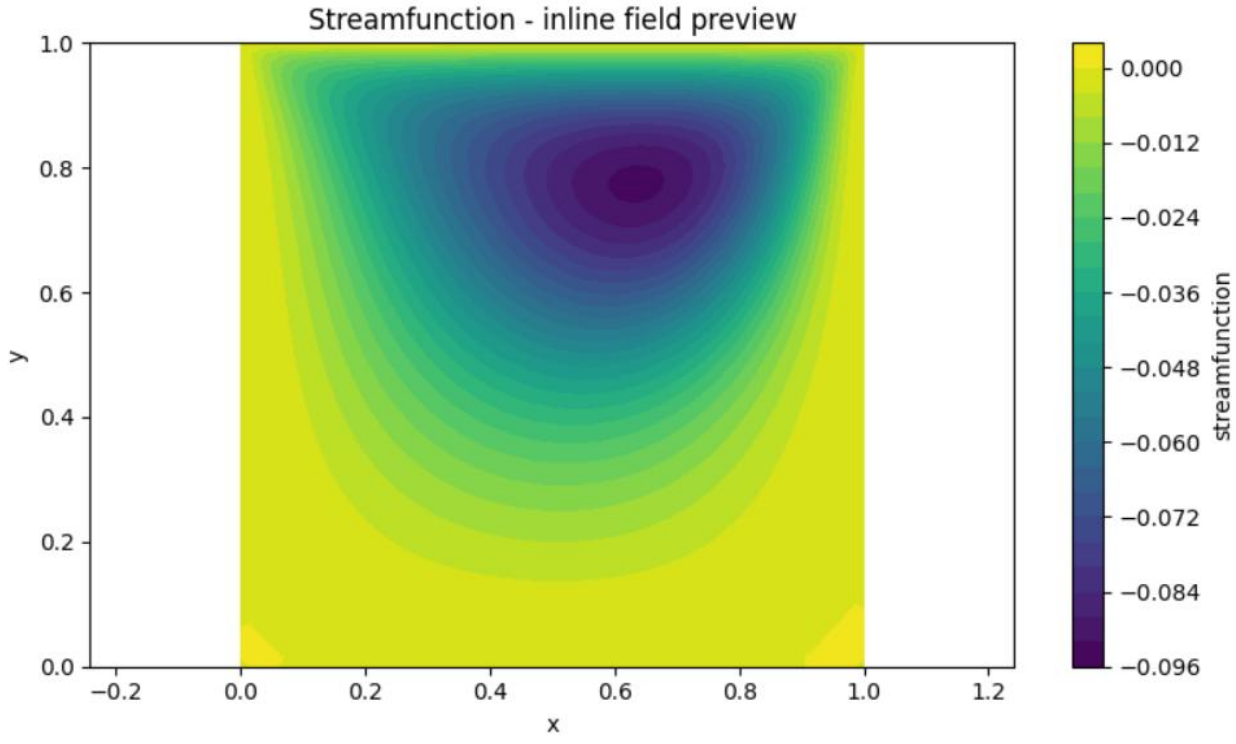
Observations: Primary vortex centre, minimum stream function, centre line velocity profiles, vorticity-change norm, Poisson residual, and grid dependence at Reynolds numbers 100 and 400. Compare centre line velocities, primary-vortex position, convergence, and secondary-corner-vortex development.

For Raynolds number 100:

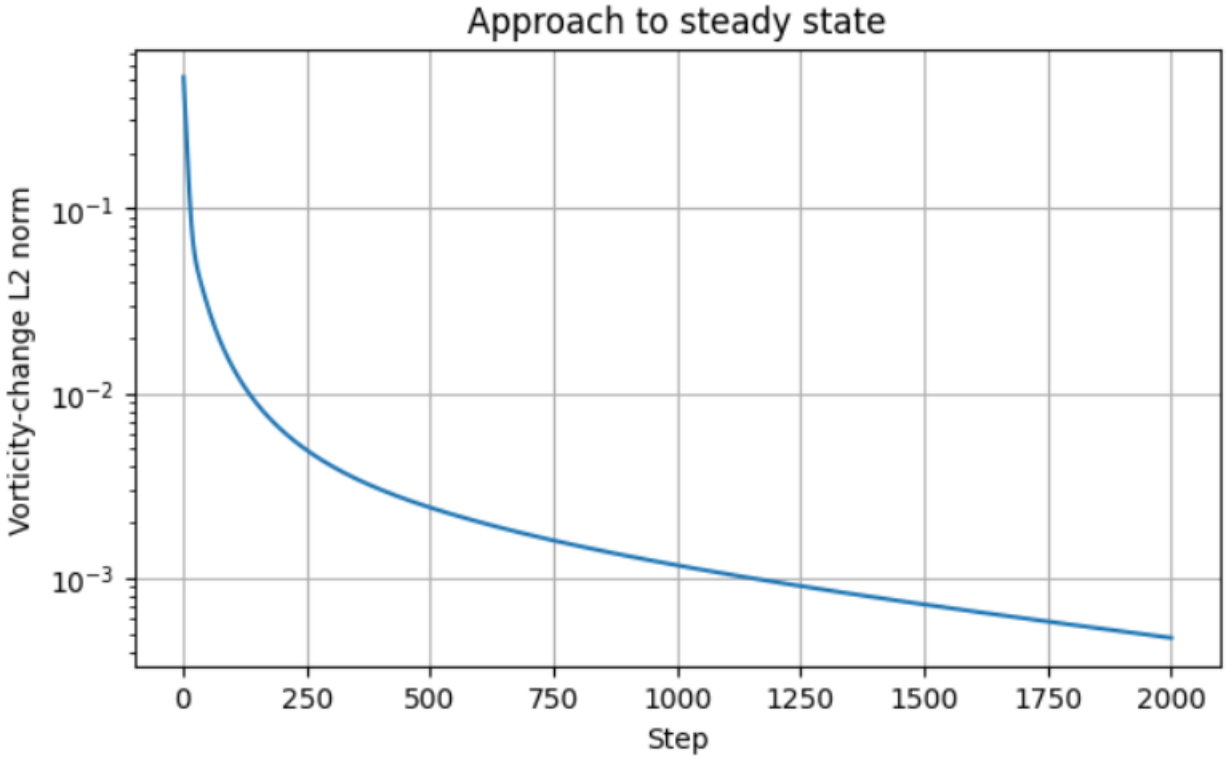


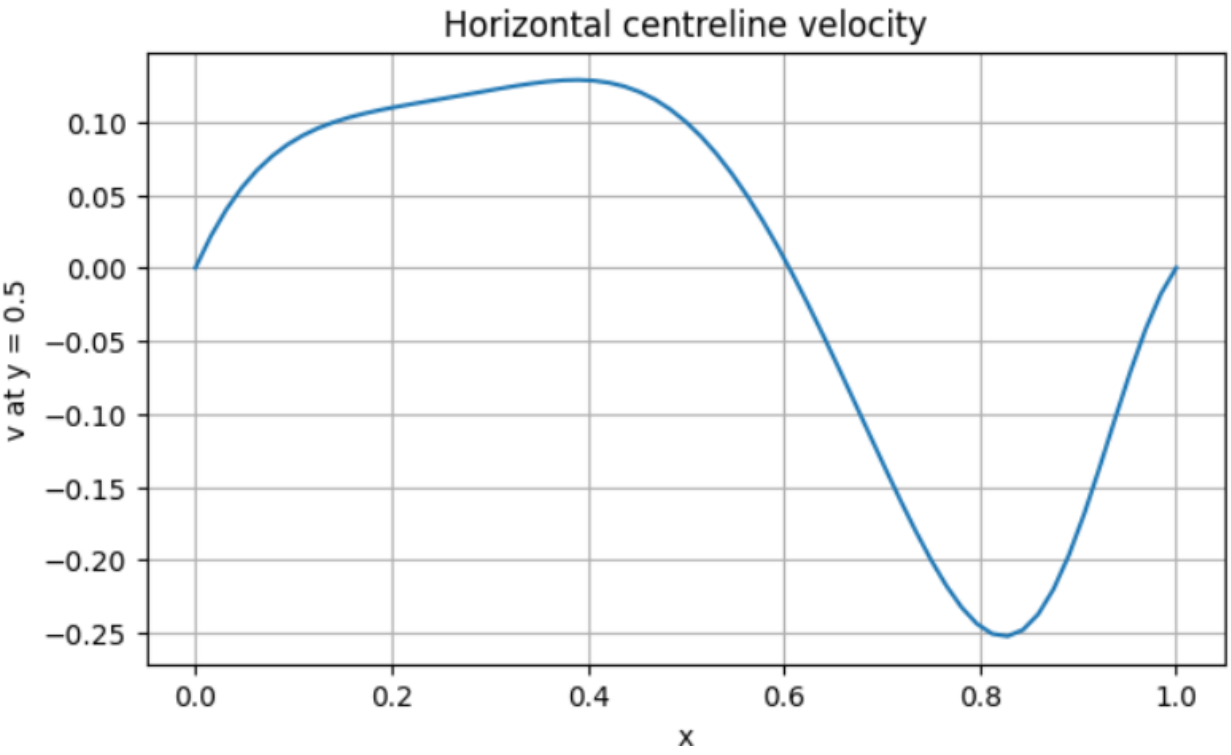


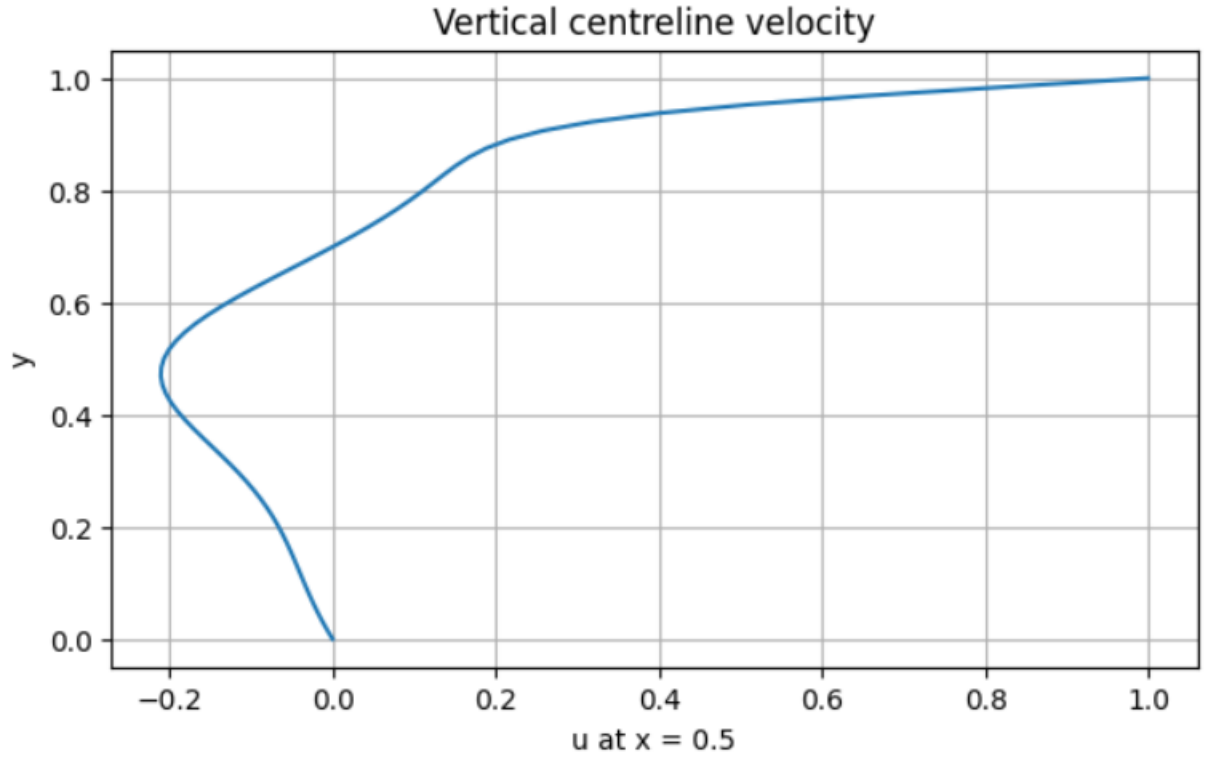

Vertical centreline velocity
y
u at x = 0.5
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

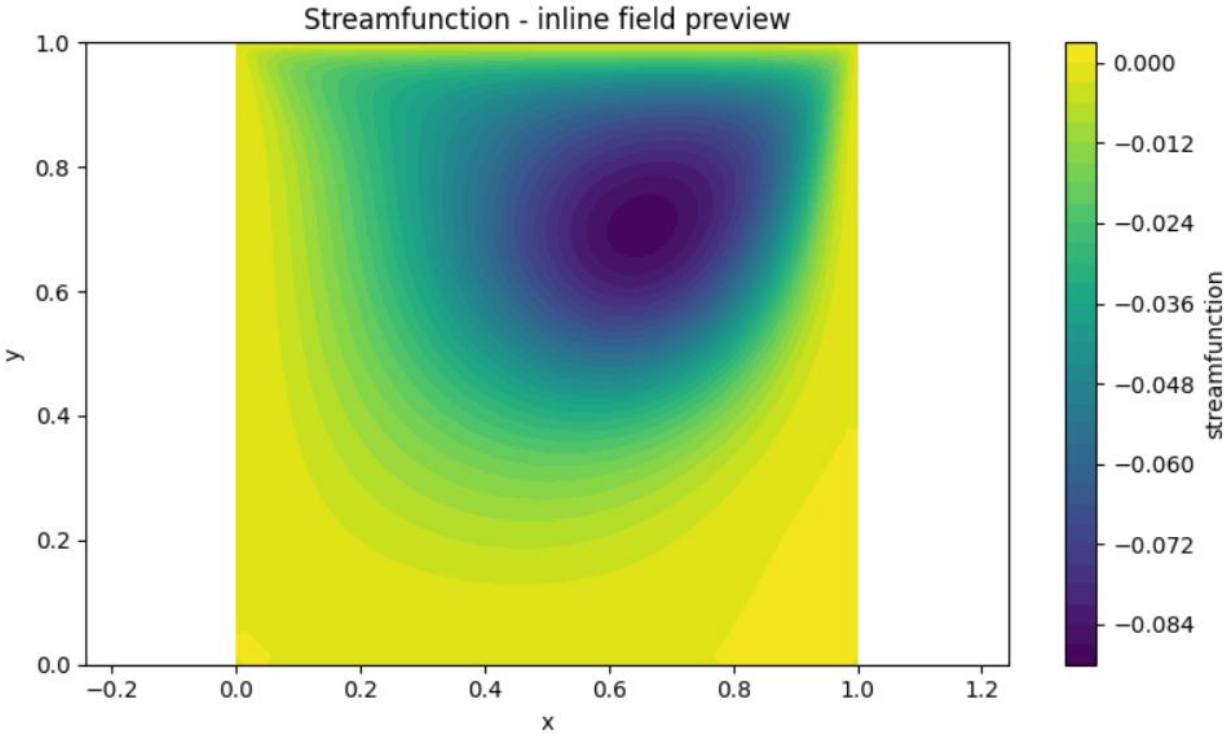

Streamfunction - inline field preview
y
x
streamfunction
0.000
−0.012
−0.024
−0.036
−0.048
−0.060
−0.072
−0.084
−0.096
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2

For Raynolds number 400:






Vertical centreline velocity
1.0
0.8
0.6
0.4
0.2
0.0
y
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
u at x = 0.5


Streamfunction - inline field preview
1.0
0.8
0.6
0.4
0.2
0.0
y
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
x
0.000
−0.012
−0.024
−0.036
−0.048
−0.060
−0.072
−0.084
streamfunction

Results:

Parameter	**Re = 100**	**Re = 400**
Grid / steps run	65 × 65 / 2000 (max)	65 × 65 / 2000 (max)
Primary vortex centre (x, y)	≈ (0.63, 0.77)	≈ (0.66, 0.70)
Minimum Stream Function, ψ_min	≈ −0.096	≈ −0.090
Min. u on vertical centreline (x = 0.5)	≈ −0.185 at y ≈ 0.55	≈ −0.205 at y ≈ 0.48
u = 0 crossing on vertical centreline	y ≈ 0.78	y ≈ 0.70
Max. v on horizontal centreline (y = 0.5)	≈ +0.123 at x ≈ 0.23	≈ +0.129 at x ≈ 0.39
Min. v on horizontal centreline (y = 0.5)	≈ −0.182 at x ≈ 0.82	≈ −0.252 at x ≈ 0.83
v = 0 crossing on horizontal centreline	x ≈ 0.57	x ≈ 0.60
Vorticity-change norm, step 0	$\approx 8 \times 10^{-1}$	$\approx 5 \times 10^{-1}$
Vorticity-change norm, step 250	$\approx 4 \times 10^{-3}$	$\approx 5 \times 10^{-3}$
Vorticity-change norm, step 2000	$\approx 2.3 \times 10^{-4}$	$\approx 4.7 \times 10^{-4}$
Benchmark ψ_min, vortex centre (Ghia et al., ≈)	≈ −0.103 at (0.62, 0.73)	≈ −0.114 at (0.55, 0.61)
Benchmark min. u on vertical centreline (≈)	≈ −0.21 at y ≈ 0.45	≈ −0.33 at y ≈ 0.28

Conclusion:

The lid-driven cavity flow in a unit square was solved with the vorticity–Stream Function formulation, using explicit upwind vorticity transport, red-black SOR for the Stream Function Poisson equation and Thom-type wall vorticity, on a 65 × 65 grid at Reynolds numbers of 100 and 400. In both cases a single primary vortex forms below the moving lid, with the Stream Function negative (clockwise circulation) in the vortex core. The primary vortex centre lies at approximately

(0.63, 0.77) for Re = 100 and (0.66, 0.70) for Re = 400, with minimum Stream Function of about −0.096 and −0.090 respectively.

Centreline velocities: On the vertical centreline the u-velocity rises from zero at the bottom wall, reaches a reverse-flow minimum of about −0.185 (Re = 100, y ≈ 0.55) and −0.205 (Re = 400, y ≈ 0.48), crosses zero at y ≈ 0.78 and 0.70, and reaches the lid speed of 1 at y = 1. On the horizontal centreline the v-velocity is positive on the left (up to ≈ 0.123 at x ≈ 0.23 for Re = 100 and ≈ 0.129 at x ≈ 0.39 for Re = 400) and negative on the right (down to ≈ −0.182 and ≈ −0.252 at x ≈ 0.82–0.83), which is the expected upward flow near the left wall and downward flow near the right wall for a clockwise vortex. The profiles are less symmetric and the right-hand downward jet is stronger at Re = 400, and the shear layer under the lid is thinner: u = 0.6 is reached at y ≈ 0.945 for Re = 100 and at y ≈ 0.965 for Re = 400. This is consistent with convection becoming more important relative to viscous diffusion as Re increases.

Convergence: The vorticity-change norm falls quickly from about 0.8 (Re = 100) and 0.5 (Re = 400) to about 4–5 × 10^{-3} by step 250, but then decays only slowly, reaching about 2.3 × 10^{-4} and 4.7 × 10^{-4} at step 2000. The curves are still falling at the end of the run and no plateau is visible, so neither case reached the specified tolerance and both runs stopped at the step limit. The solution is therefore a close-to-steady transient state and not a fully converged steady state. The vorticity-change norm at step 2000 is about twice as high at Re = 400 as at Re = 100, so the higher-Re case is further from steady state. The Poisson residual was not plotted, so the accuracy of the SOR solve cannot be assessed from the results.

Comparison with benchmark data: The well-known benchmark solution of Ghia et al. for the same problem gives a minimum stream function of about −0.103 at (0.62, 0.73) for Re = 100 and about −0.114 at (0.55, 0.61) for Re = 400, and the minimum u on the vertical centreline is about −0.21 at Re = 100 and about −0.33 at Re = 400. The present results are in qualitative agreement at Re = 100, with the vortex centre and the u-minimum close to the benchmark, but the vortex strength is lower by about 7%, and the velocity extrema are lower. At Re = 400 the differences are larger, the stream function minimum is about 20% too weak, the u-minimum of about −0.21 is far from about −0.33, and the vortex centre at (0.66, 0.70) lies to the right of and above the benchmark position.

First-order upwind transport of vorticity introduces numerical diffusion that acts like an increased viscosity, which weakens the vortex and thins the effective Reynolds number, and this effect grows with Re on a coarse 65 × 65 grid. Overall, the method reproduces the qualitative structure of the lid-driven cavity flow where a single clockwise primary vortex below the lid, reverse flow along the lower part of the vertical centre line, a thinner lid shear layer and stronger asymmetry at higher Re. However, the quantitative accuracy is limited, especially at Re = 400, by first-order upwind diffusion, the coarse grid and incomplete convergence

Turbulent Pipe-Flow Model Comparison

Aim: To compare radial eddy-viscosity closures using mean velocity, friction factor, pressure drop, and turbulent viscosity.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
vtk, nx	160
vtk, ny	61
Re_min	10^4, 10^5
Re_max	10^4,10^5
N_Re	3
Max iterations	1500
Tolerance	1e-07

Governing Equation:

For fully developed axisymmetric mean flow,

$$\frac{1}{r}\frac{d}{dr}\left[r(\nu+\nu_t)\frac{dU}{dr}\right]=\frac{1}{\rho}\frac{dp}{dx}.$$

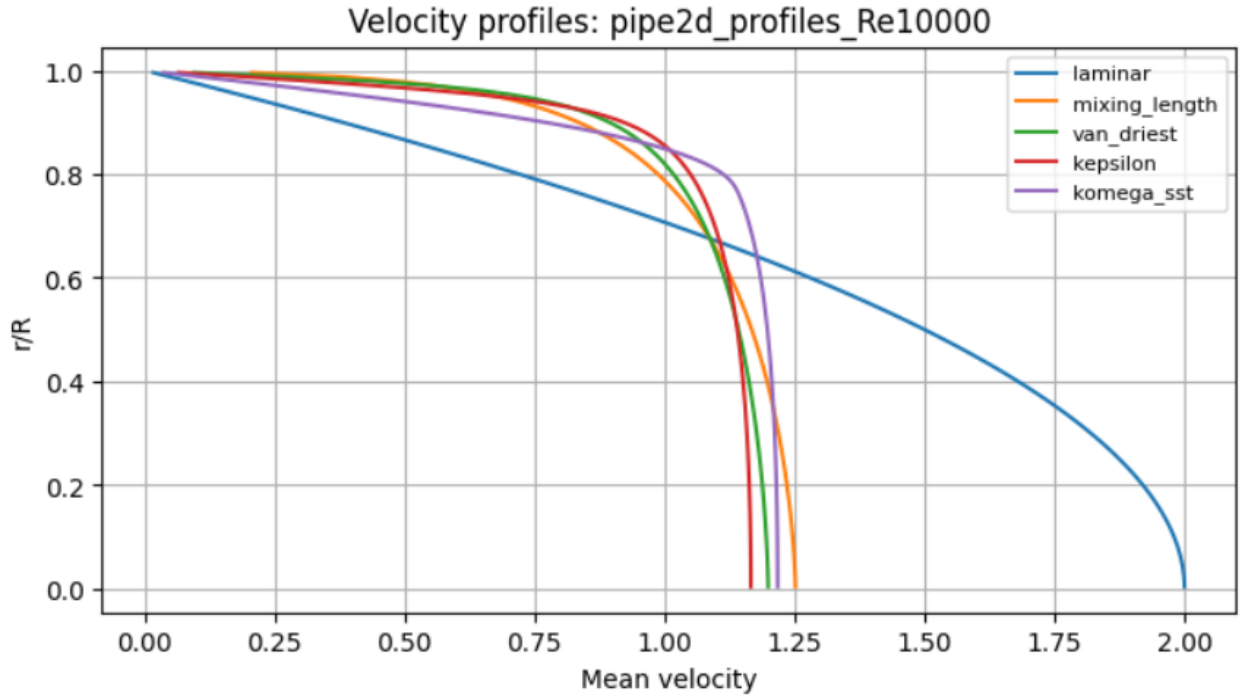
The closures differ through the model for νt

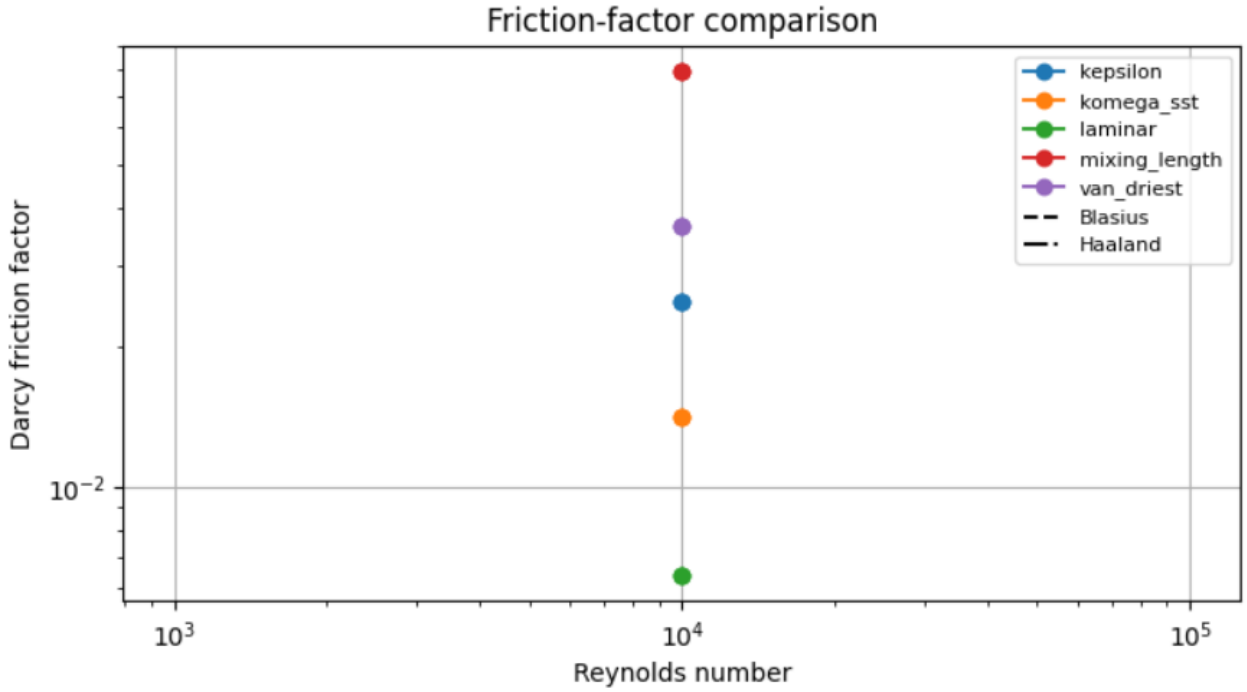
Boundary Conditions: Axis symmetry at the centreline, no-slip at the wall, and pressure gradient adjusted to obtain unit bulk velocity.

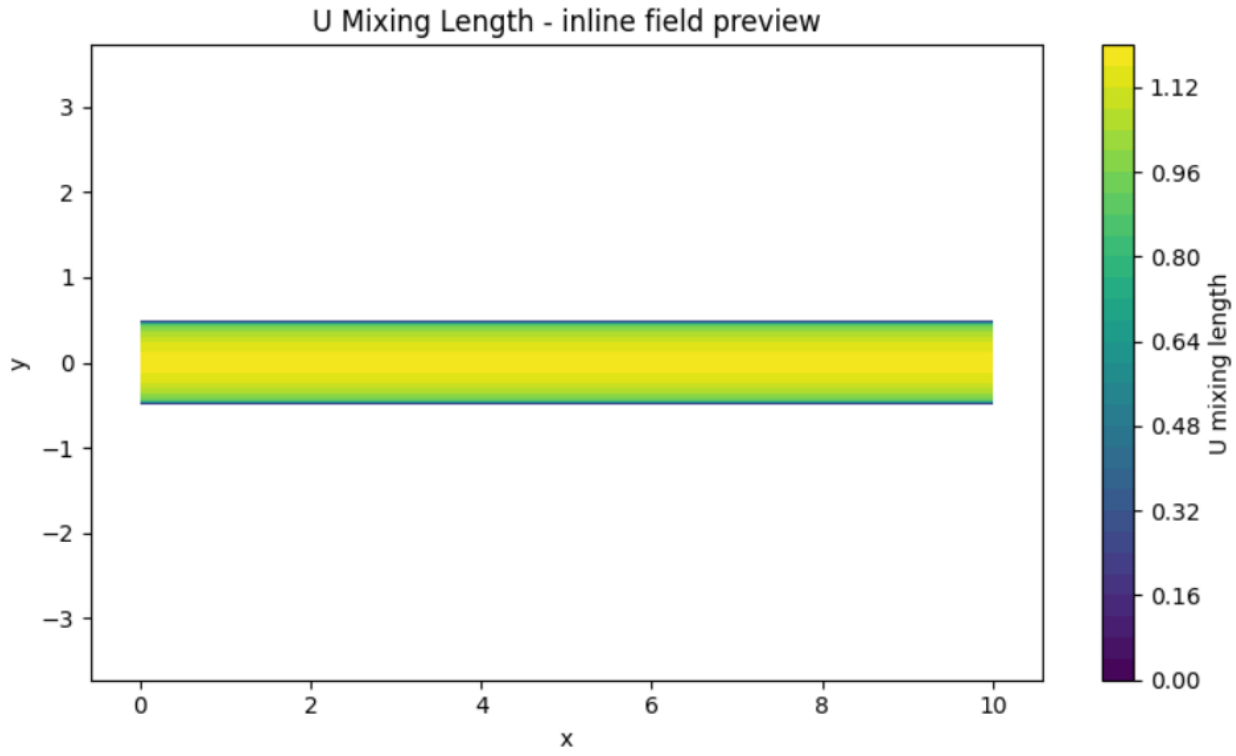
Numerical Method: Radial finite-volume momentum solution for laminar, mixing-length, Van-Driest-damped, equilibrium k-ε, and equilibrium k-ω/SST surrogate closures.

Observations: Velocity profile fullness, friction factor, Blasius and Haaland references, wall shear, pressure drop, and maximum turbulent-viscosity ratio at Re = 10^4, approximately 3.16×10^4, and 10^5. Explain which closure gives the fullest profile and compare friction factors with reference correlations.

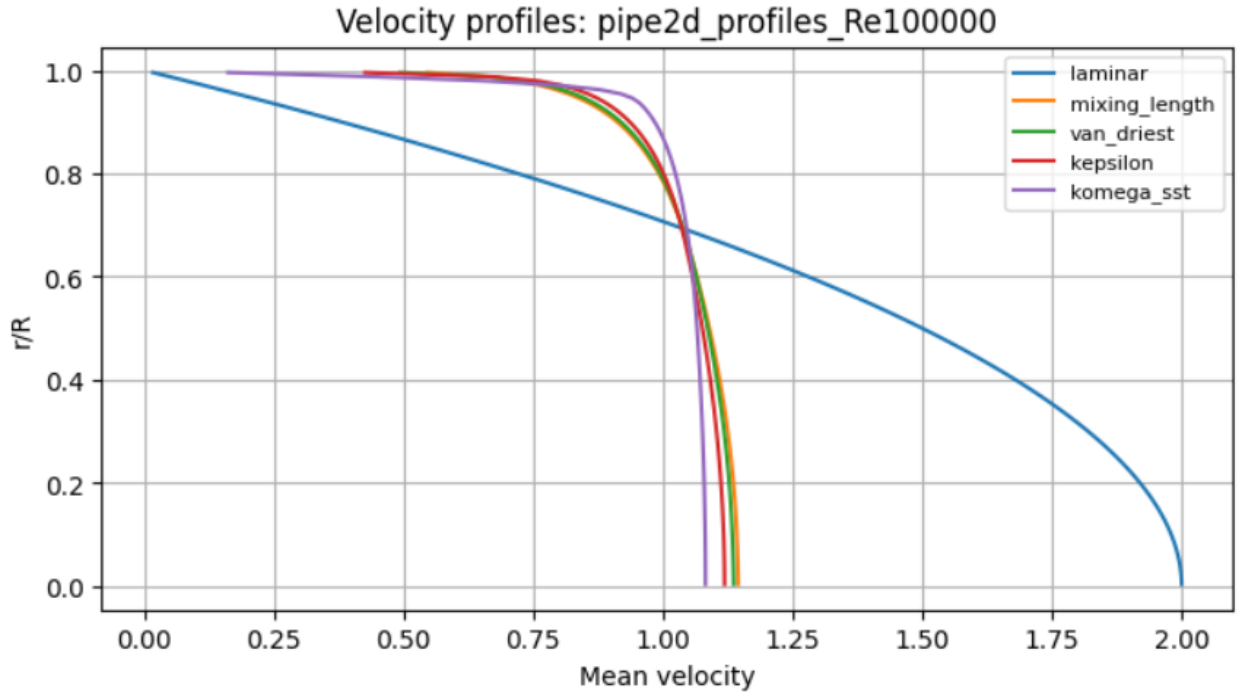
For Raynolds number 10^4:

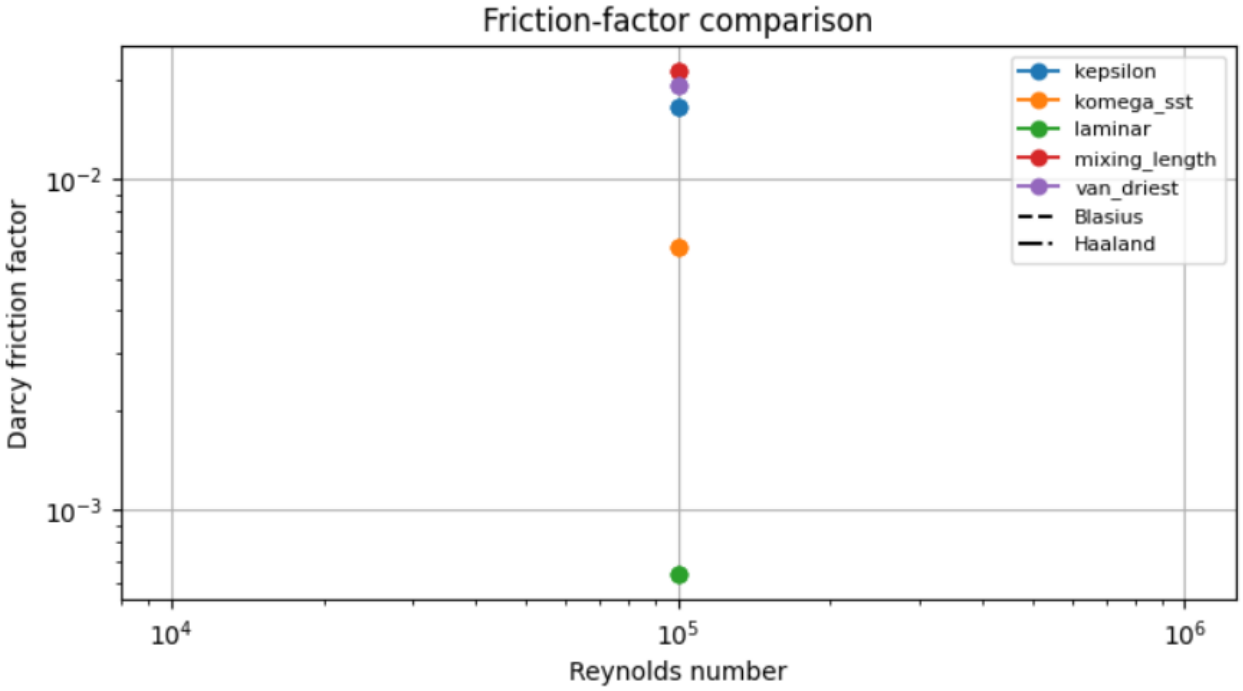


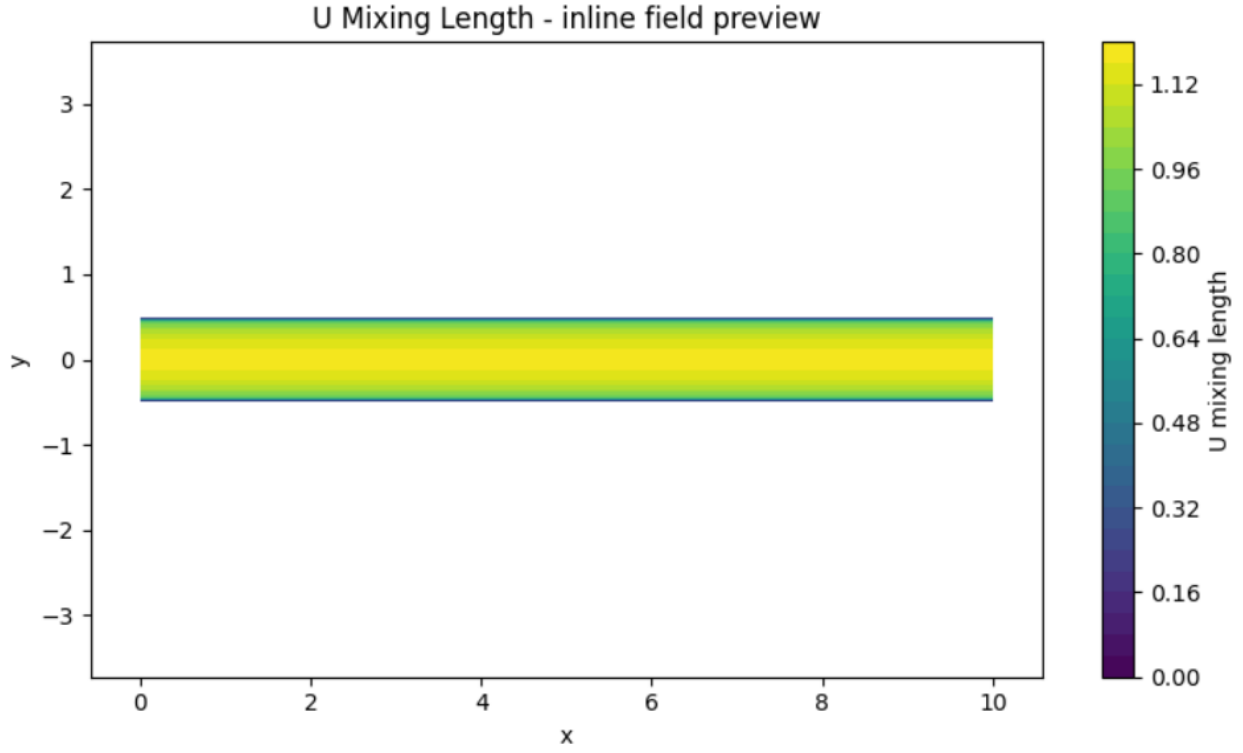




For Raynolds number 10^5:




Friction-factor comparison
kepsilon
komega_sst
laminar
mixing_length
van_driest
Blasius
Haaland
Darcy friction factor
10^{-2}
10^{-3}
10^{4}
10^{5}
10^{6}
Reynolds number


U Mixing Length - inline field preview
y
x
U mixing length
1.12
0.96
0.80
0.64
0.48
0.32
0.16
0.00

Results:

Parameter	**Re = 10^4**	**Re = 10^5**
Blasius friction factor (reference)	0.0316	0.0178
Haaland friction factor, smooth pipe (reference)	0.0309	0.0178
Centreline velocity, U_cl (fullness U_b / U_cl): laminar	2.00 (0.50)	2.00 (0.50)
Mixing length	≈ 1.25 (0.80)	≈ 1.14 (0.88)
Van Driest	≈ 1.20 (0.83)	≈ 1.14 (0.88)
k–ε	≈ 1.165 (0.86)	≈ 1.115 (0.90)
k–ω / SST surrogate	≈ 1.20 (0.83)	≈ 1.08 (0.93)
Darcy friction factor (ratio to Blasius): laminar	≈ 0.0064 (0.20)	≈ 0.00064 (0.04)
Mixing length	≈ 0.083 (2.6)	≈ 0.021 (1.20)
Van Driest	≈ 0.038 (1.19)	≈ 0.019 (1.09)
k–ε	≈ 0.026 (0.81)	≈ 0.0165 (0.93)
k–ω / SST surrogate	≈ 0.014 (0.45)	≈ 0.0062 (0.35)
Wall shear and pressure drop	proportional to f at fixed U_b	proportional to f at fixed U_b

Conclusion:

All turbulent models get the flatter, fuller velocity profile right, but having a fuller profile doesn't mean the friction factor is accurate. Van Driest gives the closest friction values to Blasius (within about 20%), while standard mixing length overshoots and the k–w/SST surrogate misses badly by more than half. To properly fix and evaluate these results, we need to check the near-wall mesh (y^+ values), use proper wall functions, and include the reference correlation lines in the plots.

Generic Convection-Diffusion Transport

Aim: To compare scalar transport in forced flat-plate, forced channel, and natural-convection surrogate velocity fields.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	65
Grid points, ny	65
Raynold's number	100,400
U_lid	2
CFL	0.8
Tolerance	1e-0.8
Max_steps	2000
SOR	1.7

Governing Equation:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = \nabla \cdot (\Gamma \nabla \phi) + S_\phi,$$

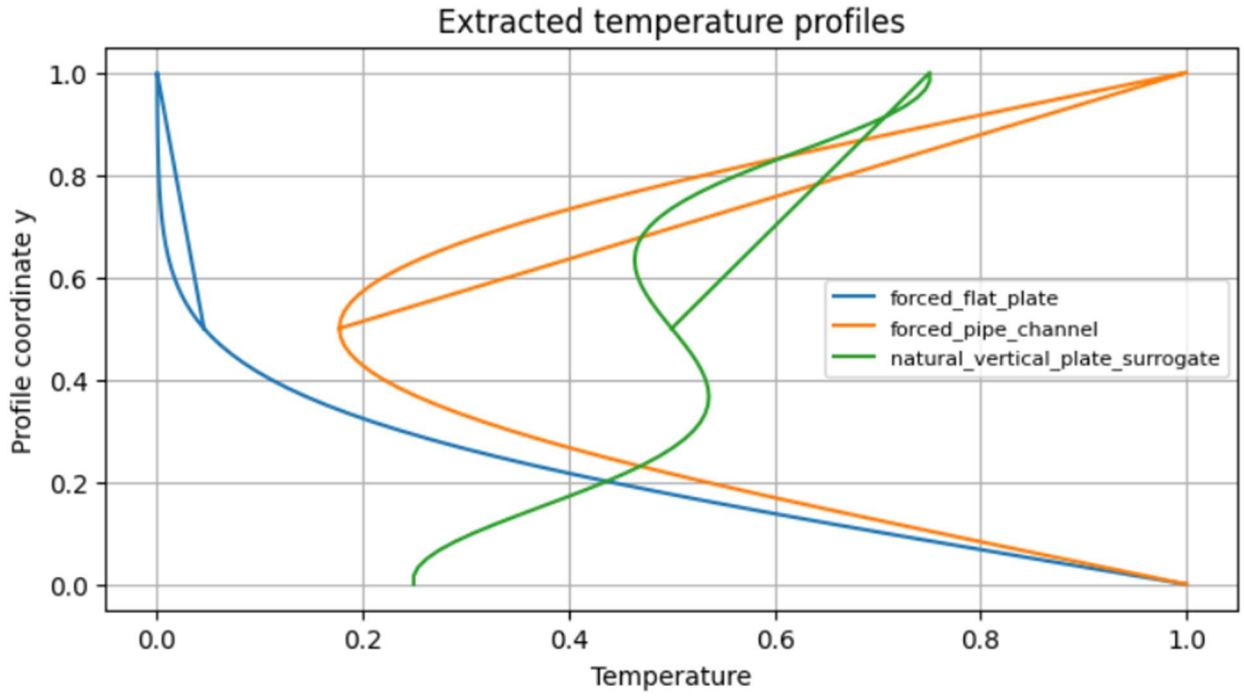
with cell Peclet number $Pe = |\mathbf{u}|\Delta x/\Gamma$.

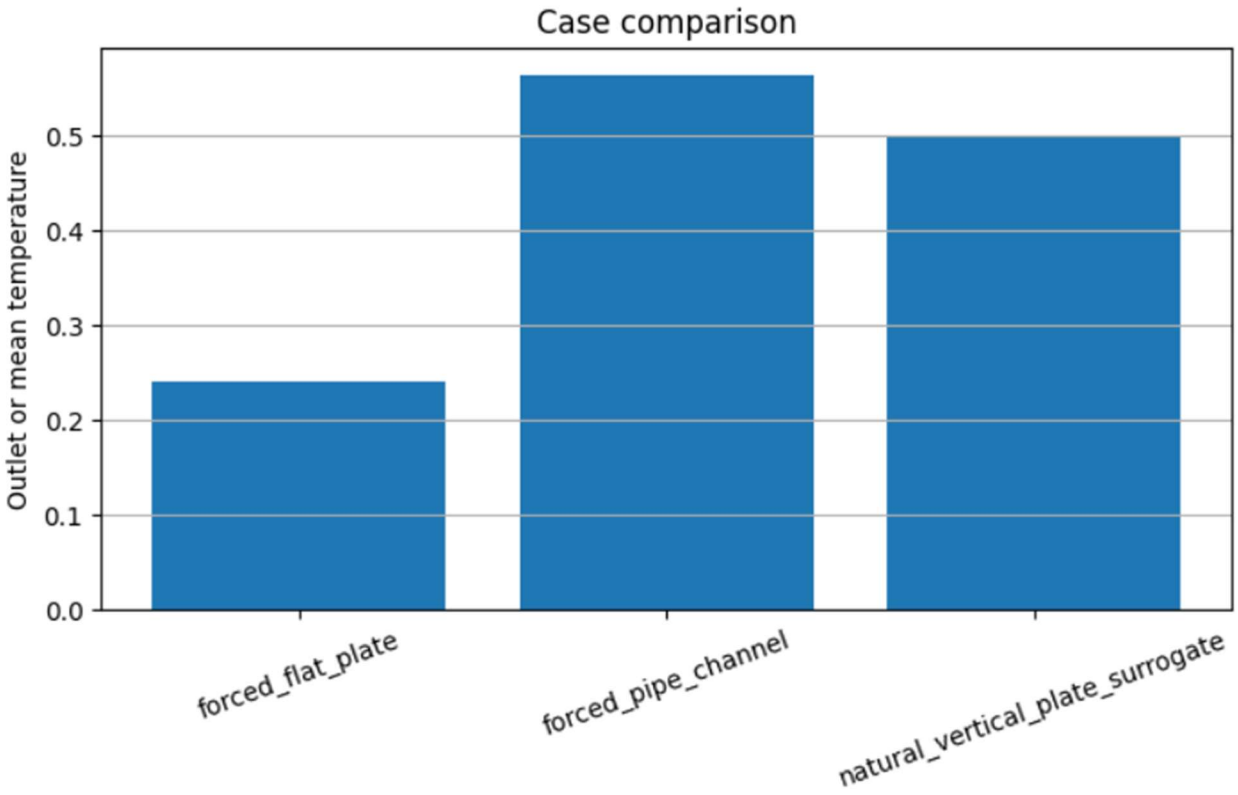
Boundary Conditions: Case-dependent thermal boundaries: heated flat plate, heated channel walls, or hot/cold vertical walls for the recirculating surrogate.

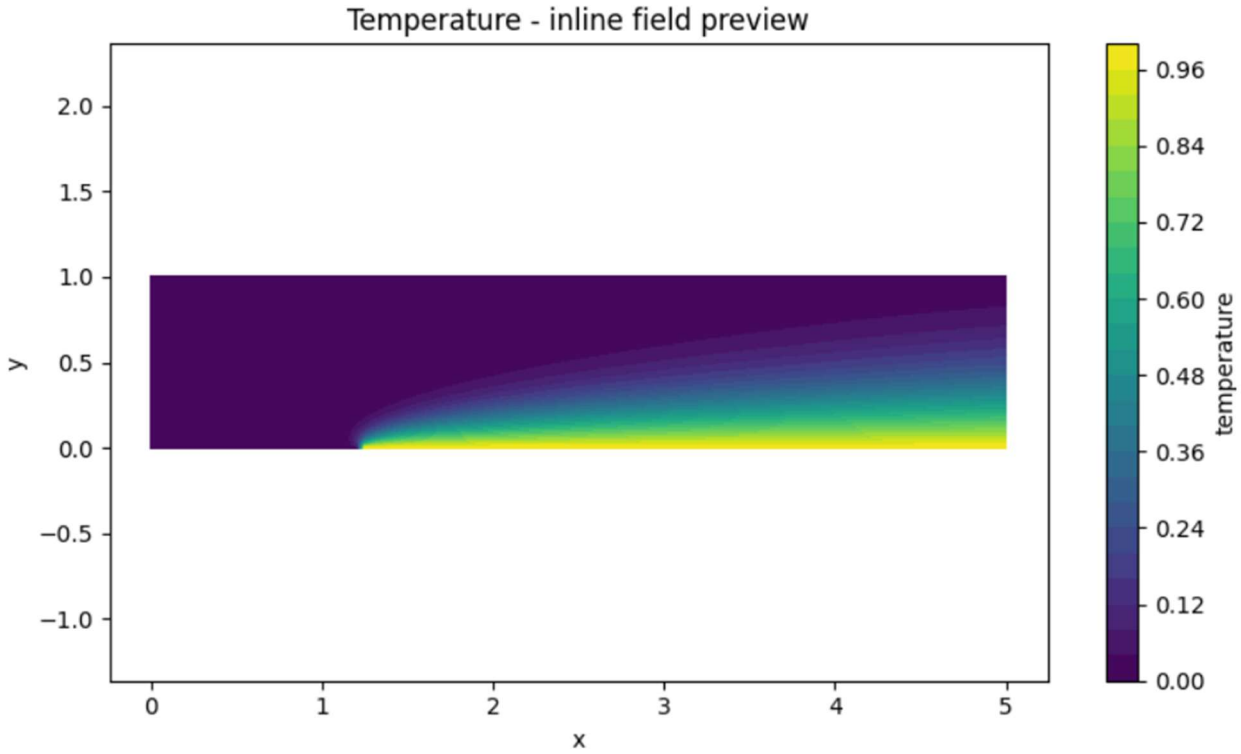
Numerical Method: Finite-volume-style transient iteration with upwind convection and central diffusion for three prescribed velocity fields.

Observations: Residual, outlet or mean temperature, average wall flux, scalar boundary-layer thickness, local Peclet number, and field shape at Peclet numbers 50, 200, and 1000. Compare field shape, boundary-layer thickness, wall flux, outlet or mean temperature, and convergence.

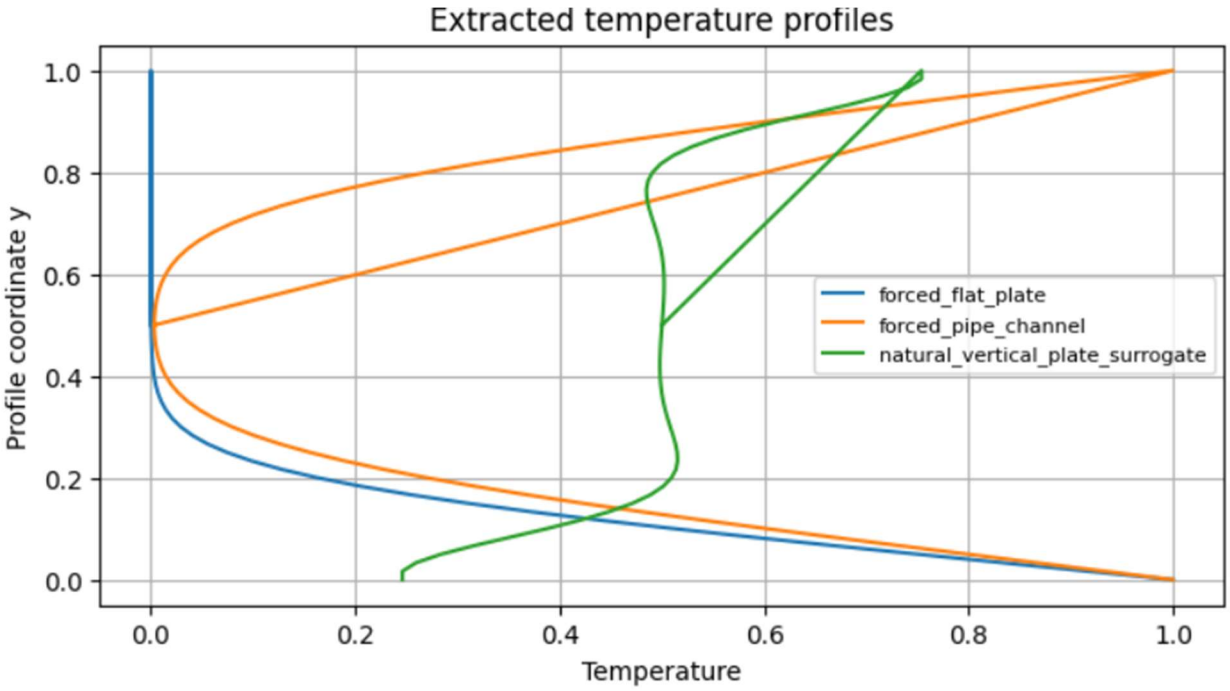
For Peclet number 50:

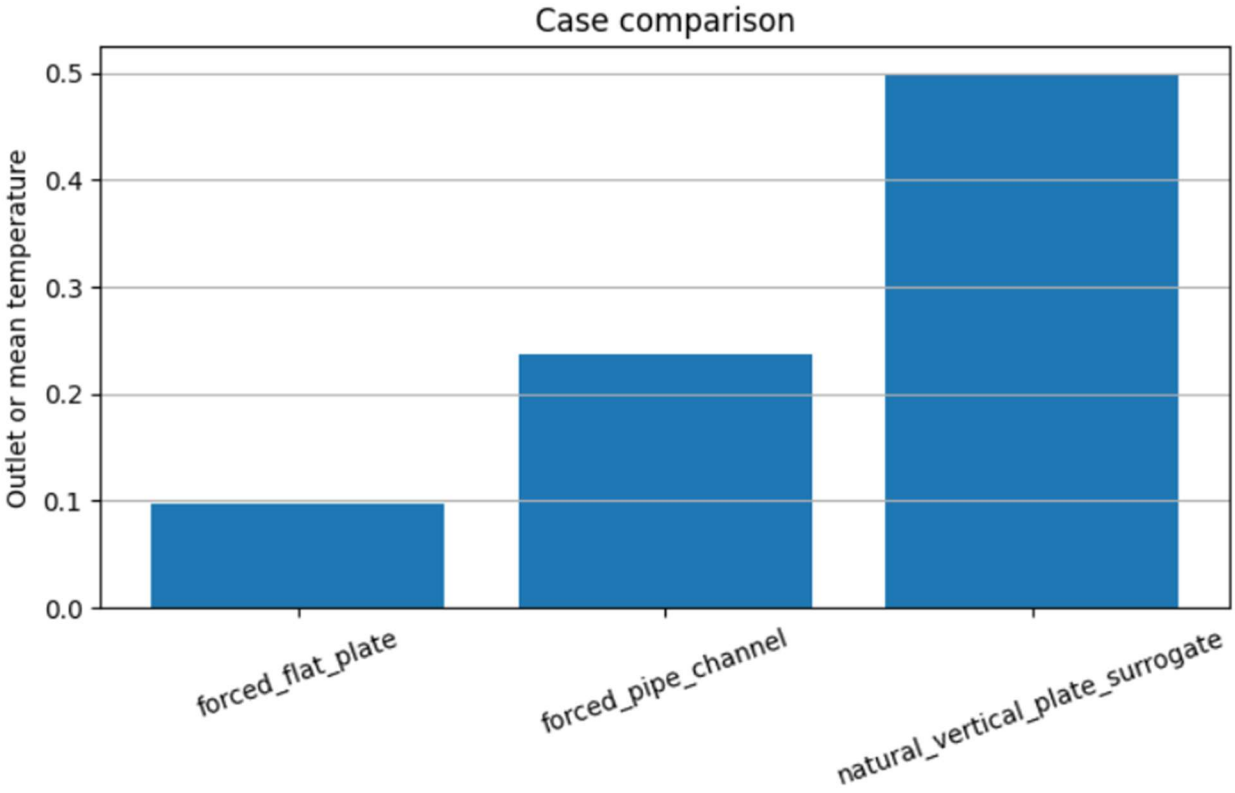


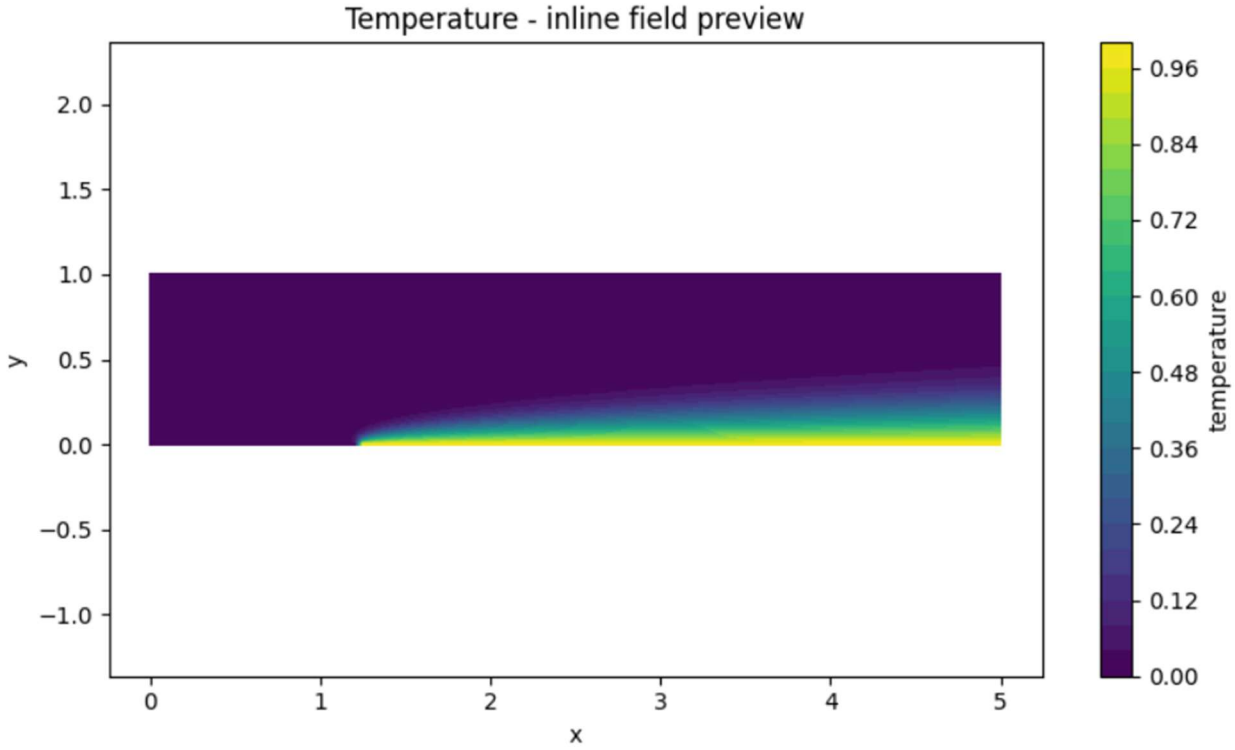




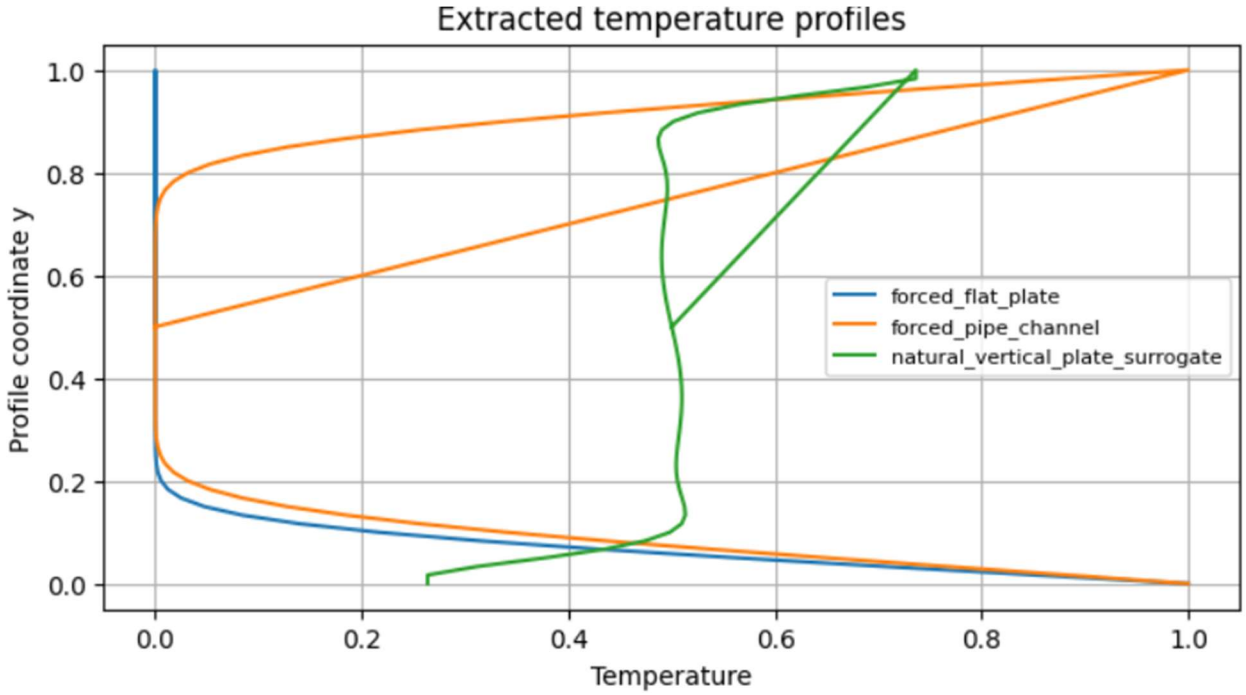
For Peclet number 200:

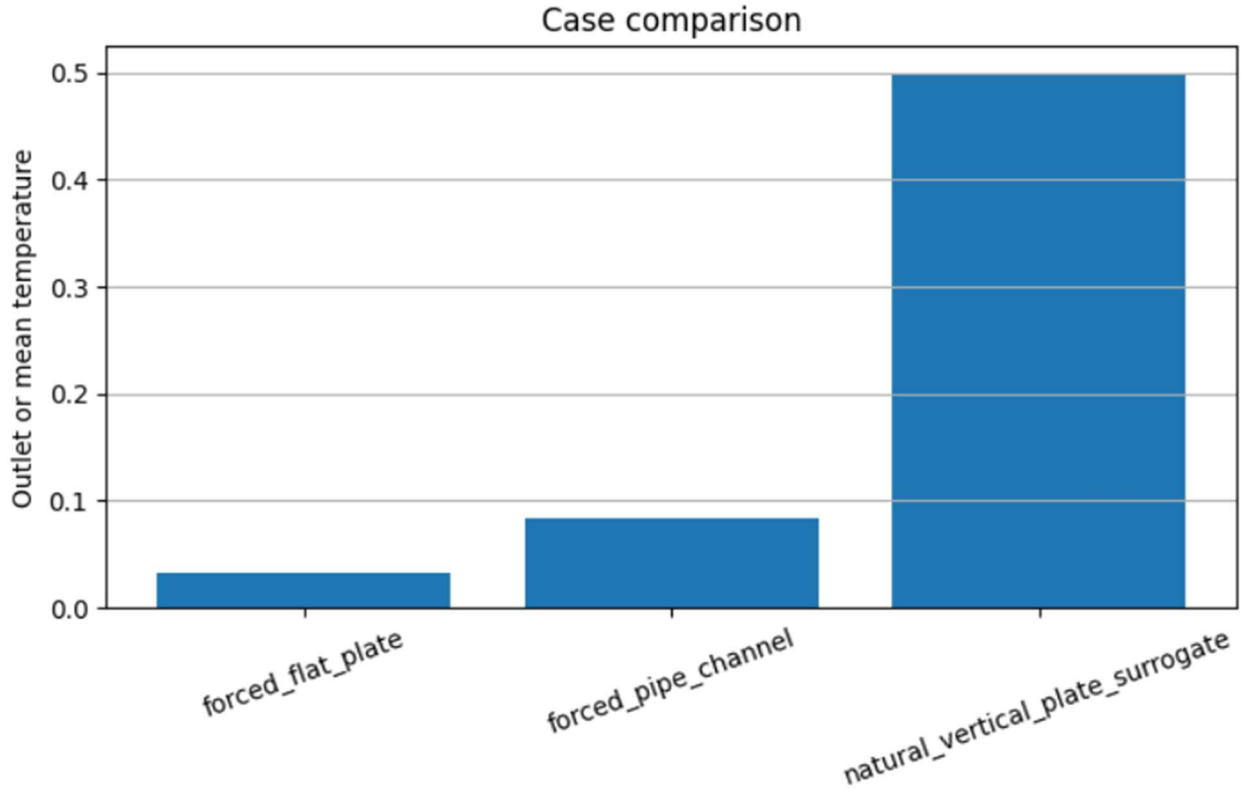


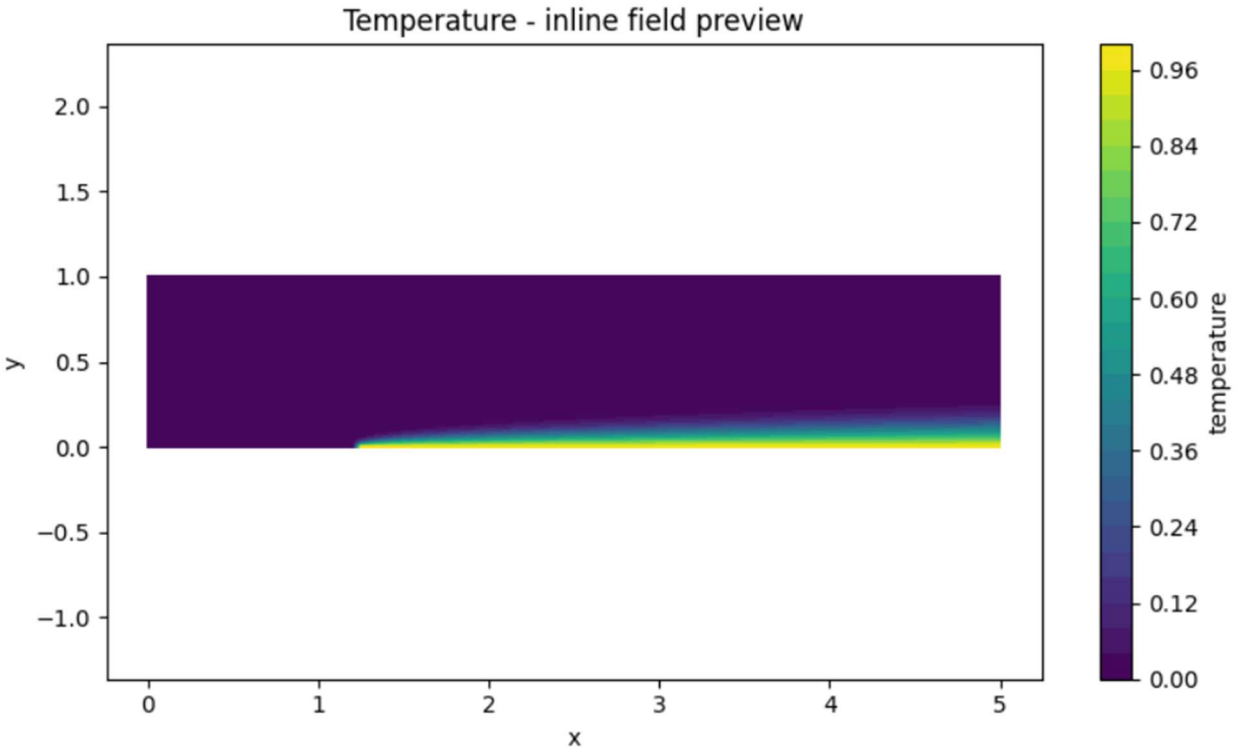

Case comparison
Outlet or mean temperature
0.0
0.1
0.2
0.3
0.4
0.5
forced_flat_plate
forced_pipe_channel
natural_vertical_plate_surrogate


Temperature - inline field preview
y
x
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
1
2
3
4
5
temperature
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

For Peclet number 1000:







Results:

Parameter	**Pe = 50**	**Pe = 200**	**Pe = 1000**
Outlet / mean temperature: forced flat plate	≈ 0.24	≈ 0.097	≈ 0.032
Outlet / mean temperature: forced channel	≈ 0.565	≈ 0.237	≈ 0.083
Outlet / mean temperature: natural-convection surrogate	≈ 0.50	≈ 0.50	≈ 0.50
Scalar boundary-layer thickness at x = 5 (flat plate)	≈ 0.8–1.0	≈ 0.45	≈ 0.2
Channel centreline temperature (profile minimum)	≈ 0.18	≈ 0	≈ 0
Field shape, forced cases	Thick, diffusive layer	Thinner layer	Thin layer hugging wall

Conclusion:

Higher Peclet numbers make convection take over, which squishes the thermal boundary layer against the wall and drops the outlet temperature for the forced plate and channel. The channel stays warmer than the plate because heat is trapped between two walls instead of escaping into open flow. Meanwhile, the natural convection case stays centered at a mean temperature of 0.5 because the recirculating loop just mixes everything in the middle regardless of Pe. To wrap this up properly, we still need the wall flux and convergence plots to check the actual heat transfer rates and solver stability.